# Supplementary material for *Autonomous Perception Through Emergent Signs: The Conditions for an Emergent Division of Labour Among Independent Learning Agents*

Anonymous authors

12 October 2026

## Contents

**1 What the measurements come to** **3**

**2 The market** **4**
2.1 The two kinds of agent . . . 4
2.2 What an agent is made of . . . 4
2.3 The names of the things that can be made . . . 5
2.4 Plans, products, and what a purchase is . . . 5
2.5 Finding a seller: the draw by sign . . . 5
2.6 Delivery, scoring and payment . . . 6
2.7 What each agent is trying to do . . . 6
2.8 Learning . . . 6
2.9 What an agent cannot see . . . 7
2.10 The control: the sign-blind twin . . . 7
2.11 Births and deaths . . . 7
2.12 Initialization, seeds and the record . . . 8
2.13 One round, in order . . . 8

**3 The cipher world** **8**
3.1 What is bought and sold . . . 8
3.2 Who can make what . . . 9
3.3 What the consumers do . . . 9
3.4 What a learner earns for a part . . . 9

**4 The machine-learning world** **10**
4.1 What is bought and sold . . . 10
4.2 Who can make what . . . 10

**5 Seeding a fresh world with a mature one's practices** **10**

**6 When buying a part pays more than making it** **11**
6.1 Staying on a part . . . 11
6.2 Building a whole . . . 11
6.3 The hiring inequality . . . 12
6.4 Finding against keeping . . . 12
6.5 Why the consumers want the smaller goods too . . . 12
6.6 A measured check of the holding ceiling . . . 13

**7 How every number was computed** **13**
7.1 A reading at a round . . . 13
7.2 What was delivered, and who made it . . . 14
7.3 Roles legible from the signs . . . 14
7.4 What the consumers obtained . . . 15
7.5 Concentration of income . . . 15
7.6 Score against income . . . 15
7.7 Trade between learners, in tokens that moved . . . 15
7.8 The statistical tests . . . 15
7.9 The words the per-world tables use . . . 16

**8 The predictions registered before the runs, and what became of them** **16**

**9 Every set, and what it produced** **18**
9.1 Cipher world, the whole at 240: 14 seed pairs . . . 18
9.2 Cipher world, the whole at 120: 10 seed pairs . . . 19
9.3 Cipher world with prices learned instead of listed: 8 seed pairs . . . 19
9.4 Cipher world without the tree of wants: 8 worlds . . . 20
9.5 Cipher world with parts paid in full: 12 seed pairs . . . 20
9.6 Machine-learning world, the main set: 12 seed pairs . . . 20
9.7 Machine-learning world, the second set: 20 seed pairs . . . 21
9.8 Machine-learning world, no learner able to reach the whole without spending its whole plan: 8 seed pairs . . . 21
9.9 Machine-learning world with every learner at 2 lines: 12 seed pairs . . . 22
9.10 Births and deaths in the machine-learning world: 8 seed pairs . . . 23
9.11 Births and deaths in the cipher world: 8 seed pairs . . . 23
9.12 Seeding a fresh world from a lone builder's practice: machine-learning world 714 . . . 24
9.13 Seeding a fresh world from a team's practice: machine-learning world 517 . . . 24
9.14 Seeding fresh cipher worlds . . . 24

**10 A table for every world** **25**

**11 Other configurations whose records are published with these** **60**

**12 What a run costs** **60**

**13 What is in the repository** **60**

# 1 What the measurements come to

Every world was run twice on the same random seed. In one run an agent finds a seller by the similarity between the sign it has learned to seek and the sign the seller has learned to display; in its *sign-blind twin* the signs are still learned and still paid for, but every offer is drawn with equal weight. Whatever differs between the two was produced by the agents' use of signs and not by the market's rules, its prices, its goods or its seed.

**A sign system formed, and roles with it.** Sellers came to be legible as kinds of worker: an outsider who sees only the other sellers' signs can tell what a seller sells, on both accuracy and mutual information against a permutation null. That holds in 12 of 14 cipher worlds with signs against 2 of 14 twins at round 20,000 ($p = 0.0002$), in 10 of 12 machine-learning worlds against 1 of 12 twins at the end ($p = 3.2 \times 10^{-4}$), in 10 of 12 cipher worlds of the second setting against 0 of 12 at 10,000, and in 12 of 20 further worlds against 0 of 20 at 20,000. Table 1 gives all nine sets: roles formed in most worlds with signs and in almost no twin. No agent was told what to make, whom to buy from, or what a sign means.

**The consumers obtained more.** Their utility was higher with signs on 12 of 12 seed pairs of the machine-learning main set ($p = 2.4 \times 10^{-4}$), on 20 of 20 of the second set ($p = 1 \times 10^{-6}$), on 8 of 8 of the capacity arm ($p = 0.0039$, the smallest value attainable at that number of pairs), and on 12 of 13 cipher pairs at 10,000 ($p = 0.0017$). The gain came through routing: the goods that few agents sell were found and paid for far more often with signs.

**Where no agent could make a whole alone, the wholes were assembled by agents who found each other through the signs.** With every learner's plan too short to hold a whole, consumers paid for wholes in 4 of 12 worlds with signs and 0 of 12 twins at 10,000 ($p = 0.047$), and every one of those wholes was assembled from parts bought from other agents. In the arm where reaching a whole alone costs an agent its entire plan, 4 of 8 worlds with signs sold wholes against 0 of 8 twins ($p = 0.038$), and in two of those four every whole was assembled while half or more of all consumer payments passed on to a supplier.

**Assembly itself is not the signs' doing; its scale is.** Twins assemble wholes too, which the hard keys and the series of smaller goods produce on their own. Pooled over the 14 cipher pairs at 20,000, the worlds with signs delivered 184,246 fully right wholes of which 97,795 were assembled, against 18,799 of which 5,723 in the twins: about twenty times as many assembled wholes, through supply chains that held.

**What formed outlived the agents that formed it.** Under deaths at 0.000125 per learner per round and births from the same distribution the founders were drawn from, about 1.3 of the sixteen founders remain at round 20,000 and 0.1 at 40,000, and a newborn inherits no sign, no want and no practice. Roles still formed in 6 of 8 machine-learning worlds with signs against 0 of 8 twins at the end, utility was still higher on 8 of 8 pairs, and two worlds grew whole-assembling teams among agents born into the market, neither of whose twins scored a whole. The order was held by the population and not by its members.

**What one agent carries can be dealt; what a society holds has to be grown again.** Dealing a mature world's practices to a fresh population replays a lone builder's work at once —

its practice is its recipe, and 5 of 5 seeded worlds rebuilt its monopoly against 1 of 5 blank starts — while a monopoly that assembled its wholes through other agents did not come back from the same deal, and came back only when half the population was dealt and the other half coevolved into the suppliers it needed. Dealt without the cipher keys, the division of labour alone was enough to rebuild wholes that the same practices *with* their keys had not produced. The distance between a dealt society and the society that grows from it is small for what one agent does alone and large for what agents do together.

**The conditions are the finding, so the failures matter as much.** Read the consumers' wants from a single number instead of a tree, so that a small change of mind lands on an unrelated good, and roles form in 0 of 8 worlds. Let the agents learn prices as bands instead of posting them, and assembly nearly vanishes — from (0/6,870) to (43/5,354) of wholes assembled in 7 of 8 worlds with signs — because a part then costs an assembler more than the whole it goes into pays. Score a bought part over a whole category of data rather than the buyer's own corpus, and the trade between agents still routes and still assembles while carrying almost none of the money: 303,848 settlements between learners for 123,579 tokens against 226 million paid by consumers. Under turnover in the cipher world the roles read by product falls away by the end of the runs, to 3 of 8, while the reading by head and the assembly of wholes hold.

**What the numbers do not settle.** Every world has 16 learners and 11 consumers, one kind of learner, and one size of data: texts of 1,000 characters in the cipher world, corpora of 1,000 documents in the machine-learning world. No set tests another learner, another scale, or agents that arrive already able to talk. Several sets are four or eight pairs, where a sign test reaches $p < 0.05$ only on a unanimous or near-unanimous result and a four-pair set cannot reach it at all, and no correction for multiple comparisons is applied across the sets and milestones.

## 2 The market

### 2.1 The two kinds of agent

A world contains 27 agents and nothing else: no bank, no auctioneer, no broker, no matchmaker, no registry of who sells what.

Eleven of them are *consumers*. A consumer has fixed wants that it never changes. It buys programs, runs each one it receives on data of its own, measures how well it performs, and pays by the result. Consumers are the source of all the money in the world and the only judges of quality.

Sixteen of them are *learners*. A learner changes what it does by trial and error on what it earns. It can make parts of programs, buy parts from other learners, and sell what it has put together. No learner is told what to make, whom to buy from, or what to sell. Nothing distinguishes one learner from another at the start except the random numbers it begins with.

Time passes in *rounds*. In each round every agent offers what it has, every purchase attempt finds at most one seller, the deliveries are run and paid, and every learner adjusts itself towards what paid. A run is 40,000 rounds.

### 2.2 What an agent is made of

An agent is one list of numbers between 0 and 1, its *genome*. There is nothing else to an agent: no memory of past rounds, no model of the other agents, no record of who paid it. Everything the agent does on a round is read off the genome it holds on that round.

The genome is read off into two things. The first is the agent's *displayed sign*, a vector of sixteen numbers. This is everything another agent can perceive of it. The second is the agent's *plan*, the list of actions it will take.

The genome's length depends on the world's settings — how many lines the plan has, how many goods the world holds, how long a sign is — and in these worlds it runs from about 150 to about 404 numbers.

### 2.3 The names of the things that can be made

Everything that can be made, bought or sold has a name, and the names form a tree four levels deep: family, kind, class and leaf. A leaf is one concrete thing, for instance the decoder that undoes one particular substitution key on one particular class of letter positions.

A name is written as one number per level. In each level's range the lowest tenth means *stop here* and the rest of the range is divided among that level's children. Two consequences matter, and they are the reason for this encoding. First, names that are close in the tree are close as numbers, so a small change in a genome moves an agent to a neighbouring thing rather than to an unrelated one: a search that proceeds by small steps can walk from one product to a related product. Second, a name may stop early and so denote a category rather than one leaf. A purchase request that stops early asks for anything in the category it names. An offer that stops early sells that category's *designated default*, the member of the category that the world's goods most often need — common, and therefore wrong for everything that needs one of the others.

### 2.4 Plans, products, and what a purchase is

A learner's plan is a list of *lines*. Its length is the learner's *capacity*, fixed by the design of the world at 2, 3 or 16 lines, and invisible to every other agent.

Each line has a kind, an item name and a sign. The kinds are *sell*, which offers a product; *construct*, which makes a part with the learner's own hands; *buy*, which asks another agent for a part; and *stop*, which ends the plan. When a genome is read off, a line is a stop with probability 0.1 and one of the other three kinds with probability 0.3 each.

A *product* is what one sell line offers. It consists of that sell line together with the construct and buy lines that follow it, up to the next sell line or stop, nested in order with the sell line outermost. A buy line inside that stretch splices in the whole product of whichever supplier the line found. So a plan reading [sell classifier, construct labeller, construct vectoriser] is the program `classifier(labeller(vectoriser(text)))`, which is then run on the buyer's data. A plan may hold several sell lines and therefore offer several products.

A purchase — one buy line — carries everything needed to ask for a part and to judge it: the category of thing wanted, the test and the data on which the delivered part will be scored, and the sign the buyer seeks in a seller. One purchase finds at most one seller per round.

A consumer's plan has buy lines only. A consumer holds one standing want per good, with its own sought sign, and makes three independent purchase attempts for each good on every round, so it can be served up to three times per good per round and a single unlucky draw does not cost it the good.

### 2.5 Finding a seller: the draw by sign

Every sell line in the world is an offer, and every offer is visible to every buyer as a displayed sign. The item the offer names is not visible.

For each purchase, one offer is drawn at random from all of them. The weight of an offer is

$$(c - c_{\min})^4,$$

where $c$ is the cosine similarity between the sign the purchase seeks and the sign the seller displays, and $c_{\min}$ is the lowest such similarity in the pool of offers on that round.

Each term of the weight has a purpose. Subtracting the pool's lowest similarity means the worst-matching offer in the world receives no tickets at all, so a sign that matches nobody's search earns nothing. Raising the difference to the fourth power turns a modest advantage in similarity into a large advantage in chance, so a sign is worth learning, while every offer that matches at all stays reachable. Drawing one offer and no more means a purchase that finds the wrong seller has lost the round.

The item the drawn offer names is compared with the category the purchase asked for only *after* the draw. If it does not lie in that category, nothing is traded by that purchase on that round. The sign therefore does all of the finding, and the category check can only refuse what the sign found.

### 2.6 Delivery, scoring and payment

A drawn offer is settled by building the seller's product: its own construct lines make their parts, and each buy line inside it is resolved to the supplier that line drew, recursively. If a supplier's own purchases are not yet settled on this round, the buyer waits for them. The finished program is run on the buyer's own data and scored by the buyer's own test.

A consumer pays the good's list price times the cube of the score. Cubing the score makes quality worth far more than mere presence: a program that is nearly right does not earn nearly the full price. A whole program that is incomplete scores zero and earns nothing, however much of it is there.

A part sold between learners is paid 4 tokens times the cube of its score, and a supplier is paid on every sale of every product its part is inside. A trade credits the seller's tokens and debits the buyer's by the same amount, so tokens are conserved between learners and enter the world only from consumers.

### 2.7 What each agent is trying to do

A consumer's reward is a Cobb–Douglas utility over the qualities of the goods it obtained,

$$U = \prod_g (0.3 + q_g)^{1/n_g},$$

over the world's $n_g$ goods, where $q_g$ is the quality obtained for good $g$ and 0.3 is what the term contributes when the good is not obtained at all. Because the terms multiply, a consumer gains more from obtaining some of every good than from obtaining a great deal of one, and a want that brings nothing drags the whole product down. Consumers do not learn; the form matters because it is the shape of the demand the learners face.

A learner's reward on a round is the tokens it earned minus the tokens it paid its suppliers. Nothing else enters it: not the quality it delivered, not how many agents it served, not whether the market as a whole did well.

### 2.8 Learning

Each agent holds its own copy of the evolution strategy CMA-ES, with the full covariance update and an initial step size of 0.1, and uses it to search its own genome. It tries one candidate genome

per round and is scored on what that candidate earned. When it has tried as many candidates as the strategy's population size — the CMA-ES default for the genome's length, which is 19 to 22 candidates for genomes of 150 to 404 numbers — the strategy ranks them, moves its centre towards what paid, and adapts its step size and covariance. A generation of learning therefore takes 19 to 22 rounds.

This is learning from reward, each agent learning only from the consequences of its own actions, carried out by an evolution strategy rather than by a gradient. No gradient, no reward and no advice is shared between agents. Because what a candidate earns depends on what every other agent happens to be doing in those rounds, the searches are coupled: the market is a coevolutionary system, and a genome that paid well last generation can be worthless in the next because the others moved.

### 2.9 What an agent cannot see

A learner perceives of another learner exactly one thing: the sign it displays. Not its plan, not its capacity, not what it sells, not what it has earned, not what it did on earlier rounds, not whether it is a learner or a consumer. Nothing passes between agents except displayed signs, delivered programs and payments.

The measurements rest on this restriction. No agent is given another agent's interpretation of a sign, so any agreement about what a sign is worth has to be built up by each side separately, from its own rewards.

### 2.10 The control: the sign-blind twin

Every world was run twice. The second run, its *sign-blind twin*, is the same world on the same random seed with one change: every offer is drawn with equal weight, as if every sign matched every search equally. The signs are still there, still learned, still paid for — they simply no longer steer the draw. Nothing else differs, and the two runs see the same random numbers in the same order.

Whatever appears in a world and not in its twin was therefore produced by the agents' use of signs and not by the rules of the market, the prices, the goods or the seed.

### 2.11 Births and deaths

In the versions of each world that test whether a structure outlives the agents that formed it, learners die and new learners are born.

On every round each learner dies with probability 0.000125, by chance alone and for no other reason: not poverty, not age, not failure. That is a mean life of 8,000 rounds. Deaths are independent, so of the sixteen learners present at the start about $16e^{-2.5} = 1.3$ are expected to remain at round 20,000 and $16e^{-5} = 0.1$ at round 40,000.

Births arrive at 0.004 per round, up to a ceiling of sixteen living learners, so a vacancy left by a death is refilled within a few hundred rounds and the population neither grows nor shrinks.

A newborn is built from a fresh draw of the same distribution the founders were drawn from at round zero: every coordinate of its genome independent and uniform between 0 and 1. Nothing is inherited and nothing is copied. It has no parent. It does not receive a sign, a want, a product, a price, a customer, a supplier, any tokens, or any part of another learner's search. Its sign is random, its wants are random, its plan is random, and it begins its own search from that random centre at the opening step size. Whatever place it comes to take in the market it has to find from the outside, through the same offers and payments as everybody else.

The eleven consumers never die, so demand is unchanged while the learners turn over.

### 2.12 Initialization, seeds and the record

Every learner's genome is drawn uniformly on $[0, 1]$ at the start. The consumers are authored: their wants, their sought signs and their list prices are set by the world's design and never change.

Two seeds are pinned, the Python hash seed and the world seed, so that a run can be repeated number for number, and a world and its twin always share the seed.

A run is 40,000 rounds unless it was stopped early to free a machine; the sets stopped early are named with their results (Section 9). Every tenth round is logged: every agent's sign and plan, every draw, every settlement, every payment and every consumer's utility. The logged rounds are the record from which every number is computed.

### 2.13 One round, in order

1. Every agent reads its current candidate genome off into its displayed sign and its plan, and posts them.
2. Every sell line in the world becomes an offer.
3. For each purchase of each agent, one offer is drawn with weight $(c - c_{\min})^4$ on the similarity between the sought and displayed signs (Section 2.5); in a sign-blind twin, with equal weight.
4. The drawn offer's item is checked against the category the purchase named. If it does not lie in it, that purchase trades nothing this round.
5. Each surviving draw is settled: the seller's product is built, following its own construct lines and resolving each of its buy lines to the supplier that line drew, waiting where a supplier's own purchases are not yet settled.
6. The delivered program is run on the buyer's data and scored by the buyer's test. Consumers pay the list price times the cube of the score; learners pay 4 times the cube of the score for a part; an incomplete whole pays nothing.
7. Each agent's earnings for the round are totalled: tokens received minus tokens paid out.
8. Every learner reports the round's earnings to its own copy of the evolution strategy, which, once it has seen a full generation of candidates, moves its centre towards what paid (Section 2.8). In the birth-and-death versions, deaths and births are drawn (Section 2.11).

## 3 The cipher world

### 3.1 What is bought and sold

The goods of this world are programs that decode enciphered text.

A text is enciphered as follows. Each letter belongs to one of four *position classes*, given by its position in its word modulo four. Each class is enciphered with one of four possible substitution keys, so a text is scrambled by four keys at once and a decoder must undo the right key on the right class. The consumers' texts are 1,000 characters long.

A *part* is a decoder that undoes one key on one class. The *whole* is a program that decodes all four classes of a text. Because each class could have been enciphered with any of its four keys, a

decoder that is right in form but wrong in key decodes nothing: this is a *hard key*, and it is the reason a whole is hard to produce by accident.

The eleven consumers each want seven goods: the four texts with one class decoded, the two with two classes decoded, and the whole text. The list prices are 10 for a one-class text, 30 for a two-class text and 240 for the whole. A second version of the world is identical except that the whole is priced at 120, which tests whether the structure depends on how much more the whole is worth.

A class counts as decoded only when at least 99 per cent of its letters come out right, so a wrong key earns nothing through lucky letters. A one-class or two-class good scores the share of its classes that are decoded. The whole scores 1 when every class is decoded and 0 otherwise: it is paid all or nothing, and three classes out of four are worth exactly as much as none.

### 3.2 Who can make what

Of the sixteen learners, eight have plans of 3 lines and eight have plans of 16. A learner with 16 lines has room to construct all four decoders and sell the whole by itself. A learner with 3 lines does not: it can sell the whole only by spending a line on buying a part from another learner. Capacity is invisible to the other agents, and the market contains no announcement of who has which.

### 3.3 What the consumers do

Each consumer wants all seven goods, with utility $U = \prod_g (0.3 + q_g)^{1/7}$ over the qualities it obtains, and makes three purchase attempts per good per round. Each want has its own sought sign, which the consumer does not change.

A consumer's want is read as a tree in the sense of Section 2.3: first how many classes the good decodes, then which good within that group. A small step in the genome therefore moves a want to a neighbouring good rather than to an unrelated one. The control for this reading is the same world with a want read from a single number cut into seven equal slices, so that neighbouring numbers need not be related goods. That control is the world *without the tree* (Section 9.4).

### 3.4 What a learner earns for a part

Two settings of the same world differ in what a learner can be paid for a part: configuration A and configuration B.

In **configuration A** a learner's test can name only a category of data, not one corpus. A part bought from another learner is therefore scored as the mean over every corpus of the category — 28 corpora, of which 21 belong to no good of this world — so a part that is exactly right for the buyer's purpose still scores a few hundredths, and is paid 4 times a few hundredths cubed. The part trade in configuration A routes buyers to sellers and assembles wholes, but carries almost no money: in world 312 of that configuration, 303,848 settlements between learners paid 123,579 tokens in total, against 226 million paid by consumers.

In **configuration B** a learner's test can name the buyer's own corpus, and the data it may name is restricted to the goods of the world. A part that is right then scores 1.0 and is paid 4.0. Under configuration B a part therefore carries real money, and the share of a consumer's payment that passes on to a supplier is a meaningful quantity.

# 4 The machine-learning world

## 4.1 What is bought and sold

The goods of this world are programs that classify documents. There are two corpora: movie-review sentiment and fake-news detection, 1,000 documents each.

Each corpus has three goods, each one built on the one before it. An *embedding* turns documents into vectors. A *labelled embedding* is an embedding with the corpus's labels attached to it. A *whole classifier* is a program that classifies documents of that corpus. Six goods in all across the two corpora, at list prices 30, 60 and 240 times the cube of the delivered score. A delivered classifier is scored by its F1 score on the buyer's data, and counts as scoring when that is above zero. The parts that a category's designated default supplies were measured offline once, so that a delivery's score does not depend on what else the machine was doing.

As in the cipher world, a consumer's want is read as a tree: first which of the three goods, then which corpus. A part bought from another learner is judged by the test and the data the buyer's purchase named and is paid 4 times the cube of its score.

## 4.2 Who can make what

In the main set eight learners have plans of 2 lines and eight have plans of 16. Building a whole classifier alone takes three lines, one for each good in the chain. A learner with 2 lines can therefore sell a labelled embedding or a whole classifier only by buying the good below it from another learner, while a learner with 16 lines can build everything by itself.

A second arm, the *capacity arm*, changes nothing but the plan lengths: eight learners with 2 lines and eight with 3. In it a learner with 3 lines can reach the whole alone, but only by spending its entire plan on that one chain, selling nothing else.

A third arm gives every learner 2 lines, so that no learner in the world can produce a whole classifier without buying the good below it. Whatever wholes appear there are necessarily assembled from bought parts, on both sides of the comparison.

# 5 Seeding a fresh world with a mature one's practices

A separate procedure asks what of a mature society one agent carries and what only the society holds. A fresh world is given the statistics measured on a mature one and nothing else. This is coevolutionary reconstruction, after the proposal in Duong, Pearman and Bladon (2013), and the difference between the shares dealt and the shares the fresh society sustains is the distance between the two.

Two things are measured over the last 2,000 rounds of the source run, for each of its learners. The first is its *practice*: the lines of the first product it sells, as (kind, item) pairs — what it makes, what it buys and what it offers, with no prices, no tests and no signs. The second is the share of learner-rounds that practice holds among the source's learners of the same capacity.

A fresh world's learners are then dealt practices in those shares. An individual learner is dealt one practice, which becomes its starting plan. It is not told the shares, it is not told that other practices exist, and it is given no price, no test and no sought sign: those it must find for itself, as must every arrangement between learners. The world is otherwise a new world, with new random genomes everywhere the practice does not reach.

There are four levels of seeding, which differ in how much is dealt:

**Presence** the practice together with the average displayed sign of the learners that held it in the source, so that the newcomers start with the source's signs as well as its work.

**Outward** the practice alone, with random signs.

**Half** the practice to every other learner, the rest of the population starting at random, so that half of a society is dealt and the other half must coevolve into whatever the dealt half needs.

**Keyless** the practice with every decoder key redrawn at random, in the cipher world. This carries the division of labour — who makes a part, who assembles, who sells — and nothing of the solution to the keys, which the new society must find for itself.

A seeded learner starts its search at a step size of 0.02 rather than the usual 0.1, so that the practice it was dealt survives its first generations instead of being lost in its first candidates; the strategy adapts the step size from there as for every other agent. The keyless worlds, which have to find keys, and the blank starts run at 0.1.

Each seeded world is read beside a *blank start*: the same world on the same seed with nothing dealt. A *powered* set is one run with five seeded worlds and five blank starts per source, so that a sign test over the pairs can reach $p < 0.05$; the single-source sets have one world per source and level and describe rather than test.

# 6 When buying a part pays more than making it

## 6.1 Staying on a part

A learner offers one candidate per round, drawn around the centre of its search with step size $\sigma$. Suppose the genome numbers that name a part have to fall within a slice of width $w$ for the part to be the right one. If the learner's centre sits in the middle of that slice, the chance that a candidate also falls inside it is

$$s = P\big(|z| < \tfrac{w}{2\sigma}\big) = \operatorname{erf}\Big(\tfrac{w}{2\sqrt{2}\,\sigma}\Big), \tag{1}$$

with $z$ standard normal. $s$ is the rate at which the learner *holds* the part. With a hundred or more numbers in a genome the full covariance update adapts slowly, so one step size governs the precision of most of the numbers, and Eq. 1 is a ceiling on any learner's holding rate rather than a prediction for a particular one.

## 6.2 Building a whole

A good made of $k$ parts is right on a round only if every one of its parts is right on that round, so its rate is the product of the parts' holding rates. A learner that makes all $k$ parts itself, with step size $\sigma_a$, delivers a right whole on a share

$$r_{\text{alone}} = \prod_i s_i(\sigma_a)$$

of rounds. An assembler that buys parts from specialists whose step size is $\sigma_p$ delivers one on a share

$$r_{\text{asm}} = \delta \prod_i s_i(\sigma_p),$$

where $\delta$ is the chance that its purchase draws a seller that actually has the part it needs — a factor $\delta$ for each bought input, and the goods of these worlds need one at most.

### 6.3 The hiring inequality

Let $P_w$ be the price of the whole and $P_p$ what the assembler pays for parts. Assembling is selected over building alone only when

$$r_{\text{asm}}\,(P_w - P_p) > r_{\text{alone}}\,P_w. \tag{2}$$

In a world where every part is easy to stay on, every $s_i$ is near 1 for anybody, so $r_{\text{asm}}$ and $r_{\text{alone}}$ are both near 1, the right-hand side approaches $P_w$, and building alone wins however well the signs work. Buying can only win where the whole needs more lines than the learner has, or where a lone builder rarely gets all the parts right at the same time.

### 6.4 Finding against keeping

At equal step sizes the two products in $r_{\text{alone}}$ and $r_{\text{asm}}$ are the same, so in the long run hiring changes the rate at which a whole appears only through precision. What hiring saves is *finding*: a part must be found before it can be held, and a part found once by a specialist serves every assembler that draws it. A part is therefore worth buying when it is hard to find and easy to keep.

Both worlds are built to make the whole hard to find. Each cipher class has several possible keys, so a whole needs the right key in all four classes at once; the whole is paid all or nothing; and the goods below the whole are paid on their own, so a learner can be paid for progress without having to reach the whole.

Consider a learner that already sells a two-class text. To reach the whole by itself it must find the two remaining decoders in a single candidate. To reach it by buying, it needs one more line, a purchase whose draw by sign must land on a seller of the other half. Replacing the two-class good by an assembled whole pays only when

$$\delta\, s_{\text{half}}\, P_w > P_{\text{two}}, \tag{3}$$

where $s_{\text{half}}$ is the chance that the supplier's half is right on the round and $P_{\text{two}}$ the price of the two-class good. With the list prices of these worlds, where the whole is worth 8 two-class texts in the cipher world and 4 labelled embeddings in the machine-learning world, this asks that $\delta\, s_{\text{half}}$ exceed 0.125 and 0.25 respectively. Those are the numbers the sign draw has to deliver, and they are the reason the signs can decide whether a supply chain forms.

### 6.5 Why the consumers want the smaller goods too

A learner's search can only learn about a good if some of its candidates are paid for that good. With $p_{\text{make}}$ the chance that a candidate makes the good at all and $p_{\text{paid}}$ the chance that a buyer draws it and pays, the expected number of paid candidates per learner per generation is

$$E = 19\, p_{\text{make}}\, p_{\text{paid}},$$

19 being the number of candidates a generation ranks (Section 2.8). A good with $E \ll 1$ is invisible to the search: no candidate that makes it is ever paid for making it, so the ranking carries no information about it and the centre never moves towards it. Such a good can be reached only through a nearer good that is paid on its own.

This is why the consumers in both worlds want every good in the series from the smallest part to the whole, and not the whole alone. A lone builder can extend one product by one good at a time, each step paid for by the buyers of the smaller goods. The question the worlds then put is whether, under hard keys, a lone builder reaches the whole before an assembler does.

### 6.6 A measured check of the holding ceiling

Eq. 1 is a ceiling, so the check is whether real learners stay under it and whether holding falls with step size as the equation says.

It was measured on three long runs of a variant of the cipher world in which an item is read as a single number over the whole tree, which makes the slice for one decoder key 0.082 wide — narrow enough that the ceiling is well below 1 and the comparison is informative. The reading covers 48 learners (three runs, sixteen each) at about round 36,000, over the last 6,000 rounds of each run, 301 logged rounds apiece, with each learner's step size taken from its saved search state. For each learner, holding is the share of its logged rounds on which its candidate sold its most frequent decoder.

Holding falls with step size (Spearman correlation $-0.48$, $p = 0.0006$). Of the 48 learners, (46/48) held at most 0.15 more than the ceiling allows, (18/48) held within 0.15 of the ceiling, and (28/48) held less than it allows, their centres lying outside any single slice (Figure 1).

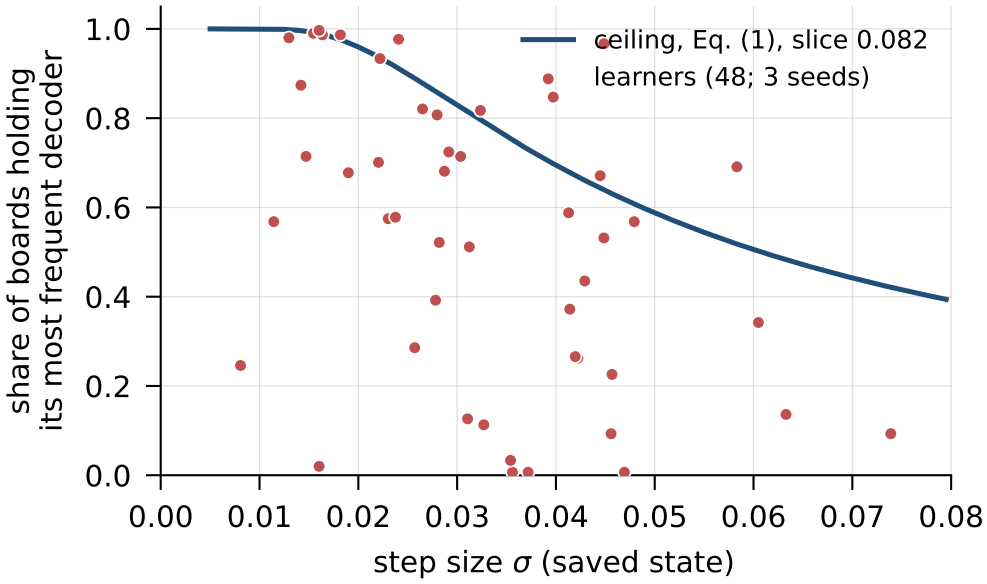



Figure 1: Holding a part against the learner's step size, in the variant of the cipher world described in Section 6.6. Each point is a learner (3 runs, 16 learners each, the last 6,000 rounds): the share of its logged rounds on which its candidate sold its most frequent decoder. The curve is Eq. 1 for that world's slice width, the rate of a learner centred in the slice.

## 7 How every number was computed

Every number is computed from the record of a run by a program published with the records. Each measurement is defined by that computation.

### 7.1 A reading at a round

A reading "at round $B$" is computed from the world's record *cut* at $B$: the logged rounds up to $B$, the settlement, payment and utility records restricted to their rows at rounds up to $B$, the world's configuration and its starting state. Nothing after round $B$ enters it.

A reading at a milestone covers the 2,000 rounds before the milestone: the reading at 10,000 covers rounds 8,000–9,999 and the reading at 20,000 covers 18,000–19,999. The reading at the end uses the whole record after the world stopped, which for a 40,000-round run is rounds 38,000–39,999.

Cutting the record matters for the comparison with the twin. Because the cut is defined by the round and not by the moment the reading was taken, a world and its twin are always read over the same span of their own histories, however long either run took in wall-clock time.

One number is not computed from a cut record: the reading behind prediction 1 (Section 8) was taken from the record as it stood up to 1,580 rounds after round 5,000, and is marked where it appears. Readings taken from a record as it stood rather than cut at a round are kept apart in the repository, under `reads/_lagged`, with the round at which each was taken in its name; no other number comes from them.

For two worlds of the control without the tree of wants (seeds 312 and 313) the raw records were overwritten by later worlds stored under the same names. Their reading files survive and are the only record of those two worlds; every other number can be recomputed from its logs.

### 7.2 What was delivered, and who made it

This reading answers, for each delivery a consumer paid for, what the delivered program was and which agents made its parts.

From the offer the buyer drew, the seller's product is rebuilt: its own construct lines, and every purchase inside it followed to the supplier's line that made the part, and every purchase that supplier made in turn, recursively. The result is the list of agents that contributed to the delivery and what each contributed.

A delivery is **assembled** when at least two learners each supplied a part the good requires: in the cipher world, a decoder of the right class and key; in the machine-learning world, one of the goods in the chain. A delivery built entirely from its seller's own construct lines is not assembled, however complicated it is.

A cipher delivery is **fully right** when it scores at least 0.99, which for a whole means all four classes decoded. A machine-learning delivery is **scoring** when its F1 score is above zero.

Counts are taken over windows of 2,000 rounds, which is 200 logged rounds.

### 7.3 Roles legible from the signs

This reading asks whether an outsider who sees only the signs can tell what a seller sells. It is the measurement behind every statement that roles formed.

Within a window of 100 rounds, a *unit* is a (seller, product) pair that appears on at least 5 logged rounds of the window, together with that seller's mean displayed sign over the window. Each unit's product is then guessed as follows: take the average sign of the units of every other seller that sold each product, and guess the product whose average is nearest to this unit's sign by cosine similarity.

The seller's own units are excluded from the averages it is guessed against. A product that only one seller sells is therefore always guessed wrong, and a sign that marks an individual rather than a kind of work earns nothing in this measurement.

Two quantities are computed from the guesses: the *accuracy*, the share of units guessed correctly, and the *mutual information* between the true and the guessed products, in bits. Both are compared with the same quantities computed after the products are permuted among the window's units, 200 permutations, so the smallest attainable $p$-value is 0.005. A span of 2,000 rounds holds 20 windows

of 100 rounds, and the span reports the mean over its 20 windows against the same mean over the permuted draws.

**Roles have formed** in a world when both the accuracy and the mutual information exceed their permuted values at $p < 0.05$. Both are required because accuracy alone can beat its null without any sign system: where a few products are much more common than the others, guessing can do better than chance from the frequencies alone, and in these runs accuracy alone exceeded its null in up to 6 of 14 sign-blind twins. Mutual information penalizes a guess that is right only because it always names the common product.

The reading is made twice: *by product*, where a unit's label is the sorted items of the whole product sold, and *by head*, where the label is the item on the sell line alone. By product is the stricter of the two and is the main measurement.

### 7.4 What the consumers obtained

The consumers' *utility* is their mean Cobb–Douglas utility per window of 1,000 rounds, computed from the qualities they actually obtained, with 0.3 contributed by a good a consumer did not obtain. A consumer may be served several times in a round for the same good, since it makes three purchase attempts per good. The lowest value the measurement can take is therefore 0.3, which is the value for a consumer that obtains nothing at all.

### 7.5 Concentration of income

The *Gini coefficient* of the learners' income from consumers is computed over the last 200 logged rounds of the span, excluding payments of zero, together with the *top earner's share*, the fraction of all learners' income from consumers that the highest-earning learner received. Income between learners is not counted here; this is the distribution of the money that entered the world.

### 7.6 Score against income

For the question whether the market pays for quality, the Spearman rank correlation is computed between a learner's mean delivery score as a seller and its income in the window, over the learners with at least five scored deliveries in a 2,000-round window, and taken as one minus that correlation, so that zero is a market that pays in the order of measured quality.

### 7.7 Trade between learners, in tokens that moved

Trade between learners is counted only where money moved: the share of consumer-paid deliveries whose seller's product contains a part bought from another learner and paid for in that same delivery. A purchase that was drawn but not settled, or settled at a price of zero, does not count.

### 7.8 The statistical tests

Paired comparisons between a world and its twin on the same seed carry a one-sided sign test over the pairs. Counts of worlds — how many worlds with signs against how many twins — carry a one-sided Fisher exact test. The roles reading carries the permutation test described in Section 7.3. The direction of every test was declared before the runs: signs above twin.

A sign test over 8 pairs cannot return less than $1/2^8 = 0.0039$, over 4 pairs less than 0.0625, and the permutation test with 200 draws cannot return less than 0.005; where a result sits at one of these floors, the floor is given beside it.

No correction for multiple comparisons is applied across sets and milestones.

### 7.9 The words the per-world tables use

The per-world tables (Section 10) use these short names.

**S, B** S is the world with signs; B is its sign-blind twin on the same seed (Section 2.10).

**right** fully right wholes delivered to consumers in the window: in the cipher world, a score of at least 0.99.

**assembled** of those, the deliveries to which at least two learners each supplied a part the good requires (Section 7.2).

**utility** the consumers' mean Cobb–Douglas utility in the last window where it was logged.

**product acc. (null)** the accuracy of guessing a seller's product from the other sellers' signs, with the permuted value in parentheses; **head acc.** the same for the item on the sell line alone.

**product MI, head MI** the mutual information of those guesses minus its permuted value, in bits.

$^{*}$, $^{+}$ accuracy above its permuted value, and mutual information above its permuted value, at $p < 0.05$. Roles have formed where both marks are present.

**configuration A, configuration B** the two settings of what a learner can earn for a part (Section 3.4): in A a test can name only a category of data, so a right part earns a few hundredths of its price; in B it can name the buyer's corpus, so a right part scores 1 and is paid 4.

**presence, outward, half, keyless** the four levels at which a mature world's practices are dealt to a fresh one (Section 5).

**blank start** the same world on the same seed with nothing dealt.

**powered** a reconstruction set run with five seeded worlds and five blank starts per source, so that a sign test over the pairs can reach $p < 0.05$.

**capacity arm** the machine-learning world with eight learners of 2 lines and eight of 3, so that reaching the whole alone costs a learner its entire plan.

**rows** the lines of a plan (Section 2); a learner's capacity is how many it has.

## 8 The predictions registered before the runs, and what became of them

Five predictions were written down before the first set of cipher runs, with the rounds at which they would be read and the thresholds that would count as holding. Two failed and two turned out not to be readable as written. Outcomes are from the readings computed from the record cut at the round named, except where noted for prediction 1.

1. **Registered:** consumer-paid two-class goods assembled by two or more learners appear by round 5,000 in at least three quarters of the worlds with signs, at the list price of 240.

   **Outcome: held, and the control shows it was not the signs.** Assembled two-class goods appeared in 14 of 14 worlds with signs — and in all 16 sign-blind twins as well. Assembling is produced by the hard keys and by the series of goods below the whole, not by the signs. This reading was taken from the record as it stood up to 1,580 rounds after round 5,000, not from the record cut at 5,000 (Section 7).

2. **Registered:** by round 10,000, purchases refused because the drawn seller's item was in the wrong category are fewer with signs than blind, and assembled goods more numerous with signs, in at least three quarters of pairs.

   **Outcome: the first half failed, the second fell short.** At the reading from the record cut at 10,000, refused draws were fewer with signs on 8 of 14 pairs, and assembled fully right wholes were more numerous with signs on 10 of 14 pairs. Neither reaches three quarters.

3. **Registered:** sellers are legible as groups from the other sellers' signs — accuracy and mutual information both above their permuted values — in at least three quarters of the worlds with signs and in fewer than a quarter of the twins, at round 10,000. This is the measurement that counts as roles having formed.

   **Outcome: short of the threshold at 10,000, met later.** At 10,000 roles had formed in 7 of 14 worlds with signs against 1 of 14 twins, short of three quarters. At 20,000 they had formed in 12 of 14 against 2 of 14, and at the stop in 11 of 14 against 0 of 14. (Counting the two twin runs on seeds 310 and 311, which have no world with signs and are excluded elsewhere, the twin count at the stop is 1 of 16.) Accuracy alone was above its permuted value in 14 of 14 worlds with signs and in 2 of 14 twins at 10,000, and in 14 of 14 against 6 of 14 at 20,000, which is why the conjunction of accuracy and mutual information is the criterion.

4. **Registered** (as uncertain): the consumers' wants read as a tree raise the consumers' routing: at 10,000 the share of consumer draws landing on a product that scores for the good the want names is higher with the tree than without, on at least 3 of 4 seed pairs.

   **Outcome: not readable as registered.** The share of draws was not measured against the control without the tree, so the prediction cannot be scored. What was read of that control is that assembled wholes appeared both with the tree and without it, and that sellers were legible as groups in 0 of the 8 worlds without it. Those readings were taken within 390 rounds of their milestones and are, for two of the eight worlds, the only surviving record (Section 7).

5. **Registered** (as uncertain): a whole assembled from three or more learners appears on at least 2 of the 4 seeds of the 240 set by round 20,000.

   **Outcome: not readable as registered.** The composition reading distinguishes a bought part from the seller's own, but does not count how many distinct suppliers a delivery had, so the number of suppliers cannot be scored. The nearest quantity that was read is wholes sold by learners with 3 lines, which hold at most two of the four decoders and so must buy the rest from one or two suppliers. At the reading cut at 20,000 these were more numerous with signs on 9 of 14 pairs: 5,417, 3,803, 862, 15,929, 70, 4,834, 1,826, 16, 4, 13,397, 14,351, 0, 0 and 9,224 with signs against 63, 396, 52, 386, 294, 379, 32, 777, 139, 0, 556, 286, 561 and 28 in the twins, on seeds 312 to 325.

# 9 Every set, and what it produced

A *set* is a group of worlds that share a design, run on a list of seeds, each with its sign-blind twin on the same seed. Table 1 gives every set at one reading; Section 10 has a table for every world at every reading.

Table 1: Every set at a glance: the world with signs against its sign-blind twin. Roles: worlds whose sellers formed roles by product (accuracy and mutual information above their shuffled nulls at $p < 0.05$), signs against twins, with Fisher's one-sided exact $p$. Utility: pairs on which the consumers' mean utility was higher with signs, with the one-sided sign-test $p$. Wholes: for the machine-learning sets the worlds with any scoring whole, signs against twins (Fisher's $p$); for the cipher sets the pairs on which more fully right wholes were delivered with signs (sign-test $p$). End = the whole record at the end of the run; otherwise the record cut at the round named.

| Set | pairs | read | roles with signs vs twins ($p$) | utility higher ($p$) | wholes |
|---|---|---|---|---|---|
| Machine-learning main set | 12 | end | 10/12 vs 1/12 (3e-4) | 12 of 12 (2e-4) | worlds selling wholes 5/12 vs 1/12 (0.077) |
| Machine-learning second set (blank starts) | 20 | 20,000 | 12/20 vs 0/20 (2e-5) | 20 of 20 (1e-6) | worlds selling wholes 9/20 vs 3/20 (0.041) |
| Capacity arm (2 and 3 lines) | 8 | end | 5/8 vs 0/8 (0.013) | 8 of 8 (0.004) | worlds selling wholes 4/8 vs 0/8 (0.038) |
| Every learner 2 lines | 12 | end | 9/12 vs 2/11 (0.009) | 11 of 12 (0.003) | worlds selling wholes 4/12 vs 1/12 (0.158) |
| Cipher, configuration A | 14 | 20,000 | 12/14 vs 2/14 (2e-4) | 11 of 13 (0.011) | right wholes higher with signs 12 of 14 (0.006) |
| Cipher, configuration B | 12 | end | 11/12 vs 3/12 (0.001) | 8 of 8 (0.004) | right wholes higher with signs 12 of 12 (2e-4) |
| Cipher, whole priced 120 | 10 | 20,000 | 8/10 vs 0/10 (4e-4) | 9 of 10 (0.011) | right wholes higher with signs 8 of 10 (0.055) |
| Machine-learning births and deaths | 8 | end | 6/8 vs 0/8 (0.003) | 8 of 8 (0.004) | worlds selling wholes 3/8 vs 1/8 (0.285) |
| Cipher births and deaths | 8 | end | 3/8 vs 0/8 (0.100) | 3 of 4 (0.312) | right wholes higher with signs 4 of 8 (0.637) |

## 9.1 Cipher world, the whole at 240: 14 seed pairs

Fourteen seed pairs, seeds 312 to 325, in the setting where a learner earns almost nothing for a part (Section 3.4). These worlds were stopped between rounds 24,000 and 31,000 to free the machines they were running on, so they have readings at 10,000, at 20,000 and at their stop. Two further twin runs, on seeds 310 and 311, have no world with signs to be compared against and are excluded from every count.

**What the consumers received.** Fully right wholes delivered to consumers in the window were more numerous with signs on 12 of 14 pairs at 10,000 and again at 20,000 (sign test $p = 0.0065$), and on 14 of 14 at the stop ($p = 0.0001$). Among the twelve, the factor by which the world with signs exceeded its twin runs from about 1.1 (seed 322) to about 1,400 (seed 318). On the two seeds where the twin delivered more wholes, 319 and 323, the world with signs was selling the smaller goods instead: 3 wholes against 216 on seed 319 and 3 against 822 on seed 323. Pooled over the fourteen pairs at 10,000, the worlds with signs delivered 71,261 fully right wholes of which 21,213

were assembled, and their twins 3,727 of which 1,075 were assembled; at 20,000, 184,246 of which 97,795 assembled, against 18,799 of which 5,723. Assembling therefore happens in the twins as well — it is produced by the hard keys and by the series of smaller goods, not by the signs — and what the signs change is its scale, about twenty times as many assembled wholes.

**What the consumers obtained.** Consumers' utility was higher with signs on 12 of 13 pairs at 10,000 ($p = 0.0017$), from 0.55 to 3.91 with signs against 0.48 to 0.84 blind, and on 11 of 13 at 20,000 ($p = 0.011$), from 0.60 to 5.16 against 0.55 to 1.64. The thirteenth pair is seed 312, whose world with signs has no utility line in its record at either reading. The two pairs where the twin was higher at 20,000 are again 319 and 323.

**Roles.** Roles formed by product in 7 of 14 worlds with signs against 1 of 14 twins at 10,000 (Fisher's exact test $p = 0.016$), in 12 of 14 against 2 of 14 at 20,000 ($p = 0.0002$), and in 11 of 14 against 0 of 14 at the stop ($p = 1.7 \times 10^{-5}$). Accuracy alone was above its permuted value in 14 of 14 worlds with signs and in 2 of 14 twins at 10,000, and in 14 of 14 against 6 of 14 at 20,000: this is the measurement that needs mutual information beside it. Read by the head item alone, roles formed in 5 of 14 worlds with signs against 2 of 14 twins at 20,000. The per-seed numbers at 10,000 are in Table 3 and the trajectory in Figure 2.

**Which kind of learner sold the wholes.** At 20,000 the set divides into two regimes. On seeds 316 and 324 a learner with 16 lines built nearly every whole by itself: (21,929/21,999) and (19,554/19,554) of the wholes sold by 16-line sellers, with only (70/21,999) and (12/19,554) assembled from a bought part. On seeds 315, 321 and 322 the wholes were sold by learners with 3 lines — (15,929/15,938), (13,397/13,397) and (14,351/14,354) — and every one of them was assembled, because a 3-line learner cannot hold a whole without buying a part. The remaining seeds are mixed (Section 10).

**Wholes sold by learners with 3 lines.** These are the deliveries that must have passed through at least one supplier. At 20,000 they were more numerous with signs on 9 of 14 pairs: 5,417, 3,803, 862, 15,929, 70, 4,834, 1,826, 16, 4, 13,397, 14,351, 0, 0 and 9,224 with signs, against 63, 396, 52, 386, 294, 379, 32, 777, 139, 0, 556, 286, 561 and 28 in the twins, on seeds 312 to 325.

### 9.2 Cipher world, the whole at 120: 10 seed pairs

The same world with the whole priced at 120 instead of 240, so that the whole is worth four two-class texts rather than eight. Ten seed pairs. At 20,000, roles formed by product in 8 of 10 worlds with signs against 0 of 10 twins ($p = 0.0004$), consumers' utility was higher with signs on 9 of 10 pairs ($p = 0.011$), and fully right wholes were more numerous with signs on 8 of 10 pairs ($p = 0.055$). Halving the whole's price leaves the roles and the welfare difference in place and weakens the difference in wholes.

### 9.3 Cipher world with prices learned instead of listed: 8 seed pairs

In this set the consumers do not post a list price. Every agent learns a price band for what it buys and sells, the agreed price is the middle of the overlap between the two bands, and a draw whose two bands do not overlap settles nothing. Eight seed pairs, seeds 306 to 313, read at the end of their runs.

Fully right wholes were more numerous with signs on 7 of 8 pairs (9,130 against 1,937; 6,870 against 2,635; 7,679 against 2,194; 5,354 against 2,547; 10,809 against 1,426; 9,856 against 1,179; 8,962 against 2,604; and on seed 311, 1,215 against 1,216, the one pair where the twin was ahead, by a single whole; sign test $p = 0.035$).

What learned prices suppressed is assembly. In 7 of the 8 worlds with signs almost nothing was assembled — (19/9,130), (0/6,870), (0/7,679), (43/5,354), (0/10,809), (0/9,856) and (1/8,962) — while seed 311 is the exception, with every one of its 1,215 wholes assembled. The twins assembled a small and steady share, from (8/2,604) to (196/2,194). Under learned prices a part cost an assembler more than the whole it went into paid, which is the condition Eq. 2 says must fail, and the market answered by building alone.

Roles still formed: by product in 4 of 8 worlds with signs against 0 of 8 twins ($p = 0.038$). Consumers' utility was not logged in this set, so there is no welfare comparison for it. The signs sort sellers into roles under learned prices as well; what learned price bands prevented is the assembly of wholes from bought parts.

### 9.4 Cipher world without the tree of wants: 8 worlds

The control for the way a consumer's want is read (Section 3): the good is read from a single number cut into seven equal slices, so that a want which moves by a small step in the genome may land on an unrelated good. Eight worlds with signs and four twins.

Assembled wholes appeared both with the tree and without it, so the tree is not what makes assembly possible. Roles, however, did not form: sellers were legible as groups in 0 of the 8 worlds without the tree. Steady demand for each good — which the tree provides, by keeping a want on or near the good it names — is therefore one of the conditions under which the signs come to mean a kind of work. The readings of this set were taken within 390 rounds of their milestones, and for two of its worlds they are the only surviving record (Section 7).

### 9.5 Cipher world with parts paid in full: 12 seed pairs

Twelve seed pairs in the setting where a learner's test can name the buyer's own corpus, so that a right part scores 1.0 and is paid 4.0 (Section 3.4). The first four pairs' records end at about 30,000 rounds, where they were stopped; the rest ran to 40,000.

At 10,000 fully right wholes were more numerous with signs on 7 of 12 pairs and utility was higher on 8 of 12, neither reaching $p < 0.05$. At 20,000 both were higher on 11 of 12 pairs ($p = 0.0032$). At the end both were higher on 12 of 12 pairs, and on the 8 pairs with an end utility reading the consumers' utility was 3.03 to 4.03 with signs against 0.50 to 1.35 blind. Of the fully right wholes delivered at the end, (86,320/135,266) were assembled in the worlds with signs against (5,280/18,436) in the twins.

Roles formed by product in 10 of 12 worlds with signs against 0 of 12 twins at 10,000 ($p < 10^{-4}$) and in 11 of 12 against 3 of 12 at the end.

Paying a part its full price therefore moves real money down the chain without changing what forms: the roles appear earlier here than where a part earns almost nothing, and by the end the worlds with signs lead their twins on wholes, on utility and on the share of wholes assembled, on every pair.

### 9.6 Machine-learning world, the main set: 12 seed pairs

Twelve seed pairs, seeds 512 to 519 and 712 to 715, eight learners with 2 lines and eight with 16, run to 40,000 rounds, read at the end, rounds 38,000–39,999; Table 2 gives every seed.

**What the consumers obtained.** Utility was higher with signs on 12 of 12 pairs (sign test $p = 2.4 \times 10^{-4}$), 0.52 to 2.06 with signs against 0.39 to 0.42 in the twins, where 0.30 is the value for a consumer that obtains nothing.

**Where the gain came from.** Labelled embeddings, which few learners sell, were bought and scored by consumers more often with signs on 12 of 12 pairs ($p = 2.4 \times 10^{-4}$). Whole classifiers were sold and scored in 5 of 12 worlds with signs and in 1 of 12 twins (Fisher's exact test $p = 0.077$). The signs do not create a whole-seller on most seeds; they raise the chance of one, and they decide what kind it is.

**Roles.** Roles formed by product in 10 of 12 worlds with signs against 1 of 12 twins ($p = 3.2 \times 10^{-4}$). Accuracy alone was above its permuted value in 12 of 12 worlds with signs and in 3 of 12 twins. Read by the head item alone, roles formed in 10 of 12 against 1 of 11 ($p = 5.5 \times 10^{-4}$).

**Two market structures from the same rules.** In the five worlds with signs that sold wholes, the learners' income from consumers concentrated: an income Gini of 0.76 to 0.90 at the end, with the top earner holding 0.37 to 0.93 of all learners' income. The twins had a Gini of 0.04 to 0.17, and the worlds with signs that sold no whole 0.36 to 0.61. The five divide into two kinds. In 516, 518 and 714 one learner with 16 lines built the wholes with its own construct lines and held 0.92 to 0.93 of the income: a lone builder's monopoly — in 516, (9,947/10,146) of the wholes delivered in the last 2,000 rounds were built by their seller's own lines. In 517 every whole was assembled from a labelled embedding bought from another learner — (4,698/4,698) and (3,155/3,155) on the two corpora at 20,000 — and in 515 about a third were, (2,173/6,397) at the end, with the top earner at 0.37: a team. Which of the two formed was settled early, by which kind of learner reached the whole's market first.

## 9.7 Machine-learning world, the second set: 20 seed pairs

Twenty further seed pairs of the same design, seeds 306 to 325, which are also the blank starts of the reconstruction sets (Section 9.12). At 20,000 roles formed by product in 12 of 20 worlds with signs against 0 of 20 twins ($p = 2 \times 10^{-5}$), consumers' utility was higher with signs on 20 of 20 pairs ($p = 1 \times 10^{-6}$), wholes were sold in 9 of 20 worlds with signs against 3 of 20 twins ($p = 0.041$), and labelled embeddings scored more often with signs on 18 of 20 pairs ($p = 0.0002$).

Taken together with the main set, the design has 32 seed pairs, over which wholes were sold in 14 worlds with signs and 5 twins at 20,000 ($p = 0.014$).

## 9.8 Machine-learning world, no learner able to reach the whole without spending its whole plan: 8 seed pairs

Eight seed pairs, seeds 320 to 323 and 716 to 719, with eight learners of 2 lines and eight of 3. A learner with 3 lines can reach the whole classifier alone only by spending its entire plan on that one chain and selling nothing else.

**At 10,000.** Consumers scored wholes in 4 of the 8 worlds with signs (716, 321, 322 and 323) and in none of the twins beyond a single whole draw, (1/3,637) on seed 320 (Fisher's exact test $p = 0.14$ counting that draw as a whole-selling world, $p = 0.038$ not counting it). How the wholes were built differs by seed. In 716 and 321 every whole was assembled from a labelled embedding bought from another learner — (2,914/2,914) and (2,422/2,422) on the two corpora in 716, (1,185/1,185)

and (747/747) in 321 — none built by the seller's own construct lines. In 322 and 323 a single learner with 3 lines spent its whole plan on the chain and built the wholes itself, (8,539/8,714) and (6,974/7,162) by its own lines, with (175/8,714) and (188/7,162) assembled. Consumers' utility was higher with signs on 8 of 8 pairs (sign test $p = 0.0039$, the smallest value attainable at this number of pairs), 0.49 to 1.52 against 0.38 to 0.40. The seeds without wholes sold labelled embeddings by the thousands against hundreds in their twins (higher with signs on 7 of 8 pairs). Roles formed by product in 7 of 8 worlds with signs against 0 of 8 twins ($p = 0.0007$), and by head in 8 of 8 against 0 of 7 ($p = 0.0002$).

**At 20,000.** The same four worlds sold wholes and no twin did (4 of 8 against 0 of 8, $p = 0.038$). In 716 and 321 every whole was still assembled — (5,702/5,702) and (4,896/4,896) in 716, (6,658/6,658) and (6,267/6,267) in 321 — and half or more of all the consumer-paid deliveries in those two worlds carried a payment to a supplier, (10,641/21,206) and (12,976/22,289), against (2/32,421) and (105/19,934) in their twins. In 322 the lone builder made every whole itself, (10,977/10,977), and in 323 nearly all, (8,174/8,278). Utility was higher with signs on 8 of 8 pairs ($p = 0.0039$), 0.51 to 2.09 against 0.38 to 0.41. Roles formed by product in 6 of 8 worlds with signs against 0 of 8 twins ($p = 0.0035$), and by head in 7 of 8 against 1 of 8 ($p = 0.0051$).

**At the end.** All eight pairs ran to 40,000 rounds. In the last 2,000 rounds consumers scored wholes in the same four worlds with signs and in none of the twins ($p = 0.038$), on (20,767/29,961) of their whole draws in 321, (24,791/29,635) in 322, (18,672/21,726) in 323 and (15,304/29,988) in 716. In 321 every whole was assembled, (10,489/10,489) and (10,278/10,278) on the two corpora, as in 716, (7,722/7,722) and (7,582/7,582); (20,793/28,296) of all consumer-paid deliveries in 321 and (15,304/22,605) in 716 carried a payment to a supplier, against (7/22,805) and (0/37,249) in their twins. In 322 and 323 the lone builder with 3 lines made every whole itself, (13,249/13,249) and (11,542/11,542) in 322 and (9,810/9,810) and (8,862/8,862) in 323, and no delivery in either world paid a supplier, (0/32,272) and (0/34,895). Utility was higher with signs on 8 of 8 pairs ($p = 0.0039$): 0.54, 2.02, 2.43, 2.04, 1.82, 0.54, 0.54 and 0.52 on seeds 320 to 323 and 716 to 719, against 0.39 to 0.41 in the twins. The income Gini was 0.90, 0.91, 0.82 and 0.88 in the four whole-selling worlds, with the top earner at 0.92, 0.94, 0.87 and 0.91 of the learners' income, and 0.54, 0.55, 0.52 and 0.56 in 320, 717, 718 and 719, against 0.03 to 0.11 in the twins. The worlds without wholes sold labelled embeddings by the thousands: (24,416/25,690) of draws scoring in 320 and (22,791/25,213), (22,722/25,270) and (18,952/24,259) in 717 to 719, against (88/1,634), (4,108/18,155), (0/329) and (1,627/16,484) in their twins. Roles formed by product in 5 of 8 worlds with signs against 0 of 8 twins ($p = 0.013$) and by head in 6 of 8 against 0 of 7 ($p = 0.0056$).

Two things separate in this arm. Whether a whole gets sold at all is decided by capacity and by which kind of agent reaches its market first; whether it is assembled or built alone follows from the same thing, and in the two worlds where the whole was taken by an agent with 3 lines nothing was ever bought, while in the two where it was taken by an agent with 2 lines everything was. The signs decide neither by themselves: what they do, on every pair, is raise what the consumers obtain.

### 9.9 Machine-learning world with every learner at 2 lines: 12 seed pairs

Twelve seed pairs, seeds 306 to 317, every learner with a plan of 2 lines. No learner in these worlds can produce a whole classifier without buying the good below it, so every whole that appears on either side of the comparison is necessarily assembled.

**At 10,000.** Consumers scored wholes in 4 of 12 worlds with signs (309, 310, 313 and 314) and in 0 of 12 twins ($p = 0.047$), and every whole in those four was assembled: (1,468/1,468) and (1,767/1,767) on the two corpora in 309, (3,728/3,728) and (2,596/2,596) in 310, (1,614/1,614) and (785/785) in 313, and (3,756/3,756) and (4,170/4,170) in 314, with (3,269/17,025) to (8,244/20,281) of all consumer-paid deliveries in those worlds carrying a payment to a supplier. Utility was higher with signs on 10 of 12 pairs ($p = 0.019$), 0.39 to 1.20 against 0.38 to 0.41; labelled embeddings scored more often with signs on 10 of 12 pairs ($p = 0.019$); roles formed by product in 8 of 12 worlds with signs against 0 of 11 twins read ($p = 0.001$) and by head in 9 of 11 against 0 of 11 ($p = 0.0001$).

**At 20,000.** Wholes were sold in the same 4 of 12 worlds with signs, on (8,995/24,278), (11,864/25,886), (11,565/24,283) and (15,209/27,608) of their whole draws in 309, 310, 313 and 314, and in 1 of 12 twins, (55/6,463) on seed 317 ($p = 0.16$). Utility was higher with signs on 10 of 12 pairs ($p = 0.019$), 0.39 to 1.60 against 0.39 to 0.42, and labelled embeddings scored more often with signs on 9 of 12 pairs. Roles formed by product in 9 of 12 worlds with signs against 0 of 11 twins read ($p = 0.0003$) and by head in 8 of 12 against 0 of 11 ($p = 0.001$).

**At the end.** Wholes were sold in 4 of 12 worlds with signs against 1 of 12 twins. Roles formed by product in 9 of 12 against 2 of 11 ($p = 0.0095$) and by head in 8 of 12 against 1 of 10. Utility was higher with signs on 11 of 12 pairs ($p = 0.0032$), 0.40 to 2.35 against 0.39 to 0.43.

This is the arm where assembly cannot be anything else. Every whole on either side had to be put together by two agents, so the comparison is between populations that found their partners through signs and populations that drew them at random, and the wholes appear only where the signs did the finding.

### 9.10 Births and deaths in the machine-learning world: 8 seed pairs

Eight seed pairs, seeds 306 to 309 and 412 to 415, with the births and deaths of Section 2.11. About 1.3 of the sixteen founders are expected to remain at round 20,000 and 0.1 at round 40,000.

Roles formed by product in 6 of 8 worlds with signs against 0 of 8 twins at 20,000 and again at the end, and by head in 7 of 8 against 0 of 8 at the end ($p = 0.0007$). Utility was higher with signs on 8 of 8 pairs at both readings ($p = 0.0039$). Two of the eight worlds with signs, 306 and 414, grew a whole-assembling team among learners that were born into the market rather than present at the start, with nearly every whole assembled at the end — (3,380/3,400) and (2,977/3,000) on the two corpora in 306. Neither of their twins scored a whole.

The roles, the routing between learners and the assembling teams were therefore held by learners who were not there when they formed.

### 9.11 Births and deaths in the cipher world: 8 seed pairs

Eight seed pairs in the setting where a part is paid in full. The first four pairs, seeds 312 to 315, were stopped near round 24,000; the second four, seeds 306 to 309, ran to 40,000.

Roles formed by product in 6 of 8 worlds with signs against 0 of 8 twins at 10,000 and again at 20,000 ($p = 0.0035$), and by head in 7 of 8 against 0 of 8 at 10,000 and at the end ($p = 0.0007$). Consumers' utility was higher with signs on 8 of 8 pairs at 10,000 ($p = 0.0039$), on 5 of 8 at 20,000 and on 3 of the 4 pairs read at the end; roles by product had fallen to 3 of 8 by the end. Every fully right whole the worlds with signs delivered at 20,000 was assembled, (1,358/1,358), against

(783/2,471) in the twins, which delivered more wholes under turnover; at the end, (1,719/2,715) against (533/2,355).

Per seed at the end, fully right wholes with signs and how many of them were assembled, against the same in the twin: 306 (0/158) against (5/7); 307 (13/13) against (1/304); 308 (80/80) against (39/39); 309 (677/1,513) against (27/485); 312 (855/855) against (59/442); 313 (11/11) against (103/481); 314 (83/83) against (113/113); 315 (0/2) against (186/484). Utility at the end, for the second four pairs, was 0.61, 0.49, 0.57 and 1.03 with signs against 0.46, 0.54, 0.47 and 0.59 in the twins.

Under turnover in the cipher world the roles read by product does not hold up to the end of the runs, while the roles read by head and the assembly of wholes do. This is the weakest of the sets at the end of the run.

## 9.12 Seeding a fresh world from a lone builder's practice: machine-learning world 714

The source, world 714, is a world in which one learner with 16 lines built every whole classifier alone and held 0.93 of the learners' income. Its learners' practices were dealt outward (the practice alone, with random signs) to five fresh worlds, read beside the same five seeds started blank.

At 10,000 every one of the five seeded worlds had rebuilt the monopoly, with its top earner at 0.89 to 0.93 of the income. Of the five blank starts, one (seed 307) sold wholes and four sold none: 5 of 5 against 1 of 5, Fisher's exact test $p = 6/252 = 0.024$, and the same at 20,000 and at the end. The seeded worlds' utility was 1.91 to 2.45 at 10,000 and 2.52 to 2.64 at the end, with (12,674/15,145) to (13,854/15,893) of whole draws scoring, against the blank start 307's (1,862/5,745) at 10,000 and (7,748/14,325) at the end.

What came back is the recipe, not a society: (134/129,163) of the seeded worlds' wholes at the end were assembled from a bought part. A lone builder's practice is its own recipe, and dealing it to a fresh learner replays it.

## 9.13 Seeding a fresh world from a team's practice: machine-learning world 517

The source, world 517, is a world in which every whole was assembled from a labelled embedding bought from another learner. Its practices were dealt outward to 18 fresh worlds and at the half level to 2 more, beside 20 blank starts on the same seeds.

At 10,000, 17 of the 18 seeded worlds sold wholes and every one was assembled, (62,287/62,287) pooled, against 7 of the 20 blank starts ($p = 1.5 \times 10^{-4}$), of whose wholes (12,191/58,783) were assembled. At the end, 15 of 18 seeded worlds sold wholes, still every one assembled, (277,995/277,995) pooled, against 6 of 18 blank starts on the same seeds ($p = 0.003$), of whose wholes (26,369/129,072) were assembled. Roles had formed in 17 of 18 seeded worlds, against 14 of the 20 blank starts, and utility was higher in the seeded world on 17 of 18 common seeds.

Among the blank starts that did sell wholes at the end, the structure differs from world to world: seed 317 assembled every whole, (17,151/17,151); seed 307 about half, (7,532/14,404); and seeds 313, 316, 319, 322 and 323 built nearly all of theirs by their sellers' own lines, (1,686/97,517) assembled. A team's practices carry its structure, where a lone builder's carry its recipe.

## 9.14 Seeding fresh cipher worlds

Four sources were used. Two are monopolies, worlds 320 and 322, in which one learner assembled every whole and held 0.89 and 0.95 of the income. Two are distributed worlds, 317 and 324, in which learners with 16 lines built wholes alone.

**Dealt outward.** The two distributed sources replayed at once: 28,836 and 32,140 fully right wholes at 10,000, with (817/28,836) and (0/32,140) assembled. The two monopoly sources did not replay: 0 and 0 fully right wholes at 10,000, and 0 and 1 at 20,000, although their roles came back. What a monopoly's practice does not carry is the arrangement between its members, which has to be coevolved again.

**Dealt to half the learners.** With half the population dealt the source's practices and half starting at random, the random half coevolved into the suppliers the dealt half needed: 4,604 and 7,416 fully right wholes at 10,000, every one assembled, and 11,934 and 15,589 at 20,000, with (11,934/11,934) and (15,335/15,589) assembled, at consumers' utility 3.16 and 3.91.

**Dealt with the signs as well.** Seeded at the presence level, which deals the practice together with the source's average displayed sign for it, world 322 delivered 2,721 fully right wholes at 20,000, every one assembled.

**Dealt without the keys.** The keyless level carries the division of labour and nothing of the solution to the cipher. Three worlds from the monopoly source delivered 269, 827 and 1,456 fully right wholes at 10,000 and 2,951, 3,085 and 7,663 at 20,000, every one assembled — where the same roles dealt *with* their keys had delivered none. Three worlds from the distributed source stayed at a blank start's scale, (0/787), (270/5,354) and (1,379/3,890) assembled at 10,000, so the keys were most of what that source's outward seed had carried.

**The set run for statistical power.** Five outward-seeded worlds and five blank starts per source. All five from the distributed source replayed, 29,841 to 31,691 fully right wholes against 1 to 4,752 in the blanks (sign test $p = 0.031$, the smallest value attainable at five pairs), with at most (287/30,852) assembled: again the recipe and not the society. From the monopoly source the assembling structure came back in three of the five worlds, (20,269/20,269), (25,366/25,366) and (434/434) assembled at 10,000, and not at all in the other two; those three rebuilt worlds delivered 5,983 to 22,444 fully right wholes at 20,000, while the blank starts delivered 0 to 5,260 at 10,000 and 2,982 to 11,557 at 20,000, mostly built alone.

## 10 A table for every world

The short names in the columns are defined in Section 7.9.

One table per set and reading: the record cut at 10,000, at 20,000, and the whole record at the end of the run, each generated from the reading files in the repository.

Table 4: Cipher reconstruction, half level, source 317 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 417 | S | 15,840 | 7,647/15,840 | 4.033 | 0.23 (0.07)* | $+0.14^{+}$ | 0.34 (0.30) | $+0.28^{+}$ |

Table 2: The machine-learning main set at the end of the run (rounds 38,000–39,999), by seed; S = the world with signs, B = its sign-blind twin on the same seed. Wholes and labelled embeddings: the consumers' purchase attempts that scored. Utility: mean Cobb–Douglas utility in the last 1,000 rounds, where 0.30 is a consumer that obtains nothing. Top-1: the top earner's share of the learners' income from consumers. Categories: the accuracy of guessing a seller's product from the other sellers' signs, with the permuted value in parentheses; $*$ above the permuted value at $p < 0.05$, $+$ mutual information above its permuted value. Roles have formed where both marks are present.

| | wholes | | labelled emb. | | utility | | top-1 | | categories | |
|---|---|---|---|---|---|---|---|---|---|---|
| seed | S | B | S | B | S | B | S | B | S | B |
| 512 | 0 | 0 | 21,174 | 642 | 0.53 | 0.39 | 0.18 | 0.07 | 0.84 (0.33)*+ | 0.25 (0.32) |
| 513 | 0 | 0 | 19,940 | 0 | 0.52 | 0.40 | 0.22 | 0.07 | 0.81 (0.45)*+ | 0.32 (0.36) |
| 514 | 0 | 0 | 22,616 | 5,682 | 0.53 | 0.42 | 0.15 | 0.08 | 0.83 (0.33)*+ | 0.46 (0.45) |
| 515 | 12,686 | 0 | 3,262 | 0 | 1.56 | 0.40 | 0.37 | 0.07 | 0.86 (0.36)*+ | 0.42 (0.45) |
| 516 | 19,970 | 0 | 3,632 | 2,566 | 2.06 | 0.40 | 0.92 | 0.07 | 0.63 (0.23)*+ | 0.31 (0.26) |
| 517 | 12,317 | 0 | 2,877 | 718 | 1.57 | 0.39 | 0.54 | 0.07 | 0.66 (0.25)*+ | 0.65 (0.55)* |
| 518 | 17,561 | 0 | 3,198 | 632 | 1.75 | 0.39 | 0.93 | 0.07 | 0.59 (0.32)*+ | 0.19 (0.28) |
| 519 | 0 | 486 | 21,646 | 504 | 0.53 | 0.42 | 0.34 | 0.15 | 0.57 (0.41)*+ | 0.32 (0.46)+ |
| 712 | 0 | 0 | 22,858 | 0 | 0.53 | 0.40 | 0.65 | 0.07 | 0.87 (0.62)* | 0.62 (0.49)* |
| 713 | 0 | 0 | 18,859 | 1,919 | 0.52 | 0.39 | 0.24 | 0.07 | 0.96 (0.57)*+ | 0.64 (0.47)*+ |
| 714 | 20,576 | 0 | 3,473 | 0 | 2.03 | 0.39 | 0.93 | 0.07 | 0.93 (0.35)*+ | 0.31 (0.45)+ |
| 715 | 0 | 0 | 22,342 | 0 | 0.53 | 0.40 | 0.62 | 0.07 | 0.92 (0.69)* | 0.53 (0.50) |

Table 5: Cipher reconstruction, half level, source 320 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, $^*$ and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 420 | S | 4,604 | 4,604/4,604 | 1.505 | 0.41 (0.13)* | +0.05 | 0.26 (0.23) | +0.26+ |

Table 6: Cipher reconstruction, half level, source 322 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, $^*$ and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 422 | S | 7,416 | 7,416/7,416 | 2.104 | 0.36 (0.14)* | +0.05 | 0.69 (0.33)* | +0.28+ |

Table 3: The cipher world with the whole at 240, at the reading cut at round 10,000 (rounds 8,000–9,999, 200 logged rounds per world). Wholes: fully right four-class texts delivered to consumers; assembled: of those, the ones to which two or more learners each supplied a decoder the good requires. Categories: the accuracy of guessing a seller's product from the other sellers' signs, the mean over the span's 20 windows of 100 rounds, with the permuted mean in parentheses; * accuracy above its permuted value and + mutual information above its permuted value at $p < 0.05$ (200 permutations, so the smallest attainable value is 0.005). Roles have formed where both marks are present: 7 of the 14 worlds with signs and 1 of the 14 twins.

| | with signs | | sign-blind twin | | categories, accuracy (null) | |
|---|---|---|---|---|---|---|
| seed | wholes | assembled | wholes | assembled | with signs | blind twin |
| 312 | 2,047 | 2,047 | 11 | 11 | 0.63 (0.17)*+ | 0.10 (0.20) |
| 313 | 13,476 | 5,422 | 48 | 48 | 0.52 (0.23)*+ | 0.10 (0.14) |
| 314 | 5,747 | 1,062 | 43 | 43 | 0.22 (0.04)* | 0.47 (0.26)* |
| 315 | 712 | 712 | 44 | 44 | 0.26 (0.16)* | 0.38 (0.35) |
| 316 | 6,120 | 53 | 59 | 59 | 0.24 (0.08)*+ | 0.21 (0.23) |
| 317 | 2,028 | 1,230 | 221 | 221 | 0.46 (0.10)*+ | 0.21 (0.23) |
| 318 | 14,291 | 216 | 10 | 10 | 0.17 (0.05)* | 0.22 (0.20) |
| 319 | 3 | 3 | 216 | 216 | 0.66 (0.25)*+ | 0.05 (0.09) |
| 320 | 6,855 | 6,855 | 419 | 25 | 0.53 (0.33)* | 0.16 (0.17) |
| 321 | 2,328 | 2,328 | 143 | 11 | 0.42 (0.12)*+ | 0.17 (0.14) |
| 322 | 719 | 719 | 646 | 166 | 0.44 (0.08)* | 0.16 (0.16) |
| 323 | 3 | 3 | 822 | 143 | 0.30 (0.11)* | 0.39 (0.13)*+ |
| 324 | 7,877 | 37 | 717 | 72 | 0.31 (0.07)* | 0.14 (0.18) |
| 325 | 9,055 | 526 | 328 | 6 | 0.28 (0.14)*+ | 0.09 (0.12) |

Table 7: Cipher reconstruction, half level, source 324 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 20,963 | 2,169/20,963 | 5.062 | 0.31 (0.19)* | +0.20+ | 0.48 (0.27)* | +0.03 |

Table 8: Cipher reconstruction, keyless level, source 322 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 269 | 269/269 | 0.627 | 0.54 (0.19)* | +0.15+ | 0.38 (0.21)* | +0.16+ |
| 307 | S | 827 | 827/827 | 0.723 | 0.61 (0.29)* | +0.31+ | 0.47 (0.32)* | +0.05 |
| 308 | S | 1,456 | 1,456/1,456 | 0.968 | 0.24 (0.15)* | +0.00 | 0.44 (0.22)* | +0.18+ |
| 312 | S | 305 | 1/305 | 0.687 | 0.80 (0.47)* | +0.22+ | 0.76 (0.37)* | +0.40+ |
| 313 | S | 1,969 | 1,969/1,969 | 1.143 | 0.55 (0.31)* | +0.08+ | 0.46 (0.37)* | -0.03 |
| 314 | S | 8 | 8/8 | 0.561 | 0.59 (0.21)* | +0.21+ | 0.51 (0.27)* | +0.01 |
| 315 | S | 271 | 271/271 | 0.687 | 0.66 (0.36)* | +0.20+ | 0.52 (0.33)* | +0.22+ |

Table 8: Cipher reconstruction, keyless level, source 322 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 316 | S | 748 | 748/748 | 0.762 | 0.63 (0.29)* | +0.41+ | 0.58 (0.31)* | +0.39+ |
| 317 | S | 77 | 77/77 | 0.575 | 0.53 (0.18)* | +0.25+ | 0.53 (0.45)* | -0.07 |
| 318 | S | 231 | 231/231 | 0.636 | 0.31 (0.19)* | +0.26+ | 0.64 (0.29)* | +0.29+ |

Table 9: Cipher reconstruction, keyless level, source 324 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 309 | S | 787 | 0/787 | 0.793 | 0.57 (0.57) | -0.03 | 0.40 (0.30)* | +0.25+ |
| 310 | S | 5,354 | 270/5,354 | 1.932 | 0.30 (0.16)* | -0.08 | 0.17 (0.21) | +0.07 |
| 311 | S | 3,890 | 1,379/3,890 | 1.626 | 0.40 (0.19)* | -0.01 | 0.37 (0.27)* | +0.04 |

Table 10: Cipher reconstruction, outward level, source 317 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 417 | S | 28,836 | 817/28,836 | 6.288 | 0.12 (0.12) | -0.10 | 0.42 (0.33)* | +0.26+ |

Table 11: Cipher reconstruction, outward level, source 320 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 420 | S | 0 | 0/0 | 0.622 | 0.53 (0.16)* | +0.35+ | 0.56 (0.15)* | +0.71+ |

Table 12: Cipher reconstruction, outward level, source 322 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 319 | S | 3 | 3/3 | 0.605 | 0.46 (0.20)* | +0.16+ | 0.42 (0.23)* | -0.03 |
| 320 | S | 7,766 | 7,766/7,766 | 2.560 | 0.13 (0.09)* | -0.13 | 0.14 (0.17) | -0.02 |
| 321 | S | 5,822 | 5,822/5,822 | 2.018 | 0.44 (0.12)* | +0.28+ | 0.60 (0.17)* | +0.55+ |
| 322 | S | 0 | 0/0 | 0.602 | 0.28 (0.12)* | +0.09+ | 0.18 (0.22) | -0.22 |
| 323 | S | 2,734 | 2,734/2,734 | 1.336 | 0.25 (0.09)* | +0.05 | 0.16 (0.12)* | +0.26+ |

Table 12: Cipher reconstruction, outward level, source 322 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 422 | S | 0 | 0/0 | 0.610 | 0.61 (0.18)* | +0.32+ | 0.44 (0.21)* | +0.24+ |

Table 13: Cipher reconstruction, outward level, source 324 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 32,140 | 0/32,140 | 6.788 | 0.31 (0.18)* | +0.20+ | 0.24 (0.23) | -0.15 |

Table 14: Powered cipher reconstruction, outward level, source 322 (five worlds) at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 430 | S | 0 | 0/0 | 0.603 | 0.59 (0.12)* | +0.36+ | 0.40 (0.12)* | +0.00 |
| 431 | S | 20,269 | 20,269/20,269 | 4.518 | 0.31 (0.09)* | +0.04 | 0.39 (0.24)* | +0.05 |
| 432 | S | 25,366 | 25,366/25,366 | 5.243 | 0.18 (0.10)* | +0.10+ | 0.26 (0.20)* | +0.10+ |
| 433 | S | 434 | 434/434 | 0.682 | 0.34 (0.14)* | +0.17+ | 0.61 (0.24)* | +0.45+ |
| 434 | S | 0 | 0/0 | 0.618 | 0.53 (0.12)* | +0.26+ | 0.44 (0.24)* | +0.35+ |

Table 15: Powered cipher reconstruction, outward level, source 324 (five worlds) at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 440 | S | 30,542 | 0/30,542 | 6.567 | 0.55 (0.17)* | +0.51+ | 0.27 (0.24) | -0.11 |
| 441 | S | 30,852 | 287/30,852 | 6.582 | 0.33 (0.17)* | +0.08+ | 0.28 (0.23)* | +0.01 |
| 442 | S | 31,691 | 0/31,691 | 6.966 | 0.45 (0.17)* | +0.48+ | 0.10 (0.23) | +0.02 |
| 443 | S | 31,494 | 0/31,494 | 6.723 | 0.50 (0.17)* | +0.60+ | 0.19 (0.23) | -0.10 |
| 444 | S | 29,841 | 0/29,841 | 6.704 | 0.36 (0.14)* | +0.57+ | 0.23 (0.23) | -0.10 |

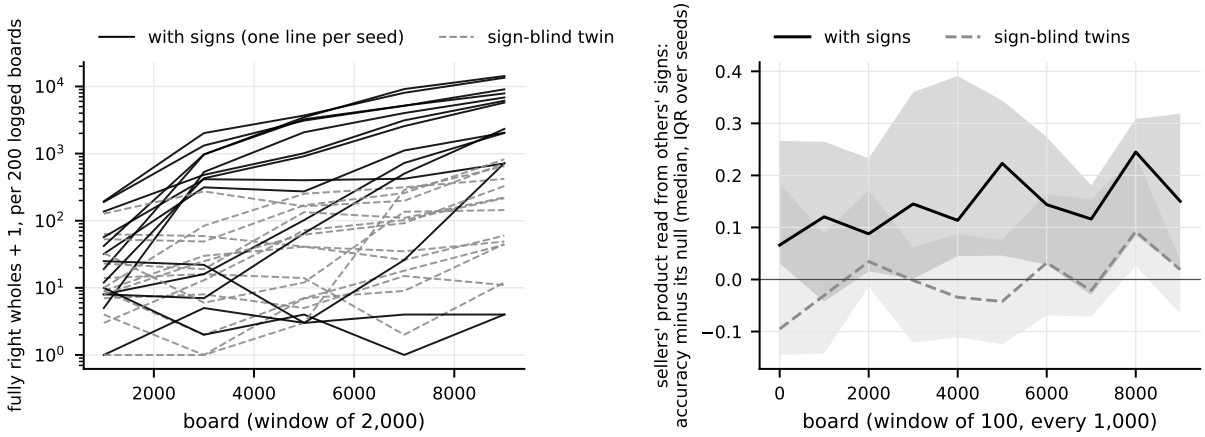



Figure 2: The cipher world with the whole at 240, rounds 0 to 10,000. Left: fully right wholes delivered to consumers per window of 2,000 rounds (200 logged rounds), one line per seed, with signs (solid, 14 worlds) and sign-blind twins (dashed, 16 worlds), on a logarithmic scale. Right: the roles reading in windows of 100 rounds taken every 1,000 rounds — the accuracy of guessing a seller's product from the other sellers' signs, minus its permuted value — as the median across seeds with the band between the first and third quartiles.

Table 16: Blank starts beside the powered cipher reconstruction at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, $^*$ and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 430 | S | 0 | 0/0 | 0.538 | 0.65 (0.18)$^*$ | +0.47$^+$ | 0.42 (0.31)$^*$ | +0.20$^+$ |
| 431 | S | 613 | 613/613 | 0.751 | 0.67 (0.39)$^*$ | +0.16$^+$ | 0.65 (0.34)$^*$ | +0.04 |
| 432 | S | 5,260 | 499/5,260 | 1.816 | 0.61 (0.14)$^*$ | +0.25$^+$ | 0.59 (0.26)$^*$ | +0.33$^+$ |
| 433 | S | 648 | 12/648 | 0.669 | 0.53 (0.22)$^*$ | +0.26$^+$ | 0.59 (0.38)$^*$ | +0.12$^+$ |
| 434 | S | 2,691 | 2,691/2,691 | 1.377 | 0.37 (0.23)$^*$ | -0.02 | 0.22 (0.24) | -0.10 |
| 440 | S | 294 | 294/294 | 0.651 | 0.33 (0.10)$^*$ | +0.09$^+$ | 0.15 (0.19) | +0.16$^+$ |
| 441 | S | 3,001 | 576/3,001 | 1.432 | 0.42 (0.09)$^*$ | +0.19$^+$ | 0.25 (0.26) | +0.06 |
| 442 | S | 1 | 1/1 | 0.543 | 0.64 (0.18)$^*$ | +0.27$^+$ | 0.35 (0.21)$^*$ | +0.17$^+$ |
| 443 | S | 1 | 1/1 | 0.560 | 0.56 (0.24)$^*$ | +0.29$^+$ | 0.37 (0.20)$^*$ | +0.11$^+$ |
| 444 | S | 4,752 | 480/4,752 | 1.919 | 0.36 (0.14)$^*$ | +0.25$^+$ | 0.29 (0.23)$^*$ | +0.21$^+$ |

Table 17: Cipher reconstruction, presence level, source 322 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, $^*$ and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 422 | S | 0 | 0/0 | 0.619 | 0.61 (0.10)$^*$ | +0.19$^+$ | 0.69 (0.26)$^*$ | +0.14$^+$ |

Table 18: Cipher reconstruction, presence level, source 324 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 31,689 | 0/31,689 | 6.952 | 0.71 (0.14)* | +0.67+ | 0.49 (0.23)* | +0.02 |

Table 19: Cipher world, configuration A, whole at 240 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 312 | S | 2,047 | 2,047/2,047 | – | 0.63 (0.17)* | +0.21+ | 0.56 (0.23)* | +0.13+ |
| 312 | B | 11 | 11/11 | 0.479 | 0.10 (0.20) | +0.03 | 0.30 (0.24)* | +0.03 |
| 313 | S | 13,476 | 5,422/13,476 | 3.906 | 0.52 (0.23)* | +0.38+ | 0.29 (0.25) | +0.16+ |
| 313 | B | 48 | 48/48 | 0.484 | 0.10 (0.14) | -0.07 | 0.21 (0.23) | -0.01 |
| 314 | S | 5,747 | 1,062/5,747 | 2.204 | 0.22 (0.04)* | +0.02 | 0.10 (0.19) | +0.27+ |
| 314 | B | 43 | 43/43 | 0.480 | 0.47 (0.26)* | +0.03 | 0.29 (0.17)* | -0.28 |
| 315 | S | 712 | 712/712 | 0.743 | 0.26 (0.16)* | -0.08 | 0.19 (0.24) | +0.00 |
| 315 | B | 44 | 44/44 | 0.490 | 0.38 (0.35) | +0.01 | 0.35 (0.31) | -0.03 |
| 316 | S | 6,120 | 53/6,120 | 2.285 | 0.24 (0.08)* | +0.11+ | 0.20 (0.16) | -0.05 |
| 316 | B | 59 | 59/59 | 0.524 | 0.21 (0.23) | -0.06 | 0.20 (0.25) | +0.01 |
| 317 | S | 2,028 | 1,230/2,028 | 1.295 | 0.46 (0.10)* | +0.34+ | 0.36 (0.24)* | +0.37+ |
| 317 | B | 221 | 221/221 | 0.527 | 0.21 (0.23) | +0.01 | 0.35 (0.27)* | +0.06 |
| 318 | S | 14,291 | 216/14,291 | 3.749 | 0.17 (0.05)* | -0.08 | 0.28 (0.23)* | -0.03 |
| 318 | B | 10 | 10/10 | 0.493 | 0.22 (0.20) | +0.06 | 0.21 (0.20) | -0.02 |
| 319 | S | 3 | 3/3 | 0.571 | 0.66 (0.25)* | +0.33+ | 0.44 (0.21)* | +0.46+ |
| 319 | B | 216 | 216/216 | 0.559 | 0.05 (0.09) | -0.03 | 0.09 (0.15) | -0.14 |
| 320 | S | 6,855 | 6,855/6,855 | 2.254 | 0.53 (0.33)* | +0.05 | 0.33 (0.22)* | -0.03 |
| 320 | B | 419 | 25/419 | 0.600 | 0.16 (0.17) | -0.18 | 0.29 (0.24) | -0.10 |
| 321 | S | 2,328 | 2,328/2,328 | 1.326 | 0.42 (0.12)* | +0.11+ | 0.56 (0.21)* | +0.28+ |
| 321 | B | 143 | 11/143 | 0.568 | 0.17 (0.14) | -0.19 | 0.29 (0.24)* | +0.01 |
| 322 | S | 719 | 719/719 | 0.912 | 0.44 (0.08)* | -0.04 | 0.34 (0.18)* | +0.14+ |
| 322 | B | 646 | 166/646 | 0.695 | 0.16 (0.16) | -0.05 | 0.36 (0.32) | -0.04 |
| 323 | S | 3 | 3/3 | 0.554 | 0.30 (0.11)* | +0.05 | 0.48 (0.17)* | +0.30+ |
| 323 | B | 822 | 143/822 | 0.835 | 0.39 (0.13)* | +0.24+ | 0.30 (0.24) | -0.05 |
| 324 | S | 7,877 | 37/7,877 | 2.667 | 0.31 (0.07)* | +0.00 | 0.40 (0.18)* | -0.04 |
| 324 | B | 717 | 72/717 | 0.776 | 0.14 (0.18) | +0.00 | 0.20 (0.31) | +0.01 |
| 325 | S | 9,055 | 526/9,055 | 2.772 | 0.28 (0.14)* | +0.09+ | 0.29 (0.19)* | -0.05 |
| 325 | B | 328 | 6/328 | 0.648 | 0.09 (0.12) | -0.10 | 0.39 (0.27)* | -0.13 |

Table 20: Cipher world, configuration A, whole at 120 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 0 | 0/0 | 0.796 | 0.73 (0.22)* | +0.23+ | 0.45 (0.21)* | +0.18+ |
| 306 | B | 249 | 249/249 | 0.683 | 0.09 (0.13) | +0.02 | 0.14 (0.25) | -0.01 |
| 307 | S | 133 | 133/133 | 0.798 | 0.53 (0.09)* | -0.05 | 0.27 (0.23) | +0.13+ |
| 307 | B | 368 | 64/368 | 0.731 | 0.09 (0.12) | +0.03 | 0.35 (0.26)* | +0.01 |
| 308 | S | 8,215 | 903/8,215 | 2.152 | 0.28 (0.06)* | +0.09+ | 0.26 (0.18)* | +0.01 |
| 308 | B | 220 | 220/220 | 0.614 | 0.13 (0.16) | +0.09+ | 0.36 (0.22)* | -0.04 |
| 309 | S | 8,086 | 911/8,086 | 2.523 | 0.30 (0.11)* | +0.05 | 0.47 (0.23)* | +0.02 |

Table 20: Cipher world, configuration A, whole at 120 at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 309 | B | 782 | 118/782 | 0.817 | 0.03 (0.13) | +0.00 | 0.43 (0.22)* | +0.08$^+$ |
| 310 | S | 6,396 | 673/6,396 | 1.830 | 0.60 (0.24)* | +0.41$^+$ | 0.62 (0.23)* | +0.26$^+$ |
| 310 | B | 456 | 82/456 | 0.744 | 0.08 (0.14) | -0.07 | 0.11 (0.22) | -0.07 |
| 311 | S | 4 | 4/4 | 0.839 | 0.58 (0.10)* | +0.04 | 0.35 (0.24)* | +0.13$^+$ |
| 311 | B | 945 | 540/945 | 0.875 | 0.08 (0.14) | +0.02 | 0.20 (0.25) | -0.10 |
| 312 | S | 7,297 | 1,386/7,297 | 1.972 | 0.06 (0.06) | +0.04 | 0.32 (0.24)* | +0.01 |
| 312 | B | 311 | 143/311 | 0.729 | 0.13 (0.16) | -0.05 | 0.25 (0.31) | -0.18 |
| 313 | S | 11,648 | 103/11,648 | 2.672 | 0.51 (0.16)* | +0.30$^+$ | 0.29 (0.22)* | -0.01 |
| 313 | B | 113 | 113/113 | 0.661 | 0.05 (0.16) | +0.07 | 0.16 (0.20) | +0.06 |
| 314 | S | 30 | 30/30 | 0.813 | 0.38 (0.06)* | +0.09 | 0.50 (0.21)* | +0.19$^+$ |
| 314 | B | 3 | 3/3 | 0.572 | 0.14 (0.19) | -0.05 | 0.16 (0.25) | -0.03 |
| 315 | S | 10,157 | 1,516/10,157 | 2.500 | 0.88 (0.50)* | +0.27$^+$ | 0.31 (0.32) | +0.03 |
| 315 | B | 260 | 260/260 | 0.620 | 0.35 (0.27)* | -0.07 | 0.31 (0.28) | -0.05 |

Table 21: Cipher world, configuration B, first four pairs at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 312 | S | 0 | 0/0 | 0.552 | 0.65 (0.22)* | +0.32$^+$ | 0.64 (0.27)* | +0.25$^+$ |
| 312 | B | 226 | 33/226 | 0.574 | 0.02 (0.17) | +0.08 | 0.41 (0.20)* | +0.03 |
| 313 | S | 5,071 | 201/5,071 | 1.560 | 0.56 (0.22)* | +0.21$^+$ | 0.33 (0.22)* | +0.26$^+$ |
| 313 | B | 100 | 97/100 | 0.498 | 0.11 (0.20) | +0.09$^+$ | 0.10 (0.24) | +0.00 |
| 314 | S | 808 | 808/808 | 0.879 | 0.44 (0.31)* | +0.06 | 0.39 (0.19)* | +0.09 |
| 314 | B | 153 | 6/153 | 0.522 | 0.26 (0.22) | +0.04 | 0.40 (0.28)* | +0.37$^+$ |
| 315 | S | 5,674 | 68/5,674 | 2.165 | 0.43 (0.14)* | +0.21$^+$ | 0.21 (0.23) | +0.43$^+$ |
| 315 | B | 244 | 7/244 | 0.552 | 0.08 (0.16) | +0.02 | 0.15 (0.25) | -0.03 |

Table 22: Cipher world, configuration B, second set (eight pairs) at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 316 | S | 0 | 0/0 | 0.568 | 0.79 (0.26)* | +0.55$^+$ | 0.59 (0.25)* | +0.41$^+$ |
| 316 | B | 319 | 0/319 | 0.557 | 0.33 (0.33) | -0.13 | 0.51 (0.40)* | -0.02 |
| 317 | S | 1,911 | 1,911/1,911 | 1.202 | 0.44 (0.14)* | +0.07 | 0.32 (0.22)* | +0.08$^+$ |
| 317 | B | 222 | 60/222 | 0.555 | 0.22 (0.21) | -0.03 | 0.15 (0.21) | -0.13 |
| 318 | S | 31 | 31/31 | 0.568 | 0.47 (0.16)* | +0.30$^+$ | 0.40 (0.24)* | +0.05 |
| 318 | B | 787 | 213/787 | 0.727 | 0.10 (0.21) | -0.02 | 0.32 (0.24)* | +0.09$^+$ |
| 319 | S | 306 | 306/306 | 0.706 | 0.64 (0.15)* | +0.34$^+$ | 0.42 (0.23)* | +0.11$^+$ |
| 319 | B | 200 | 58/200 | 0.549 | 0.12 (0.16) | +0.01 | 0.21 (0.32) | +0.05 |
| 320 | S | 3,146 | 283/3,146 | 1.367 | 0.40 (0.14)* | +0.09$^+$ | 0.57 (0.26)* | +0.16$^+$ |
| 320 | B | 1 | 1/1 | 0.458 | 0.14 (0.13) | +0.02 | 0.05 (0.22) | +0.13$^+$ |
| 321 | S | 0 | 0/0 | 0.538 | 0.39 (0.11)* | +0.12$^+$ | 0.51 (0.21)* | +0.18$^+$ |
| 321 | B | 934 | 10/934 | 0.763 | 0.17 (0.14) | -0.05 | 0.25 (0.23) | -0.16 |
| 322 | S | 389 | 389/389 | 0.656 | 0.38 (0.07)* | +0.19$^+$ | 0.52 (0.29)* | +0.72$^+$ |
| 322 | B | 21 | 21/21 | 0.458 | 0.26 (0.20)* | +0.06 | 0.17 (0.28) | +0.02 |
| 323 | S | 106 | 106/106 | 0.591 | 0.64 (0.14)* | +0.43$^+$ | 0.47 (0.24)* | +0.55$^+$ |

Table 22: Cipher world, configuration B, second set (eight pairs) at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 323 | B | 322 | 124/322 | 0.638 | 0.05 (0.12) | $+0.12^{+}$ | 0.12 (0.19) | +0.02 |

Table 23: Cipher world, births and deaths (configuration B), first four pairs at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 312 | S | 225 | 77/225 | 0.653 | 0.57 (0.20)* | $+0.43^{+}$ | 0.50 (0.25)* | $+0.26^{+}$ |
| 312 | B | 554 | 25/554 | 0.580 | 0.15 (0.31) | -0.04 | 0.40 (0.36) | -0.04 |
| 313 | S | 0 | 0/0 | 0.530 | 0.47 (0.15)* | $+0.18^{+}$ | 0.44 (0.22)* | $+0.41^{+}$ |
| 313 | B | 155 | 49/155 | 0.474 | 0.02 (0.06) | +0.06 | 0.20 (0.17) | -0.01 |
| 314 | S | 390 | 168/390 | 0.623 | 0.33 (0.11)* | +0.08 | 0.26 (0.22) | $+0.23^{+}$ |
| 314 | B | 43 | 43/43 | 0.456 | 0.37 (0.28)* | -0.04 | 0.09 (0.17) | +0.03 |
| 315 | S | 50 | 50/50 | 0.540 | 0.59 (0.18)* | $+0.20^{+}$ | 0.73 (0.25)* | $+0.41^{+}$ |
| 315 | B | 22 | 22/22 | 0.460 | 0.41 (0.32)* | -0.02 | 0.30 (0.24) | -0.02 |

Table 24: Cipher world, births and deaths (configuration B), second four pairs at 10,000, rounds 8,000–9,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 11 | 11/11 | 0.535 | 0.35 (0.20)* | $+0.27^{+}$ | 0.61 (0.30)* | $+0.44^{+}$ |
| 306 | B | 87 | 8/87 | 0.478 | 0.27 (0.22) | -0.02 | 0.43 (0.36)* | +0.01 |
| 307 | S | 419 | 419/419 | 0.625 | 0.73 (0.21)* | $+0.47^{+}$ | 0.69 (0.32)* | $+0.28^{+}$ |
| 307 | B | 2 | 2/2 | 0.434 | 0.24 (0.38) | +0.05 | 0.27 (0.25) | -0.06 |
| 308 | S | 15 | 15/15 | 0.540 | 0.71 (0.23)* | $+0.44^{+}$ | 0.43 (0.28)* | $+0.30^{+}$ |
| 308 | B | 127 | 0/127 | 0.507 | 0.22 (0.24) | $+0.11^{+}$ | 0.31 (0.31) | -0.09 |
| 309 | S | 778 | 45/778 | 0.859 | 0.29 (0.12)* | -0.04 | 0.33 (0.22)* | $+0.07^{+}$ |
| 309 | B | 120 | 61/120 | 0.489 | 0.15 (0.23) | +0.01 | 0.14 (0.30) | $+0.15^{+}$ |

Table 25: Cipher reconstruction, half level, source 317 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 417 | S | 18,078 | 10,061/18,078 | 4.329 | 0.35 (0.07)* | $+0.20^{+}$ | 0.21 (0.27) | $+0.24^{+}$ |

Table 26: Cipher reconstruction, half level, source 320 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 420 | S | 11,934 | 11,934/11,934 | 3.161 | 0.44 (0.10)* | +0.05 | 0.27 (0.20)* | -0.10 |

Table 27: Cipher reconstruction, half level, source 322 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 422 | S | 15,589 | 15,335/15,589 | 3.905 | 0.54 (0.12)* | +0.32$^{+}$ | 0.65 (0.23)* | +0.43$^{+}$ |

Table 28: Cipher reconstruction, half level, source 324 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 22,460 | 3,972/22,460 | 5.155 | 0.26 (0.17)* | +0.38$^{+}$ | 0.47 (0.27)* | +0.03 |

Table 29: Cipher reconstruction, keyless level, source 322 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 2,951 | 2,951/2,951 | 1.239 | 0.52 (0.17)* | +0.29$^{+}$ | 0.26 (0.19)* | +0.13$^{+}$ |
| 307 | S | 3,085 | 3,085/3,085 | 1.222 | 0.38 (0.28)* | -0.05 | 0.46 (0.31)* | -0.03 |
| 308 | S | 7,663 | 7,663/7,663 | 2.572 | 0.31 (0.19)* | +0.01 | 0.47 (0.28)* | +0.16$^{+}$ |
| 312 | S | 9,056 | 136/9,056 | 2.480 | 0.64 (0.25)* | +0.23$^{+}$ | 0.71 (0.26)* | +0.25$^{+}$ |
| 313 | S | 5,787 | 4,868/5,787 | 2.263 | 0.33 (0.20)* | +0.22$^{+}$ | 0.23 (0.24) | -0.08 |
| 314 | S | 1,585 | 1,585/1,585 | 1.035 | 0.37 (0.15)* | +0.10$^{+}$ | 0.45 (0.29)* | +0.13$^{+}$ |
| 315 | S | 4,602 | 4,602/4,602 | 1.605 | 0.69 (0.24)* | +0.23$^{+}$ | 0.61 (0.34)* | +0.24$^{+}$ |
| 316 | S | 9,239 | 4,234/9,239 | 2.571 | 0.74 (0.24)* | +0.56$^{+}$ | 0.47 (0.23)* | +0.22$^{+}$ |
| 317 | S | 110 | 110/110 | 0.629 | 0.53 (0.26)* | +0.19$^{+}$ | 0.31 (0.32) | -0.06 |
| 318 | S | 5,729 | 3,642/5,729 | 1.761 | 0.47 (0.22)* | +0.23$^{+}$ | 0.64 (0.24)* | +0.19$^{+}$ |

Table 30: Cipher reconstruction, keyless level, source 324 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 309 | S | 4,951 | 2,645/4,951 | 1.814 | 0.66 (0.47)* | +0.11+ | 0.51 (0.22)* | +0.21+ |
| 310 | S | 9,333 | 929/9,333 | 2.789 | 0.25 (0.11)* | -0.08 | 0.15 (0.24) | +0.01 |
| 311 | S | 9,160 | 6,190/9,160 | 2.463 | 0.34 (0.13)* | +0.12+ | 0.47 (0.30)* | +0.10+ |

Table 31: Cipher reconstruction, outward level, source 317 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 417 | S | 26,103 | 5,697/26,103 | 5.585 | 0.20 (0.11)* | +0.10+ | 0.62 (0.30)* | +0.20+ |

Table 32: Cipher reconstruction, outward level, source 320 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 420 | S | 0 | 0/0 | 0.639 | 0.58 (0.14)* | +0.28+ | 0.59 (0.14)* | +0.40+ |

Table 33: Cipher reconstruction, outward level, source 322 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 319 | S | 0 | 0/0 | 0.615 | 0.44 (0.21)* | +0.24+ | 0.60 (0.25)* | +0.09+ |
| 320 | S | 14,641 | 14,641/14,641 | 3.510 | 0.19 (0.09)* | +0.00 | 0.34 (0.23)* | -0.16 |
| 321 | S | 9,575 | 9,575/9,575 | 2.219 | 0.43 (0.11)* | +0.26+ | 0.49 (0.15)* | +0.51+ |
| 322 | S | 0 | 0/0 | 0.619 | 0.44 (0.13)* | +0.34+ | 0.50 (0.23)* | -0.04 |
| 323 | S | 9,708 | 9,708/9,708 | 2.773 | 0.32 (0.08)* | -0.05 | 0.25 (0.11)* | +0.30+ |
| 422 | S | 1 | 1/1 | 0.625 | 0.53 (0.18)* | +0.12+ | 0.34 (0.25)* | +0.29+ |

Table 34: Cipher reconstruction, outward level, source 324 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 32,482 | 0/32,482 | 6.992 | 0.38 (0.18)* | +0.30+ | 0.35 (0.23)* | -0.18 |

Table 35: Powered cipher reconstruction, outward level, source 322 (five worlds) at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 430 | S | 0 | 0/0 | 0.625 | 0.58 (0.09)* | +0.28$^{+}$ | 0.50 (0.13)* | +0.30$^{+}$ |
| 431 | S | 22,444 | 22,444/22,444 | 5.486 | 0.43 (0.10)* | +0.25$^{+}$ | 0.41 (0.25)* | +0.15$^{+}$ |
| 432 | S | 21,961 | 21,961/21,961 | 4.999 | 0.36 (0.14)* | +0.00 | 0.42 (0.23)* | +0.31$^{+}$ |
| 433 | S | 5,983 | 5,983/5,983 | 1.955 | 0.53 (0.15)* | +0.03 | 0.58 (0.24)* | +0.41$^{+}$ |
| 434 | S | 0 | 0/0 | 0.626 | 0.75 (0.16)* | +0.55$^{+}$ | 0.62 (0.26)* | +0.56$^{+}$ |

Table 36: Powered cipher reconstruction, outward level, source 324 (five worlds) at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 440 | S | 32,136 | 0/32,136 | 6.884 | 0.62 (0.18)* | +0.46$^{+}$ | 0.29 (0.24) | -0.13 |
| 441 | S | 30,684 | 438/30,684 | 6.706 | 0.57 (0.17)* | +0.55$^{+}$ | 0.29 (0.23)* | +0.06 |
| 442 | S | 31,408 | 0/31,408 | 6.826 | 0.49 (0.16)* | +0.36$^{+}$ | 0.25 (0.23) | -0.01 |
| 443 | S | 31,355 | 160/31,355 | 6.736 | 0.69 (0.26)* | +0.81$^{+}$ | 0.36 (0.23)* | -0.10 |
| 444 | S | 32,627 | 0/32,627 | 7.043 | 0.45 (0.17)* | +0.70$^{+}$ | 0.38 (0.23)* | -0.07 |

Table 37: Blank starts beside the powered cipher reconstruction at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 430 | S | 3,111 | 35/3,111 | 1.579 | 0.63 (0.15)* | +0.25$^{+}$ | 0.66 (0.43)* | -0.06 |
| 431 | S | 3,149 | 3,149/3,149 | 1.506 | 0.58 (0.18)* | -0.04 | 0.50 (0.36)* | -0.07 |
| 432 | S | 11,557 | 13/11,557 | 3.222 | 0.51 (0.13)* | +0.20$^{+}$ | 0.53 (0.24)* | +0.37$^{+}$ |
| 433 | S | 2,982 | 1,351/2,982 | 1.277 | 0.42 (0.16)* | +0.05 | 0.55 (0.26)* | +0.24$^{+}$ |
| 434 | S | 8,699 | 8,699/8,699 | 2.660 | 0.62 (0.21)* | +0.34$^{+}$ | 0.29 (0.25) | +0.46$^{+}$ |
| 440 | S | 5,811 | 1,916/5,811 | 1.891 | 0.46 (0.13)* | +0.25$^{+}$ | 0.23 (0.20) | -0.07 |
| 441 | S | 7,493 | 912/7,493 | 2.494 | 0.15 (0.11) | -0.08 | 0.36 (0.24)* | +0.11$^{+}$ |
| 442 | S | 0 | 0/0 | 0.562 | 0.60 (0.10)* | +0.14$^{+}$ | 0.47 (0.21)* | +0.21$^{+}$ |
| 443 | S | 3,507 | 3,507/3,507 | 1.721 | 0.37 (0.13)* | +0.01 | 0.22 (0.12)* | +0.08 |
| 444 | S | 8,763 | 225/8,763 | 2.606 | 0.19 (0.10)* | +0.02 | 0.52 (0.24)* | +0.10$^{+}$ |

Table 38: Cipher reconstruction, presence level, source 322 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 422 | S | 2,721 | 2,721/2,721 | 1.291 | 0.62 (0.11)* | -0.06 | 0.62 (0.20)* | +0.32$^{+}$ |

Table 39: Cipher reconstruction, presence level, source 324 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 32,396 | 0/32,396 | 7.124 | 0.68 (0.11)* | +0.59+ | 0.41 (0.23)* | -0.06 |

Table 40: Cipher world, configuration A, whole at 240 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 312 | S | 10,876 | 10,231/10,876 | – | 0.52 (0.18)* | +0.34+ | 0.33 (0.24)* | +0.36+ |
| 312 | B | 372 | 63/372 | 0.677 | 0.19 (0.19) | +0.00 | 0.20 (0.25) | +0.00 |
| 313 | S | 16,275 | 6,634/16,275 | 4.158 | 0.48 (0.17)* | +0.26+ | 0.18 (0.20) | +0.04 |
| 313 | B | 450 | 450/450 | 0.643 | 0.21 (0.13)* | +0.07+ | 0.35 (0.22)* | +0.04 |
| 314 | S | 11,386 | 3,415/11,386 | 3.225 | 0.08 (0.04)* | +0.21+ | 0.13 (0.17) | -0.11 |
| 314 | B | 218 | 218/218 | 0.550 | 0.19 (0.17) | -0.13 | 0.31 (0.21)* | +0.00 |
| 315 | S | 15,938 | 15,938/15,938 | 3.941 | 0.48 (0.12)* | +0.15+ | 0.21 (0.24) | +0.56+ |
| 315 | B | 398 | 398/398 | 0.595 | 0.18 (0.23) | +0.11+ | 0.44 (0.30)* | -0.09 |
| 316 | S | 21,999 | 70/21,999 | 5.155 | 0.17 (0.07)* | +0.00 | 0.16 (0.22) | +0.06 |
| 316 | B | 2,355 | 373/2,355 | 1.233 | 0.15 (0.09)* | +0.02 | 0.30 (0.20)* | +0.02 |
| 317 | S | 17,974 | 5,813/17,974 | 4.456 | 0.48 (0.05)* | +0.33+ | 0.25 (0.24) | +0.00 |
| 317 | B | 457 | 457/457 | 0.651 | 0.20 (0.18) | +0.03 | 0.15 (0.25) | -0.03 |
| 318 | S | 13,626 | 6,620/13,626 | 3.818 | 0.23 (0.08)* | +0.29+ | 0.18 (0.19) | -0.19 |
| 318 | B | 192 | 192/192 | 0.593 | 0.11 (0.06)* | -0.03 | 0.14 (0.23) | -0.11 |
| 319 | S | 16 | 16/16 | 0.622 | 0.86 (0.31)* | +0.56+ | 0.55 (0.22)* | +0.50+ |
| 319 | B | 1,419 | 877/1,419 | 0.997 | 0.12 (0.09) | +0.00 | 0.48 (0.24)* | +0.41+ |
| 320 | S | 11,092 | 11,092/11,092 | 2.650 | 0.31 (0.14)* | +0.06 | 0.26 (0.14)* | +0.21+ |
| 320 | B | 1,421 | 577/1,421 | 0.917 | 0.08 (0.13) | -0.13 | 0.45 (0.24)* | +0.08+ |
| 321 | S | 13,397 | 13,397/13,397 | 3.790 | 0.44 (0.09)* | +0.07+ | 0.33 (0.16)* | +0.13+ |
| 321 | B | 1,173 | 232/1,173 | 0.831 | 0.04 (0.09) | +0.04 | 0.10 (0.21) | -0.03 |
| 322 | S | 14,354 | 14,354/14,354 | 3.629 | 0.39 (0.12)* | +0.14+ | 0.34 (0.15)* | -0.01 |
| 322 | B | 1,571 | 586/1,571 | 0.992 | 0.36 (0.23)* | -0.06 | 0.32 (0.32) | -0.05 |
| 323 | S | 0 | 0/0 | 0.598 | 0.19 (0.08)* | +0.15+ | 0.30 (0.14)* | +0.15+ |
| 323 | B | 2,262 | 381/2,262 | 1.198 | 0.15 (0.10)* | -0.18 | 0.16 (0.20) | -0.20 |
| 324 | S | 19,554 | 12/19,554 | 4.874 | 0.54 (0.10)* | +0.45+ | 0.50 (0.25)* | -0.11 |
| 324 | B | 2,950 | 805/2,950 | 1.298 | 0.16 (0.12) | -0.08 | 0.13 (0.23) | -0.10 |
| 325 | S | 17,759 | 10,203/17,759 | 4.852 | 0.36 (0.11)* | +0.46+ | 0.34 (0.20)* | +0.06 |
| 325 | B | 3,561 | 114/3,561 | 1.642 | 0.39 (0.17)* | +0.08+ | 0.48 (0.25)* | +0.05 |

Table 41: Cipher world, configuration A, whole at 120 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 12 | 12/12 | 0.890 | 0.62 (0.19)* | +0.27+ | 0.51 (0.23)* | +0.20+ |
| 306 | B | 721 | 721/721 | 0.766 | 0.05 (0.08) | -0.05 | 0.14 (0.20) | +0.12+ |
| 307 | S | 7,184 | 7,184/7,184 | 2.326 | 0.20 (0.08)* | +0.14+ | 0.28 (0.24) | +0.03 |
| 307 | B | 1,499 | 743/1,499 | 0.986 | 0.07 (0.11) | +0.14+ | 0.20 (0.29) | -0.08 |
| 308 | S | 13,683 | 705/13,683 | 2.944 | 0.22 (0.04)* | +0.04 | 0.23 (0.17)* | -0.05 |
| 308 | B | 724 | 724/724 | 0.798 | 0.14 (0.11) | -0.07 | 0.33 (0.24)* | +0.02 |
| 309 | S | 15,431 | 5,671/15,431 | 3.306 | 0.26 (0.10)* | +0.13+ | 0.25 (0.21) | +0.04 |

Table 41: Cipher world, configuration A, whole at 120 at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 309 | B | 3,182 | 326/3,182 | 1.429 | 0.16 (0.15) | -0.12 | 0.17 (0.28) | +0.43$^+$ |
| 310 | S | 13,040 | 1,168/13,040 | 3.180 | 0.20 (0.09)* | +0.21$^+$ | 0.38 (0.15)* | +0.11$^+$ |
| 310 | B | 1,152 | 367/1,152 | 0.930 | 0.10 (0.19) | +0.15$^+$ | 0.05 (0.23) | -0.05 |
| 311 | S | 183 | 183/183 | 0.964 | 0.40 (0.08)* | +0.09$^+$ | 0.56 (0.18)* | +0.25$^+$ |
| 311 | B | 2,201 | 1,004/2,201 | 1.180 | 0.12 (0.12) | +0.03 | 0.22 (0.24) | -0.09 |
| 312 | S | 10,725 | 3,016/10,725 | 2.746 | 0.07 (0.06) | +0.19$^+$ | 0.52 (0.33)* | +0.07 |
| 312 | B | 1,283 | 544/1,283 | 0.911 | 0.02 (0.08) | +0.10$^+$ | 0.18 (0.24) | -0.01 |
| 313 | S | 15,066 | 4,804/15,066 | 3.388 | 0.43 (0.12)* | +0.39$^+$ | 0.36 (0.22)* | -0.14 |
| 313 | B | 701 | 278/701 | 0.850 | 0.46 (0.27)* | -0.07 | 0.38 (0.23)* | -0.09 |
| 314 | S | 7,833 | 7,833/7,833 | 2.263 | 0.49 (0.07)* | +0.18$^+$ | 0.38 (0.17)* | +0.12$^+$ |
| 314 | B | 61 | 61/61 | 0.664 | 0.16 (0.20) | -0.04 | 0.20 (0.20) | -0.14 |
| 315 | S | 17,900 | 8,578/17,900 | 3.532 | 0.77 (0.26)* | +0.26$^+$ | 0.49 (0.23)* | +0.26$^+$ |
| 315 | B | 1,013 | 504/1,013 | 0.828 | 0.15 (0.19) | -0.07 | 0.28 (0.30) | -0.05 |

Table 42: Cipher world, configuration B, first four pairs at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 312 | S | 1,706 | 1,706/1,706 | 1.138 | 0.61 (0.24)* | +0.52$^+$ | 0.34 (0.34) | -0.07 |
| 312 | B | 1,595 | 131/1,595 | 0.998 | 0.13 (0.15) | +0.06 | 0.31 (0.20)* | -0.02 |
| 313 | S | 8,465 | 2,487/8,465 | 2.600 | 0.06 (0.05) | +0.06 | 0.28 (0.22)* | +0.00 |
| 313 | B | 121 | 121/121 | 0.528 | 0.30 (0.30) | +0.00 | 0.16 (0.24) | +0.06 |
| 314 | S | 5,755 | 5,755/5,755 | 1.960 | 0.49 (0.22)* | +0.27$^+$ | 0.34 (0.23)* | +0.15$^+$ |
| 314 | B | 591 | 94/591 | 0.666 | 0.18 (0.18) | -0.07 | 0.35 (0.26)* | -0.04 |
| 315 | S | 10,258 | 3,426/10,258 | 2.530 | 0.08 (0.08) | +0.34$^+$ | 0.40 (0.24)* | +0.05 |
| 315 | B | 1,562 | 232/1,562 | 0.951 | 0.03 (0.14) | +0.03 | 0.16 (0.23) | -0.03 |

Table 43: Cipher world, configuration B, second set (eight pairs) at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 316 | S | 9,910 | 9,910/9,910 | 2.726 | 0.90 (0.32)* | +0.57$^+$ | 0.38 (0.24)* | +0.36$^+$ |
| 316 | B | 762 | 240/762 | 0.740 | 0.14 (0.13) | -0.07 | 0.36 (0.28)* | -0.06 |
| 317 | S | 8,450 | 8,450/8,450 | 2.994 | 0.37 (0.11)* | -0.01 | 0.40 (0.24)* | +0.10$^+$ |
| 317 | B | 385 | 213/385 | 0.614 | 0.09 (0.21) | +0.01 | 0.24 (0.24) | -0.06 |
| 318 | S | 121 | 121/121 | 0.572 | 0.54 (0.11)* | +0.18$^+$ | 0.30 (0.23)* | +0.09$^+$ |
| 318 | B | 1,342 | 405/1,342 | 0.853 | 0.01 (0.17) | +0.14$^+$ | 0.25 (0.17)* | +0.04 |
| 319 | S | 1,193 | 1,193/1,193 | 0.915 | 0.22 (0.07)* | +0.19$^+$ | 0.45 (0.18)* | +0.12$^+$ |
| 319 | B | 550 | 321/550 | 0.692 | 0.08 (0.15) | -0.06 | 0.15 (0.32) | +0.10$^+$ |
| 320 | S | 12,787 | 118/12,787 | 3.261 | 0.32 (0.09)* | +0.09$^+$ | 0.42 (0.23)* | +0.03 |
| 320 | B | 5 | 5/5 | 0.477 | 0.05 (0.12) | -0.13 | 0.12 (0.23) | -0.08 |
| 321 | S | 5,932 | 5,797/5,932 | 2.183 | 0.16 (0.07)* | +0.14$^+$ | 0.21 (0.25) | +0.57$^+$ |
| 321 | B | 1,318 | 55/1,318 | 0.837 | 0.00 (0.10) | +0.04 | 0.20 (0.19) | +0.04 |
| 322 | S | 180 | 16/180 | 0.662 | 0.51 (0.11)* | +0.11$^+$ | 0.45 (0.20)* | +0.32$^+$ |
| 322 | B | 129 | 129/129 | 0.511 | 0.12 (0.19) | -0.05 | 0.26 (0.24) | -0.02 |
| 323 | S | 6,054 | 6,054/6,054 | 2.198 | 0.73 (0.14)* | +0.44$^+$ | 0.63 (0.25)* | +0.50$^+$ |

Table 43: Cipher world, configuration B, second set (eight pairs) at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 323 | B | 993 | 755/993 | 0.800 | 0.19 (0.16) | $+0.12^+$ | 0.24 (0.24) | -0.03 |

Table 44: Cipher world, births and deaths (configuration B), first four pairs at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 312 | S | 82 | 82/82 | 0.559 | 0.74 (0.23)* | $+0.39^+$ | 0.52 (0.21)* | $+0.14^+$ |
| 312 | B | 543 | 82/543 | 0.604 | 0.20 (0.21) | -0.02 | 0.28 (0.26) | +0.00 |
| 313 | S | 6 | 6/6 | 0.550 | 0.49 (0.08)* | $+0.13^+$ | 0.57 (0.25)* | $+0.37^+$ |
| 313 | B | 558 | 144/558 | 0.661 | 0.08 (0.11) | -0.05 | 0.28 (0.25) | +0.01 |
| 314 | S | 232 | 232/232 | 0.606 | 0.58 (0.17)* | -0.03 | 0.40 (0.23)* | $+0.25^+$ |
| 314 | B | 177 | 177/177 | 0.509 | 0.26 (0.27) | +0.04 | 0.27 (0.24) | +0.04 |
| 315 | S | 0 | 0/0 | 0.521 | 0.69 (0.21)* | $+0.43^+$ | 0.68 (0.31)* | $+0.43^+$ |
| 315 | B | 537 | 218/537 | 0.613 | 0.37 (0.20)* | +0.07 | 0.45 (0.23)* | $+0.18^+$ |

Table 45: Cipher world, births and deaths (configuration B), second four pairs at 20,000, rounds 18,000–19,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 496 | 496/496 | 0.720 | 0.55 (0.19)* | $+0.37^+$ | 0.70 (0.36)* | $+0.34^+$ |
| 306 | B | 521 | 89/521 | 0.611 | 0.10 (0.18) | +0.07 | 0.37 (0.24)* | $+0.17^+$ |
| 307 | S | 388 | 388/388 | 0.730 | 0.47 (0.12)* | $+0.24^+$ | 0.50 (0.23)* | $+0.30^+$ |
| 307 | B | 54 | 51/54 | 0.487 | 0.04 (0.12) | -0.04 | 0.24 (0.21) | $+0.10^+$ |
| 308 | S | 106 | 106/106 | 0.541 | 0.46 (0.13)* | $+0.23^+$ | 0.23 (0.23) | $+0.31^+$ |
| 308 | B | 81 | 22/81 | 0.469 | 0.11 (0.18) | -0.05 | 0.30 (0.32) | -0.08 |
| 309 | S | 48 | 48/48 | 0.508 | 0.30 (0.20)* | +0.03 | 0.26 (0.25) | -0.02 |
| 309 | B | 0 | 0/0 | 0.438 | 0.30 (0.20)* | -0.06 | 0.28 (0.27) | -0.07 |

Table 46: Cipher world, configuration A, whole at 240 at the end of the run, rounds the last 2,000 before the stop. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^+$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 310 | S | not read | | | | | | |
| 310 | B | 2,070 | 932/2,070 | | – 0.16 (0.09)* | $+0.22^+$ | 0.23 (0.24) | -0.04 |
| 311 | S | not read | | | | | | |
| 311 | B | 2,573 | 1,606/2,573 | | – 0.01 (0.08) | +0.06 | 0.36 (0.24)* | $+0.13^+$ |
| 312 | S | 9,075 | 5,185/9,075 | | – 0.40 (0.11)* | $+0.14^+$ | 0.40 (0.24)* | $+0.40^+$ |
| 312 | B | 1,069 | 109/1,069 | | – 0.29 (0.21)* | -0.04 | 0.32 (0.29) | -0.07 |
| 313 | S | 13,990 | 6,857/13,990 | | – 0.36 (0.13)* | -0.01 | 0.21 (0.18) | $+0.15^+$ |

Table 46: Cipher world, configuration A, whole at 240 at the end of the run, rounds the last 2,000 before the stop. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 313 | B | 1,069 | 650/1,069 | – | 0.23 (0.10)* | -0.03 | 0.39 (0.25)* | -0.11 |
| 314 | S | 12,425 | 4,430/12,425 | – | 0.05 (0.06) | +0.12+ | 0.11 (0.18) | -0.10 |
| 314 | B | 266 | 266/266 | – | 0.07 (0.15) | -0.02 | 0.11 (0.25) | -0.11 |
| 315 | S | 14,387 | 14,387/14,387 | – | 0.32 (0.10)* | +0.06 | 0.20 (0.22) | +0.39+ |
| 315 | B | 1,110 | 1,110/1,110 | – | 0.14 (0.17) | -0.01 | 0.27 (0.22) | -0.05 |
| 316 | S | 26,626 | 1/26,626 | – | 0.28 (0.08)* | +0.09+ | 0.20 (0.16) | +0.07+ |
| 316 | B | 3,056 | 543/3,056 | – | 0.00 (0.09) | -0.03 | 0.17 (0.26) | +0.08 |
| 317 | S | 19,246 | 10,977/19,246 | – | 0.34 (0.05)* | +0.15+ | 0.13 (0.17) | -0.04 |
| 317 | B | 696 | 696/696 | – | 0.04 (0.15) | -0.09 | 0.00 (0.22) | +0.15+ |
| 318 | S | 16,202 | 11,892/16,202 | – | 0.22 (0.07)* | +0.24+ | 0.41 (0.26)* | -0.15 |
| 318 | B | 1,670 | 319/1,670 | – | 0.12 (0.10) | +0.16+ | 0.25 (0.23) | +0.32+ |
| 319 | S | 8,889 | 8,889/8,889 | – | 0.57 (0.12)* | +0.47+ | 0.35 (0.21)* | +0.41+ |
| 319 | B | 1,337 | 660/1,337 | – | 0.07 (0.07) | +0.10+ | 0.37 (0.24)* | +0.12+ |
| 320 | S | 14,970 | 14,970/14,970 | – | 0.41 (0.20)* | +0.21+ | 0.36 (0.24)* | +0.11+ |
| 320 | B | 1,261 | 420/1,261 | – | 0.16 (0.11)* | -0.18 | 0.31 (0.21)* | -0.12 |
| 321 | S | 14,806 | 14,583/14,806 | – | 0.14 (0.06)* | +0.16+ | 0.45 (0.23)* | +0.10+ |
| 321 | B | 1,374 | 342/1,374 | – | 0.07 (0.08) | -0.17 | 0.12 (0.22) | +0.06 |
| 322 | S | 15,922 | 15,922/15,922 | – | 0.52 (0.10)* | +0.24+ | 0.47 (0.18)* | +0.33+ |
| 322 | B | 2,880 | 567/2,880 | – | 0.12 (0.14) | -0.19 | 0.25 (0.26) | -0.09 |
| 323 | S | 13,319 | 0/13,319 | – | 0.34 (0.10)* | +0.10+ | 0.13 (0.13) | +0.05 |
| 323 | B | 4,211 | 761/4,211 | – | 0.06 (0.09) | +0.08+ | 0.17 (0.25) | -0.15 |
| 324 | S | 22,738 | 827/22,738 | – | 0.44 (0.12)* | +0.33+ | 0.29 (0.19)* | -0.27 |
| 324 | B | 1,620 | 345/1,620 | – | 0.13 (0.10) | -0.05 | 0.15 (0.21) | +0.00 |
| 325 | S | 20,468 | 13,278/20,468 | – | 0.33 (0.09)* | +0.20+ | 0.56 (0.18)* | +0.28+ |
| 325 | B | 4,672 | 62/4,672 | – | 0.11 (0.16) | -0.04 | 0.41 (0.26)* | -0.06 |

Table 47: Cipher world, configuration A, whole at 120 at the end of the run, rounds the last 2,000 before the stop. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and + above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 203 | 203/203 | – | 0.41 (0.09)* | +0.09+ | 0.33 (0.16)* | +0.02 |
| 306 | B | 1,046 | 1,046/1,046 | – | 0.02 (0.12) | +0.01 | 0.28 (0.25) | +0.37+ |
| 307 | S | 11,598 | 11,598/11,598 | – | 0.32 (0.11)* | +0.09+ | 0.44 (0.22)* | -0.02 |
| 307 | B | 1,382 | 737/1,382 | – | 0.28 (0.08)* | +0.03 | 0.33 (0.23)* | -0.10 |
| 308 | S | 13,912 | 1,349/13,912 | – | 0.22 (0.07)* | +0.08+ | 0.18 (0.21) | -0.11 |
| 308 | B | 962 | 962/962 | – | 0.04 (0.12) | +0.04 | 0.15 (0.22) | +0.04 |
| 309 | S | 8,668 | 1,552/8,668 | – | 0.06 (0.06) | +0.25+ | 0.33 (0.19)* | -0.04 |
| 309 | B | 2,014 | 228/2,014 | – | 0.04 (0.13) | +0.05 | 0.10 (0.24) | +0.46+ |
| 310 | S | 9,472 | 675/9,472 | – | 0.38 (0.07)* | +0.09+ | 0.34 (0.14)* | +0.10+ |
| 310 | B | 1,251 | 494/1,251 | – | 0.09 (0.17) | -0.08 | 0.05 (0.24) | +0.04 |
| 311 | S | 2,329 | 2,329/2,329 | – | 0.42 (0.09)* | +0.15+ | 0.42 (0.24)* | +0.06 |
| 311 | B | 3,257 | 1,153/3,257 | – | 0.17 (0.13)* | -0.02 | 0.22 (0.22) | -0.09 |
| 312 | S | 12,442 | 4,508/12,442 | – | 0.19 (0.06)* | +0.01 | 0.72 (0.33)* | +0.28+ |
| 312 | B | 1,138 | 521/1,138 | – | 0.05 (0.10) | +0.04 | 0.10 (0.33) | +0.21+ |
| 313 | S | 10,492 | 2,974/10,492 | – | 0.33 (0.13)* | +0.58+ | 0.51 (0.24)* | +0.27+ |
| 313 | B | 996 | 419/996 | – | 0.07 (0.13) | +0.07 | 0.19 (0.25) | -0.04 |
| 314 | S | 1,937 | 1,937/1,937 | – | 0.44 (0.11)* | +0.24+ | 0.35 (0.22)* | +0.30+ |
| 314 | B | not read | | | | | | |
| 315 | S | 17,490 | 8,815/17,490 | – | 0.71 (0.32)* | +0.21+ | 0.42 (0.25)* | +0.24+ |
| 315 | B | not read | | | | | | |

Table 48: Cipher world, configuration A, learned prices at the end of the run, rounds the last 2,000 before the stop. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 9,130 | 19/9,130 | – | 0.07 (0.07) | -0.09 | 0.09 (0.25) | -0.09 |
| 306 | B | 1,937 | 22/1,937 | – | 0.05 (0.08) | -0.09 | 0.43 (0.24)* | +0.24$^{+}$ |
| 307 | S | 6,870 | 0/6,870 | – | 0.04 (0.03) | +0.09$^{+}$ | 0.24 (0.23) | +0.06 |
| 307 | B | 2,635 | 101/2,635 | – | 0.00 (0.06) | +0.06 | 0.27 (0.25) | +0.10$^{+}$ |
| 308 | S | 7,679 | 0/7,679 | – | 0.19 (0.09)* | +0.17$^{+}$ | 0.15 (0.24) | -0.18 |
| 308 | B | 2,194 | 196/2,194 | – | 0.06 (0.07) | -0.03 | 0.15 (0.22) | -0.12 |
| 309 | S | 5,354 | 43/5,354 | – | 0.08 (0.06) | +0.04 | 0.24 (0.23) | +0.05 |
| 309 | B | 2,547 | 28/2,547 | – | 0.03 (0.08) | -0.07 | 0.17 (0.24) | +0.00 |
| 310 | S | 10,809 | 0/10,809 | – | 0.10 (0.05)* | +0.04 | 0.20 (0.18) | -0.08 |
| 310 | B | 1,426 | 68/1,426 | – | 0.15 (0.11) | +0.16$^{+}$ | 0.20 (0.29) | +0.00 |
| 311 | S | 1,215 | 1,215/1,215 | – | 0.10 (0.04)* | +0.11$^{+}$ | 0.34 (0.20)* | +0.07 |
| 311 | B | 1,216 | 67/1,216 | – | 0.08 (0.08) | -0.07 | 0.14 (0.18) | +0.01 |
| 312 | S | 9,856 | 0/9,856 | – | 0.32 (0.10)* | +0.11$^{+}$ | 0.17 (0.25) | -0.08 |
| 312 | B | 1,179 | 60/1,179 | – | 0.03 (0.10) | -0.10 | 0.10 (0.24) | +0.18$^{+}$ |
| 313 | S | 8,962 | 1/8,962 | – | 0.25 (0.12)* | +0.13$^{+}$ | 0.37 (0.20)* | +0.10$^{+}$ |
| 313 | B | 2,604 | 8/2,604 | – | 0.02 (0.08) | +0.00 | 0.17 (0.23) | +0.13$^{+}$ |

Table 49: Cipher reconstruction, half level, source 317 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 417 | S | 18,555 | 10,912/18,555 | – | 0.14 (0.07)* | +0.26$^{+}$ | 0.26 (0.29) | +0.05 |

Table 50: Cipher reconstruction, half level, source 320 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 420 | S | 11,736 | 9,794/11,736 | – | 0.54 (0.08)* | +0.09$^{+}$ | 0.40 (0.20)* | -0.16 |

Table 51: Cipher reconstruction, half level, source 322 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 422 | S | 10,091 | 8,730/10,091 | – | 0.55 (0.16)* | +0.30$^{+}$ | 0.57 (0.34)* | +0.23$^{+}$ |

Table 52: Cipher reconstruction, half level, source 324 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 21,236 | 4,070/21,236 | – | 0.39 (0.16)* | +0.47$^{+}$ | 0.64 (0.25)* | +0.17$^{+}$ |

Table 53: Cipher reconstruction, keyless level, source 322 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 12,087 | 12,087/12,087 | – | 0.45 (0.19)* | +0.22$^{+}$ | 0.35 (0.17)* | +0.24$^{+}$ |
| 307 | S | 4,807 | 4,807/4,807 | – | 0.49 (0.20)* | +0.06 | 0.45 (0.33)* | +0.33$^{+}$ |
| 308 | S | 9,307 | 9,307/9,307 | – | 0.39 (0.13)* | -0.01 | 0.71 (0.33)* | +0.52$^{+}$ |
| 312 | S | 16,432 | 10,443/16,432 | 4.322 | 0.60 (0.18)* | +0.20$^{+}$ | 0.26 (0.27) | +0.36$^{+}$ |
| 313 | S | 13,092 | 10,633/13,092 | 3.862 | 0.32 (0.08)* | +0.20$^{+}$ | 0.43 (0.24)* | +0.11$^{+}$ |
| 314 | S | 12,973 | 9,185/12,973 | 3.663 | 0.30 (0.08)* | +0.27$^{+}$ | 0.42 (0.24)* | -0.08 |
| 315 | S | 13,850 | 13,850/13,850 | 3.661 | 0.56 (0.14)* | +0.41$^{+}$ | 0.67 (0.26)* | +0.19$^{+}$ |
| 316 | S | 15,558 | 7,868/15,558 | 4.112 | 0.33 (0.13)* | +0.34$^{+}$ | 0.28 (0.24) | -0.03 |
| 317 | S | 8,399 | 8,308/8,399 | 2.274 | 0.69 (0.17)* | +0.14$^{+}$ | 0.49 (0.30)* | +0.08$^{+}$ |
| 318 | S | 12,751 | 6,003/12,751 | 3.743 | 0.49 (0.15)* | +0.37$^{+}$ | 0.61 (0.24)* | +0.09$^{+}$ |

Table 54: Cipher reconstruction, keyless level, source 324 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 309 | S | 5,236 | 2,338/5,236 | – | 0.61 (0.36)* | +0.05 | 0.37 (0.21)* | +0.00 |
| 310 | S | 10,822 | 1,406/10,822 | – | 0.33 (0.08)* | +0.23$^{+}$ | 0.16 (0.19) | -0.15 |
| 311 | S | 8,266 | 4,270/8,266 | – | 0.22 (0.08)* | +0.12$^{+}$ | 0.35 (0.24)* | +0.12$^{+}$ |

Table 55: Cipher reconstruction, outward level, source 317 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 417 | S | 20,355 | 7,609/20,355 | – | 0.33 (0.10)* | +0.36$^{+}$ | 0.65 (0.33)* | +0.33$^{+}$ |

Table 56: Cipher reconstruction, outward level, source 320 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 420 | S | 0 | 0/0 | – | 0.62 (0.09)* | +0.24$^{+}$ | 0.49 (0.12)* | +0.35$^{+}$ |

Table 57: Cipher reconstruction, outward level, source 322 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 319 | S | 0 | 0/0 | 0.631 | 0.58 (0.15)* | +0.12$^{+}$ | 0.55 (0.18)* | -0.01 |
| 320 | S | 10,107 | 10,107/10,107 | 2.680 | 0.31 (0.06)* | +0.07$^{+}$ | 0.47 (0.23)* | +0.16$^{+}$ |
| 321 | S | 10,032 | 10,032/10,032 | 2.896 | 0.46 (0.11)* | +0.57$^{+}$ | 0.79 (0.24)* | +0.41$^{+}$ |
| 322 | S | 0 | 0/0 | 0.623 | 0.60 (0.10)* | +0.26$^{+}$ | 0.74 (0.23)* | +0.41$^{+}$ |
| 323 | S | 11,116 | 11,116/11,116 | 3.204 | 0.37 (0.07)* | +0.18$^{+}$ | 0.30 (0.11)* | -0.10 |
| 422 | S | 2 | 2/2 | – | 0.50 (0.16)* | +0.12$^{+}$ | 0.56 (0.24)* | +0.01 |

Table 58: Cipher reconstruction, outward level, source 324 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 31,888 | 0/31,888 | – | 0.37 (0.18)* | +0.42$^{+}$ | 0.36 (0.24)* | -0.24 |

Table 59: Powered cipher reconstruction, outward level, source 322 (five worlds) at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 430 | S | 0 | 0/0 | 0.629 | 0.56 (0.07)* | +0.11$^{+}$ | 0.56 (0.14)* | +0.10$^{+}$ |
| 431 | S | 22,700 | 22,700/22,700 | 4.985 | 0.44 (0.06)* | +0.23$^{+}$ | 0.69 (0.26)* | +0.42$^{+}$ |
| 432 | S | 21,201 | 21,201/21,201 | 4.618 | 0.50 (0.22)* | +0.47$^{+}$ | 0.68 (0.26)* | +0.35$^{+}$ |
| 433 | S | 11 | 11/11 | 0.606 | 0.56 (0.11)* | +0.02 | 0.55 (0.16)* | +0.34$^{+}$ |
| 434 | S | 0 | 0/0 | 0.624 | 0.80 (0.19)* | +0.31$^{+}$ | 0.85 (0.25)* | +0.50$^{+}$ |

Table 60: Powered cipher reconstruction, outward level, source 324 (five worlds) at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 440 | S | 31,668 | 0/31,668 | 6.895 | 0.76 (0.18)* | $+0.60^{+}$ | 0.38 (0.24)* | -0.15 |
| 441 | S | 29,868 | 4,243/29,868 | 6.580 | 0.56 (0.17)* | $+0.39^{+}$ | 0.31 (0.22)* | -0.10 |
| 442 | S | 31,794 | 0/31,794 | 6.834 | 0.50 (0.17)* | $+0.77^{+}$ | 0.25 (0.23) | $+0.16^{+}$ |
| 443 | S | 31,754 | 745/31,754 | 6.813 | 0.40 (0.17)* | $+0.43^{+}$ | 0.41 (0.23)* | -0.11 |
| 444 | S | 32,280 | 0/32,280 | 7.051 | 0.44 (0.12)* | $+0.53^{+}$ | 0.41 (0.24)* | -0.16 |

Table 61: Blank starts beside the powered cipher reconstruction at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 430 | S | 15,295 | 80/15,295 | 3.933 | 0.39 (0.13)* | $+0.50^{+}$ | 0.50 (0.34)* | $+0.12^{+}$ |
| 431 | S | 11,280 | 8,636/11,280 | 3.267 | 0.50 (0.27)* | $+0.20^{+}$ | 0.50 (0.48) | -0.10 |
| 432 | S | 14,019 | 2,397/14,019 | 3.863 | 0.76 (0.17)* | $+0.56^{+}$ | 0.45 (0.26)* | $+0.16^{+}$ |
| 433 | S | 13,983 | 3,732/13,983 | 3.469 | 0.24 (0.13)* | $+0.08^{+}$ | 0.51 (0.20)* | +0.04 |
| 434 | S | 13,496 | 10,906/13,496 | 3.592 | 0.47 (0.13)* | $+0.41^{+}$ | 0.35 (0.24)* | $+0.10^{+}$ |
| 440 | S | 13,583 | 6,615/13,583 | 3.667 | 0.40 (0.09)* | $+0.16^{+}$ | 0.33 (0.23)* | -0.16 |
| 441 | S | 12,928 | 2,690/12,928 | 3.531 | 0.36 (0.09)* | +0.02 | 0.30 (0.24) | +0.02 |
| 442 | S | 9,264 | 5,506/9,264 | 2.920 | 0.38 (0.07)* | $+0.29^{+}$ | 0.16 (0.16) | $+0.21^{+}$ |
| 443 | S | 9,207 | 9,202/9,207 | 3.102 | 0.32 (0.05)* | $+0.16^{+}$ | 0.50 (0.26)* | $+0.22^{+}$ |
| 444 | S | 12,145 | 7,008/12,145 | 3.483 | 0.18 (0.07)* | $+0.25^{+}$ | 0.58 (0.24)* | $+0.24^{+}$ |

Table 62: Cipher reconstruction, presence level, source 322 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 422 | S | 18,552 | 18,552/18,552 | – | 0.52 (0.07)* | -0.01 | 0.61 (0.18)* | $+0.30^{+}$ |

Table 63: Cipher reconstruction, presence level, source 324 at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 424 | S | 15,830 | 0/15,830 | – | 0.62 (0.12)* | $+0.57^{+}$ | 0.49 (0.24)* | +0.04 |

Table 64: Cipher world, configuration B, first four pairs at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 312 | S | 6,721 | 3,607/6,721 | – | 0.31 (0.13)* | +0.20$^{+}$ | 0.25 (0.25) | +0.16$^{+}$ |
| 312 | B | 2,487 | 70/2,487 | – | 0.09 (0.09) | +0.06 | 0.23 (0.19) | +0.03 |
| 313 | S | 5,865 | 1,609/5,865 | – | 0.31 (0.07)* | +0.11$^{+}$ | 0.27 (0.24) | -0.10 |
| 313 | B | 694 | 297/694 | – | 0.15 (0.23) | -0.09 | 0.19 (0.24) | -0.08 |
| 314 | S | 8,978 | 8,978/8,978 | – | 0.39 (0.17)* | +0.27$^{+}$ | 0.39 (0.24)* | +0.14$^{+}$ |
| 314 | B | 859 | 130/859 | – | 0.12 (0.14) | +0.11$^{+}$ | 0.07 (0.25) | +0.07$^{+}$ |
| 315 | S | 10,913 | 4,236/10,913 | – | 0.17 (0.06)* | +0.08$^{+}$ | 0.31 (0.24)* | -0.05 |
| 315 | B | 2,226 | 550/2,226 | – | 0.13 (0.15) | -0.05 | 0.18 (0.23) | -0.11 |

Table 65: Cipher world, configuration B, second set (eight pairs) at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 316 | S | 14,320 | 8,066/14,320 | 3.942 | 0.70 (0.16)* | +0.58$^{+}$ | 0.38 (0.25)* | +0.13$^{+}$ |
| 316 | B | 1,313 | 482/1,313 | 0.902 | 0.26 (0.15)* | +0.08$^{+}$ | 0.32 (0.27) | -0.06 |
| 317 | S | 11,189 | 11,189/11,189 | 3.030 | 0.41 (0.07)* | +0.31$^{+}$ | 0.37 (0.25)* | +0.35$^{+}$ |
| 317 | B | 1,829 | 595/1,829 | 1.037 | 0.12 (0.12) | -0.12 | 0.26 (0.24) | -0.10 |
| 318 | S | 14,438 | 10,681/14,438 | 3.695 | 0.42 (0.11)* | +0.26$^{+}$ | 0.26 (0.20)* | +0.08$^{+}$ |
| 318 | B | 2,125 | 418/2,125 | 1.161 | 0.03 (0.12) | +0.03 | 0.06 (0.23) | +0.02 |
| 319 | S | 11,588 | 8,261/11,588 | 3.609 | 0.03 (0.03) | +0.11$^{+}$ | 0.38 (0.16)* | +0.03 |
| 319 | B | 908 | 547/908 | 0.827 | 0.06 (0.11) | -0.10 | 0.30 (0.25)* | -0.26 |
| 320 | S | 13,482 | 6,671/13,482 | 3.609 | 0.51 (0.16)* | +0.43$^{+}$ | 0.18 (0.29) | +0.29$^{+}$ |
| 320 | B | 16 | 16/16 | 0.503 | 0.17 (0.13) | -0.16 | 0.23 (0.23) | +0.04 |
| 321 | S | 15,607 | 7,988/15,607 | 4.025 | 0.46 (0.10)* | +0.23$^{+}$ | 0.22 (0.23) | +0.17$^{+}$ |
| 321 | B | 2,900 | 105/2,900 | 1.354 | 0.10 (0.18) | +0.11$^{+}$ | 0.07 (0.24) | +0.02 |
| 322 | S | 10,916 | 5,180/10,916 | 3.227 | 0.55 (0.13)* | +0.35$^{+}$ | 0.27 (0.19)* | +0.25$^{+}$ |
| 322 | B | 1,663 | 1,166/1,663 | 0.975 | 0.46 (0.25)* | +0.18$^{+}$ | 0.59 (0.22)* | +0.42$^{+}$ |
| 323 | S | 11,249 | 9,854/11,249 | 3.217 | 0.62 (0.12)* | +0.30$^{+}$ | 0.22 (0.25) | +0.09$^{+}$ |
| 323 | B | 1,416 | 904/1,416 | 0.960 | 0.34 (0.16)* | +0.09$^{+}$ | 0.16 (0.24) | +0.10$^{+}$ |

Table 66: Cipher world, births and deaths (configuration B), first four pairs at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 312 | S | 855 | 855/855 | – | 0.45 (0.10)* | +0.14$^{+}$ | 0.57 (0.24)* | +0.23$^{+}$ |
| 312 | B | 442 | 59/442 | – | 0.20 (0.17) | +0.06 | 0.49 (0.28)* | +0.08 |
| 313 | S | 11 | 11/11 | – | 0.26 (0.06)* | -0.04 | 0.36 (0.24)* | +0.18$^{+}$ |
| 313 | B | 481 | 103/481 | – | 0.22 (0.10)* | -0.15 | 0.18 (0.24) | -0.13 |
| 314 | S | 83 | 83/83 | – | 0.50 (0.16)* | +0.03 | 0.37 (0.21)* | +0.27$^{+}$ |
| 314 | B | 113 | 113/113 | – | 0.29 (0.28) | +0.04 | 0.34 (0.35) | -0.07 |
| 315 | S | 2 | 0/2 | – | 0.24 (0.10)* | +0.03 | 0.41 (0.23)* | +0.18$^{+}$ |
| 315 | B | 484 | 186/484 | – | 0.26 (0.20)* | -0.02 | 0.19 (0.29) | -0.08 |

Table 67: Cipher world, births and deaths (configuration B), second four pairs at the end of the run, rounds 38,000–39,999. Right: fully right wholes delivered to consumers in the window; assembled: of those, by two or more learners; utility: consumers' mean utility in the last window where logged; categories over the last 2,000 boards by product and by head, accuracy (null) and mutual-information excess in bits, * and $^{+}$ above the null at $p < 0.05$.

| seed | side | right | assembled | utility | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|
| 306 | S | 158 | 0/158 | 0.605 | 0.50 (0.19)* | +0.05 | 0.54 (0.28)* | +0.22$^{+}$ |
| 306 | B | 7 | 5/7 | 0.464 | 0.17 (0.16) | -0.02 | 0.17 (0.17) | -0.03 |
| 307 | S | 13 | 13/13 | 0.486 | 0.39 (0.18)* | +0.32$^{+}$ | 0.42 (0.23)* | +0.32$^{+}$ |
| 307 | B | 304 | 1/304 | 0.544 | 0.27 (0.33) | -0.02 | 0.61 (0.49)* | +0.01 |
| 308 | S | 80 | 80/80 | 0.571 | 0.23 (0.07)* | +0.16$^{+}$ | 0.32 (0.19)* | -0.10 |
| 308 | B | 39 | 39/39 | 0.474 | 0.26 (0.31) | +0.06 | 0.38 (0.29)* | -0.07 |
| 309 | S | 1,513 | 677/1,513 | 1.028 | 0.25 (0.08)* | -0.04 | 0.62 (0.23)* | +0.33$^{+}$ |
| 309 | B | 485 | 27/485 | 0.594 | 0.20 (0.18) | +0.13$^{+}$ | 0.10 (0.21) | -0.03 |

Table 68: Machine-learning reconstruction, outward from the team world 517 at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (1374/6400) | (2301/9820) | (3787/6484) | 0.711 | 0.55/0.45 | 0.16 (0.10)* | +0.10$^{+}$ | 0.25 (0.18)* | -0.12 |
| 307 | S | (3278/10442) | (3283/10858) | (2324/5072) | 0.959 | 0.7/0.6 | 0.18 (0.11)* | -0.18 | 0.31 (0.20)* | +0.01 |
| 308 | S | (1/302) | (0/274) | (11761/13048) | 0.537 | 0.58/0.32 | 0.52 (0.17)* | +0.21$^{+}$ | 0.69 (0.27)* | +0.55$^{+}$ |
| 309 | S | (0/345) | (0/359) | (11376/12816) | 0.534 | 0.52/0.25 | 0.33 (0.12)* | +0.34$^{+}$ | 0.71 (0.32)* | +0.29$^{+}$ |
| 310 | S | (3274/9574) | (4584/11540) | (3174/5225) | 1.035 | 0.78/0.74 | 0.20 (0.13)* | -0.03 | 0.28 (0.28) | -0.06 |
| 311 | S | (6/518) | (1/387) | (10065/12521) | 0.523 | 0.57/0.38 | 0.33 (0.11)* | +0.02 | 0.38 (0.30)* | -0.11 |
| 312 | S | (2/281) | (3/271) | (10838/12368) | 0.533 | 0.51/0.21 | 0.20 (0.17) | +0.25$^{+}$ | 0.53 (0.32)* | +0.36$^{+}$ |
| 315 | S | (1069/4540) | (1039/4649) | (6197/8492) | 0.664 | 0.52/0.32 | 0.53 (0.15)* | +0.51$^{+}$ | 0.67 (0.25)* | +0.43$^{+}$ |
| 316 | S | (7797/14102) | (5318/9874) | (2540/4456) | 1.563 | 0.83/0.86 | 0.56 (0.24)* | +0.22$^{+}$ | 0.64 (0.29)* | +0.14$^{+}$ |
| 317 | S | (1179/4024) | (1323/4838) | (5822/7839) | 0.703 | 0.53/0.36 | 0.43 (0.16)* | +0.13$^{+}$ | 0.75 (0.32)* | +0.20$^{+}$ |
| 318 | S | (8/725) | (5/391) | (10404/12876) | 0.522 | 0.55/0.37 | 0.15 (0.20) | +0.01 | 0.40 (0.38) | -0.12 |
| 319 | S | (1532/4933) | (1232/3991) | (5777/7556) | 0.78 | 0.54/0.39 | 0.10 (0.12) | +0.35$^{+}$ | 0.73 (0.30)* | +0.46$^{+}$ |
| 320 | S | (42/1016) | (206/2393) | (9248/11625) | 0.549 | 0.49/0.31 | 0.33 (0.16)* | +0.04 | 0.47 (0.32)* | +0.18$^{+}$ |
| 321 | S | (3776/11562) | (3308/10272) | (1623/3958) | 1.018 | 0.74/0.72 | 0.18 (0.14)* | +0.12$^{+}$ | 0.22 (0.30) | +0.09$^{+}$ |
| 322 | S | (5049/12208) | (5397/12824) | (2593/4641) | 1.176 | 0.81/0.79 | 0.10 (0.11) | -0.03 | 0.35 (0.19)* | -0.01 |
| 323 | S | (19/604) | (84/1204) | (9548/12141) | 0.527 | 0.52/0.25 | 0.44 (0.12)* | +0.12$^{+}$ | 0.52 (0.21)* | +0.22$^{+}$ |
| 324 | S | (633/4557) | (264/2546) | (6924/9046) | 0.592 | 0.43/0.18 | 0.06 (0.12) | +0.02 | 0.56 (0.30)* | +0.16$^{+}$ |
| 325 | S | (2443/9551) | (2457/9793) | (5048/6479) | 0.956 | 0.72/0.61 | 0.33 (0.15)* | +0.34$^{+}$ | 0.59 (0.29)* | +0.30$^{+}$ |

Table 69: Machine-learning reconstruction, half level from the team world 517 at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 313 | S | (849/5685) | (1225/5753) | (4483/7862) | 0.671 | 0.51/0.36 | 0.50 (0.24)* | +0.12$^{+}$ | 0.60 (0.35)* | +0.01 |
| 314 | S | (3687/10252) | (4333/11548) | (2421/4679) | 1.068 | 0.76/0.78 | 0.51 (0.23)* | -0.02 | 0.67 (0.32)* | +0.12$^{+}$ |

Table 70: Machine-learning reconstruction, presence level from world 712 at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 310 | S | (0/258) | (0/236) | (12348/13239) | 0.54 | 0.58/0.61 | 0.85 (0.65)* | -0.02 | 0.85 (0.65)* | -0.02 |

Table 71: Machine-learning reconstruction, outward from world 714 at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (8925/12906) | (9769/13539) | (3216/4447) | 1.919 | 0.87/0.89 | 0.63 (0.22)* | $+0.33^{+}$ | 0.63 (0.21)* | $+0.34^{+}$ |
| 307 | S | (12215/14702) | (11759/14036) | (2100/3377) | 2.446 | 0.89/0.93 | 0.60 (0.27)* | -0.02 | 0.60 (0.27)* | -0.02 |
| 308 | S | (9845/12658) | (11604/14738) | (2389/3763) | 2.047 | 0.88/0.91 | 0.54 (0.27)* | -0.04 | 0.54 (0.27)* | -0.04 |
| 309 | S | (10975/14692) | (9518/12798) | (1547/3069) | 1.907 | 0.88/0.92 | 0.23 (0.30) | $+0.12^{+}$ | 0.23 (0.30) | $+0.12^{+}$ |
| 315 | S | (10259/12358) | (9926/11751) | (3921/4988) | 2.064 | 0.86/0.89 | 0.87 (0.36)* | $+0.45^{+}$ | 0.87 (0.36)* | $+0.45^{+}$ |

Table 72: Machine-learning reconstruction, presence level from world 714 at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (12328/14185) | (12817/14786) | (3125/3573) | 2.411 | 0.9/0.93 | 0.94 (0.40)* | $+0.47^{+}$ | 0.94 (0.40)* | $+0.47^{+}$ |

Table 73: Machine-learning main set, seeds 512–519 at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 512 | S | (0/1457) | (0/710) | (7422/12062) | 0.486 | 0.3/0.14 | 0.75 (0.35)* | $+0.48^{+}$ | 0.96 (0.49)* | $+0.70^{+}$ |
| 512 | B | (2/2636) | (2/1659) | (726/9286) | 0.383 | 0.15/0.08 | 0.25 (0.24) | +0.01 | 0.42 (0.37)* | -0.07 |
| 513 | S | (0/663) | (0/495) | (7354/12676) | 0.485 | 0.35/0.19 | 0.96 (0.60)* | $+0.49^{+}$ | 0.97 (0.56)* | $+0.61^{+}$ |
| 513 | B | (0/1080) | (0/1521) | (0/865) | 0.39 | 0.06/0.07 | 0.47 (0.41) | -0.01 | 0.47 (0.43) | -0.01 |
| 514 | S | (0/618) | (0/639) | (8972/12336) | 0.5 | 0.29/0.13 | 0.88 (0.42)* | $+0.67^{+}$ | 0.94 (0.45)* | $+0.73^{+}$ |
| 514 | B | (2/1156) | (0/1395) | (3111/10538) | 0.411 | 0.12/0.08 | 0.23 (0.25) | -0.03 | 0.43 (0.41) | -0.03 |
| 515 | S | (3507/10929) | (3703/11178) | (1601/4446) | 1.009 | 0.66/0.53 | 0.80 (0.39)* | $+0.49^{+}$ | 0.76 (0.26)* | $+0.50^{+}$ |
| 515 | B | (0/2236) | (0/1114) | (315/6426) | 0.382 | 0.15/0.08 | 0.28 (0.27) | +0.02 | 0.28 (0.27) | +0.02 |
| 516 | S | (4955/11366) | (5333/12298) | (2119/3982) | 1.201 | 0.82/0.84 | 0.56 (0.32)* | +0.04 | 0.79 (0.36)* | $+0.23^{+}$ |
| 516 | B | (0/2276) | (0/2220) | (1287/10569) | 0.388 | 0.06/0.07 | 0.19 (0.24) | -0.11 | 0.45 (0.50) | -0.06 |
| 517 | S | (2131/8775) | (164/2439) | (4350/7805) | 0.601 | 0.54/0.45 | 0.81 (0.58)* | $+0.09^{+}$ | 0.78 (0.51)* | $+0.10^{+}$ |
| 517 | B | (0/1777) | (0/1619) | (444/8030) | 0.384 | 0.07/0.08 | 0.44 (0.46) | $+0.03^{+}$ | 0.43 (0.49) | +0.01 |
| 518 | S | (3425/9677) | (3154/9101) | (2532/5292) | 0.956 | 0.73/0.74 | 0.49 (0.27)* | -0.04 | 0.68 (0.37)* | $+0.16^{+}$ |
| 518 | B | (0/2900) | (0/1834) | (1082/9880) | 0.388 | 0.12/0.09 | 0.48 (0.28)* | +0.05 | 0.52 (0.40)* | +0.01 |
| 519 | S | (0/705) | (0/890) | (8536/12266) | 0.495 | 0.5/0.3 | 0.84 (0.48)* | $+0.37^{+}$ | 0.87 (0.46)* | $+0.49^{+}$ |
| 519 | B | (199/7018) | (243/6507) | (350/5617) | 0.397 | 0.17/0.17 | 0.29 (0.34) | -0.05 | 0.44 (0.32)* | -0.11 |

Table 74: Machine-learning world, whole at 120 at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 512 | S | (0/583) | (0/683) | (8367/12194) | 0.648 | 0.46/0.29 | 0.81 (0.54)* | -0.01 | 0.89 (0.60)* | +0.01 |
| 512 | B | (0/2388) | (0/2304) | (1991/10330) | 0.467 | 0.05/0.08 | 0.71 (0.49)* | $+0.13^{+}$ | 0.70 (0.48)* | $+0.11^{+}$ |
| 513 | S | (0/913) | (0/631) | (7690/11978) | 0.667 | 0.38/0.2 | 0.94 (0.49)* | $+0.57^{+}$ | 0.95 (0.50)* | $+0.64^{+}$ |
| 513 | B | (0/1747) | (0/1061) | (0/629) | 0.456 | 0.09/0.07 | 0.50 (0.39)* | -0.03 | 0.70 (0.50)* | -0.04 |
| 514 | S | (0/471) | (0/511) | (10034/12813) | 0.682 | 0.46/0.22 | 0.81 (0.38)* | $+0.48^{+}$ | 0.88 (0.44)* | $+0.60^{+}$ |
| 514 | B | (0/2676) | (0/1818) | (2175/10435) | 0.472 | 0.07/0.07 | 0.46 (0.43) | -0.04 | 0.44 (0.43) | -0.07 |
| 515 | S | (0/814) | (0/437) | (6427/12798) | 0.61 | 0.44/0.46 | 0.77 (0.57)* | $+0.06^{+}$ | 0.91 (0.73)* | -0.00 |
| 515 | B | (2/2836) | (1/1950) | (1446/10013) | 0.459 | 0.12/0.08 | 0.63 (0.50)* | -0.05 | 0.64 (0.50)* | -0.06 |

Table 75: Machine-learning world, births and deaths at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (0/1837) | (0/1778) | (7465/11641) | 0.484 | 0.55/0.33 | 0.80 (0.40)* | $+0.47^{+}$ | 0.89 (0.45)* | $+0.47^{+}$ |
| 306 | B | (74/2985) | (91/2980) | (0/1354) | 0.404 | 0.39/0.09 | 0.70 (0.43) | -0.08 | 0.70 (0.43) | -0.08 |
| 307 | S | (0/2120) | (0/984) | (0/847) | 0.387 | 0.3/0.09 | 0.69 (0.66) | -0.01 | 0.89 (0.69)* | -0.03 |
| 307 | B | (0/1693) | (0/2133) | (1175/8843) | 0.389 | 0.47/0.11 | 0.39 (0.44) | -0.07 | 0.36 (0.38) | -0.06 |
| 308 | S | (0/1094) | (0/1595) | (2172/6540) | 0.383 | 0.43/0.17 | 0.76 (0.54)* | $+0.15^{+}$ | 0.79 (0.56)* | $+0.07^{+}$ |
| 308 | B | (0/3194) | (0/2955) | (0/1210) | 0.382 | 0.23/0.09 | 0.32 (0.30) | -0.06 | 0.45 (0.47) | -0.02 |
| 309 | S | (0/1909) | (0/1937) | (0/3319) | 0.384 | 0.37/0.11 | 0.70 (0.61)* | -0.03 | 0.86 (0.64)* | -0.04 |
| 309 | B | (2/1877) | (1/2598) | (194/4273) | 0.375 | 0.48/0.12 | 0.41 (0.36) | -0.01 | 0.41 (0.36) | +0.01 |
| 412 | S | (680/5888) | (302/3467) | (2604/7647) | 0.538 | 0.59/0.26 | 0.58 (0.37)* | +0.03 | 0.60 (0.35)* | $+0.05^{+}$ |
| 412 | B | (2/3008) | (1/1568) | (1720/9543) | 0.392 | 0.26/0.08 | 0.55 (0.39)* | +0.02 | 0.55 (0.41)* | -0.01 |
| 413 | S | (0/1046) | (0/1186) | (6306/11450) | 0.465 | 0.6/0.41 | 0.44 (0.26)* | $+0.09^{+}$ | 0.78 (0.39)* | $+0.29^{+}$ |
| 413 | B | (253/6642) | (214/5954) | (297/5539) | 0.394 | 0.45/0.19 | 0.10 (0.24) | +0.04 | 0.13 (0.28) | $+0.13^{+}$ |
| 414 | S | (1/1722) | (1/1511) | (2754/8388) | 0.38 | 0.43/0.19 | 0.67 (0.52)* | +0.02 | 0.84 (0.53)* | $+0.26^{+}$ |
| 414 | B | (0/1594) | (0/1808) | (0/813) | 0.381 | 0.51/0.11 | 0.15 (0.23) | +0.04 | 0.36 (0.31) | -0.04 |
| 415 | S | (2074/10794) | (1451/9085) | (2743/6025) | 0.605 | 0.74/0.54 | 0.47 (0.22)* | +0.06 | 0.66 (0.26)* | $+0.24^{+}$ |
| 415 | B | (0/1828) | (0/2146) | (2699/10639) | 0.399 | 0.37/0.1 | 0.41 (0.31)* | +0.01 | 0.60 (0.45)* | -0.01 |

Table 76: Machine-learning capacity arm (8 learners with 2 rows, 8 with 3) at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 320 | S | (0/1037) | (0/476) | (8366/11806) | 0.494 | 0.49/0.51 | 0.84 (0.54)* | +0.03 | 0.94 (0.60)* | $+0.19^{+}$ |
| 320 | B | (1/1762) | (0/1875) | (350/6018) | 0.385 | 0.22/0.09 | 0.17 (0.26) | +0.01 | 0.41 (0.42) | -0.03 |
| 321 | S | (1185/7691) | (747/6507) | (4231/7446) | 0.648 | 0.55/0.37 | 0.74 (0.32)* | $+0.37^{+}$ | 0.82 (0.33)* | $+0.40^{+}$ |
| 321 | B | (0/1431) | (0/2146) | (688/9172) | 0.38 | 0.18/0.09 | 0.42 (0.35)* | -0.05 | 0.43 (0.41) | -0.05 |
| 322 | S | (8714/14545) | (6616/11188) | (2502/4020) | 1.497 | 0.86/0.88 | 0.66 (0.34)* | $+0.32^{+}$ | 0.66 (0.33)* | $+0.32^{+}$ |
| 322 | B | (0/2307) | (0/2190) | (1035/9017) | 0.392 | 0.12/0.08 | 0.51 (0.44)* | +0.00 | 0.52 (0.44)* | +0.03 |
| 323 | S | (7162/10540) | (6066/9152) | (0/1033) | 1.519 | 0.8/0.84 | 0.74 (0.43)* | $+0.45^{+}$ | 0.75 (0.44)* | $+0.45^{+}$ |
| 323 | B | (0/1727) | (0/1900) | (0/882) | 0.388 | 0.13/0.08 | 0.34 (0.39) | +0.00 | 0.44 (0.47) | -0.00 |
| 716 | S | (2914/10018) | (2422/8249) | (2722/5934) | 0.878 | 0.72/0.69 | 0.85 (0.39)* | $+0.53^{+}$ | 0.86 (0.39)* | $+0.52^{+}$ |
| 716 | B | (0/2752) | (0/1737) | (0/626) | 0.389 | 0.12/0.08 | 0.29 (0.33) | +0.01 | – | – |
| 717 | S | (0/874) | (0/832) | (8828/12057) | 0.5 | 0.54/0.32 | 0.71 (0.34)* | $+0.34^{+}$ | 0.89 (0.48)* | $+0.35^{+}$ |
| 717 | B | (0/1736) | (0/2251) | (1905/10363) | 0.4 | 0.1/0.08 | 0.34 (0.33) | +0.01 | 0.50 (0.48) | -0.06 |
| 718 | S | (0/997) | (0/709) | (9215/12674) | 0.495 | 0.5/0.32 | 0.92 (0.54)* | $+0.43^{+}$ | 0.97 (0.57)* | $+0.50^{+}$ |
| 718 | B | (0/626) | (0/914) | (0/830) | 0.389 | 0.1/0.08 | 0.48 (0.45) | -0.01 | 0.42 (0.45) | +0.01 |

Table 76: Machine-learning capacity arm (8 learners with 2 rows, 8 with 3) at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 719 | S | (0/579) | (0/357) | (9140/13341) | 0.489 | 0.56/0.58 | 0.74 (0.54)* | +0.08+ | 0.74 (0.54)* | +0.08+ |
| 719 | B | (0/1252) | (0/1570) | (273/5307) | 0.383 | 0.22/0.09 | 0.24 (0.26) | +0.03 | 0.34 (0.42) | +0.00 |

Table 77: Machine-learning capacity arm, every learner with 2 rows at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (0/2017) | (0/2055) | (0/419) | 0.389 | 0.2/0.09 | 0.71 (0.52)* | -0.01 | 0.85 (0.56)* | +0.13+ |
| 306 | B | (0/2215) | (0/3048) | (1931/10054) | 0.398 | 0.12/0.08 | 0.25 (0.27) | -0.05 | 0.26 (0.44) | +0.04 |
| 307 | S | (0/2504) | (0/1856) | (0/662) | 0.388 | 0.25/0.1 | 0.74 (0.62)* | -0.07 | – | – |
| 307 | B | (0/1751) | (0/1476) | (0/701) | 0.389 | 0.13/0.08 | 0.54 (0.45)* | -0.05 | 0.41 (0.43) | +0.00 |
| 308 | S | (0/1468) | (0/1253) | (2813/8534) | 0.428 | 0.27/0.22 | 0.88 (0.59)* | +0.07+ | 0.90 (0.58)* | +0.10+ |
| 308 | B | (0/2411) | (0/2083) | (0/985) | 0.386 | 0.16/0.08 | 0.46 (0.41) | -0.02 | 0.44 (0.39) | -0.02 |
| 309 | S | (1468/7581) | (1767/7966) | (2464/6434) | 0.657 | 0.65/0.54 | 0.56 (0.32)* | -0.04 | 0.81 (0.43)* | -0.02 |
| 309 | B | (0/2117) | (0/1699) | (274/5154) | 0.384 | 0.19/0.09 | 0.32 (0.34) | +0.04+ | 0.33 (0.33) | +0.02 |
| 310 | S | (3728/12214) | (2596/9621) | (2135/4921) | 1.02 | 0.67/0.48 | 0.72 (0.24)* | +0.33+ | 0.81 (0.28)* | +0.62+ |
| 310 | B | (0/1907) | (0/1959) | (1305/10122) | 0.389 | 0.19/0.09 | 0.53 (0.43)* | +0.04 | 0.56 (0.50)* | +0.00 |
| 311 | S | (0/1048) | (0/751) | (6821/11606) | 0.473 | 0.41/0.32 | 0.69 (0.45)* | +0.16+ | 0.97 (0.61)* | +0.39+ |
| 311 | B | (0/2966) | (0/2142) | (1289/10040) | 0.39 | 0.17/0.08 | 0.57 (0.35)* | -0.01 | 0.60 (0.44)* | -0.11 |
| 312 | S | (0/871) | (0/361) | (10126/13000) | 0.504 | 0.52/0.3 | 0.53 (0.26)* | +0.20+ | 0.82 (0.44)* | +0.42+ |
| 312 | B | (0/2155) | (0/2411) | (1348/10149) | 0.389 | 0.06/0.07 | 0.37 (0.38) | -0.02 | 0.46 (0.47) | -0.07 |
| 313 | S | (1614/6542) | (785/3957) | (5611/8204) | 0.684 | 0.63/0.38 | 0.49 (0.27)* | +0.30+ | 0.49 (0.28)* | +0.32+ |
| 313 | B | (0/1763) | (0/1986) | (1794/10539) | 0.397 | 0.13/0.08 | 0.49 (0.45) | -0.01 | 0.49 (0.44) | +0.01 |
| 314 | S | (3756/9061) | (4170/10344) | (3136/5636) | 1.204 | 0.77/0.77 | 0.74 (0.24)* | +0.28+ | 0.87 (0.32)* | +0.53+ |
| 314 | B | (0/1520) | (0/1423) | (2287/10237) | 0.407 | 0.07/0.07 | 0.13 (0.37) | +0.17+ | 0.21 (0.44) | +0.10+ |
| 315 | S | (0/614) | (0/555) | (8596/12211) | 0.499 | 0.49/0.3 | 0.77 (0.44)* | +0.37+ | 0.99 (0.60)* | +0.46+ |
| 315 | B | (0/1763) | (0/1873) | (1464/9656) | 0.392 | 0.13/0.08 | 0.27 (0.32) | -0.03 | 0.27 (0.36) | +0.03 |
| 316 | S | (0/709) | (0/610) | (6843/10911) | 0.472 | 0.46/0.47 | 0.80 (0.60)* | -0.04 | 0.80 (0.60)* | -0.04 |
| 316 | B | (0/1431) | (0/1358) | (0/1130) | 0.389 | 0.05/0.07 | 0.47 (0.47) | -0.02 | 0.49 (0.51) | +0.00 |
| 317 | S | (0/889) | (0/680) | (7602/11637) | 0.489 | 0.43/0.19 | 0.83 (0.36)* | +0.40+ | 0.91 (0.37)* | +0.51+ |
| 317 | B | (0/2293) | (0/1568) | (1061/10278) | 0.387 | 0.31/0.1 | 0.52 (0.42)* | -0.02 | 0.52 (0.43)* | -0.01 |

Table 78: Machine-learning main set, seeds 712–715 at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 712 | S | (0/748) | (0/930) | (7166/10721) | 0.48 | 0.48/0.5 | 0.74 (0.55)* | -0.06 | 0.82 (0.59)* | -0.06 |
| 712 | B | (0/2854) | (0/2138) | (790/9726) | 0.382 | 0.06/0.07 | 0.35 (0.35) | +0.00 | 0.40 (0.48) | +0.02+ |
| 713 | S | (0/1183) | (1/836) | (7311/11520) | 0.479 | 0.4/0.21 | 0.66 (0.36)* | +0.29+ | 1.00 (0.58)* | +0.60+ |
| 713 | B | (0/2144) | (0/1711) | (406/7155) | 0.383 | 0.15/0.08 | 0.32 (0.22)* | -0.04 | 0.26 (0.20)* | -0.06 |
| 714 | S | (6908/14127) | (4935/10862) | (1773/4153) | 1.331 | 0.83/0.87 | 0.82 (0.36)* | +0.43+ | 0.83 (0.32)* | +0.46+ |
| 714 | B | (189/4607) | (175/4619) | (133/3806) | 0.401 | 0.12/0.1 | 0.45 (0.38)* | -0.02 | 0.34 (0.31) | -0.03 |
| 715 | S | (0/473) | (0/583) | (8570/11576) | 0.488 | 0.58/0.56 | 0.61 (0.41)* | +0.14+ | 0.73 (0.49)* | +0.18+ |
| 715 | B | (0/1490) | (0/1610) | (0/857) | 0.39 | 0.04/0.07 | 0.12 (0.36) | +0.11+ | – | – |

Table 79: Blank starts of the machine-learning world beside the reconstructions (seeds 306–325), with their twins at 10,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (0/1329) | (0/478) | (7969/11714) | 0.491 | 0.4/0.21 | 0.97 (0.54)* | +0.57+ | 0.95 (0.52)* | +0.56+ |
| 306 | B | (650/8000) | (558/7067) | (0/3021) | 0.471 | 0.3/0.18 | 0.27 (0.26) | -0.01 | 0.24 (0.34) | +0.01 |
| 307 | S | (1862/5745) | (1374/5242) | (5071/7805) | 0.757 | 0.6/0.46 | 0.74 (0.45)* | +0.23+ | 0.73 (0.40)* | +0.34+ |
| 307 | B | (0/2149) | (0/1828) | (992/9311) | 0.388 | 0.12/0.08 | 0.27 (0.44) | +0.01 | 0.34 (0.51) | +0.03 |
| 308 | S | (0/961) | (0/541) | (6476/11841) | 0.466 | 0.5/0.48 | 0.83 (0.64)* | +0.11+ | 0.96 (0.68)* | +0.31+ |
| 308 | B | (0/2877) | (0/2574) | (0/744) | 0.387 | 0.04/0.08 | 0.45 (0.44) | -0.03 | 0.34 (0.43) | +0.04+ |
| 309 | S | (0/548) | (0/714) | (7980/12238) | 0.469 | 0.51/0.51 | 0.85 (0.58)* | -0.05 | 0.85 (0.59)* | -0.05 |
| 309 | B | (0/2090) | (0/2309) | (405/7448) | 0.382 | 0.08/0.08 | 0.26 (0.44) | +0.04+ | 0.30 (0.50) | +0.05+ |
| 310 | S | (0/530) | (0/430) | (9002/12520) | 0.504 | 0.48/0.31 | 0.91 (0.55)* | +0.38+ | 0.92 (0.54)* | +0.42+ |
| 310 | B | (0/1816) | (0/1769) | (1680/10140) | 0.394 | 0.15/0.08 | 0.71 (0.38)* | +0.27+ | 0.76 (0.42)* | +0.31+ |
| 311 | S | (0/441) | (0/350) | (8501/12748) | 0.49 | 0.53/0.57 | 0.88 (0.69)* | -0.02 | 0.88 (0.69)* | -0.02 |
| 311 | B | (0/1864) | (0/1832) | (1140/10581) | 0.389 | 0.09/0.08 | 0.34 (0.49) | +0.01 | 0.43 (0.50) | -0.01 |
| 312 | S | (0/838) | (0/768) | (7096/11474) | 0.486 | 0.42/0.27 | 0.49 (0.29)* | +0.30+ | 0.90 (0.50)* | +0.41+ |
| 312 | B | (0/1799) | (0/1709) | (1680/10310) | 0.396 | 0.13/0.08 | 0.29 (0.35) | -0.02 | 0.58 (0.50)* | -0.10 |
| 313 | S | (4405/10230) | (4808/9947) | (446/2780) | 1.167 | 0.72/0.46 | 0.73 (0.37)* | +0.09+ | 0.77 (0.35)* | +0.23+ |
| 313 | B | (1/1951) | (3/2167) | (0/1088) | 0.386 | 0.1/0.07 | 0.32 (0.32) | -0.02 | 0.31 (0.37) | +0.00 |
| 314 | S | (0/906) | (0/1091) | (7274/11966) | 0.484 | 0.37/0.19 | 0.73 (0.41)* | +0.24+ | 0.99 (0.58)* | +0.53+ |
| 314 | B | (0/2542) | (0/2337) | (902/9638) | 0.385 | 0.07/0.07 | 0.16 (0.25) | -0.05 | 0.27 (0.40) | +0.03+ |
| 315 | S | (0/839) | (0/832) | (4731/10317) | 0.45 | 0.42/0.39 | 0.40 (0.38) | -0.06 | 0.82 (0.59)* | -0.05 |
| 315 | B | (0/1429) | (0/2137) | (0/808) | 0.388 | 0.08/0.07 | 0.49 (0.50) | +0.01 | 0.50 (0.51) | +0.01 |
| 316 | S | (4364/11199) | (4135/10774) | (779/3095) | 1.138 | 0.75/0.79 | 0.86 (0.54)* | +0.20+ | 0.83 (0.46)* | +0.35+ |
| 316 | B | (0/958) | (0/823) | (34/2133) | 0.39 | 0.06/0.07 | 0.47 (0.41) | -0.02 | 0.47 (0.41) | -0.02 |
| 317 | S | (3900/12126) | (2774/9376) | (2997/5541) | 1.005 | 0.63/0.29 | 0.53 (0.18)* | +0.48+ | 0.69 (0.23)* | +0.79+ |
| 317 | B | (219/6484) | (279/7517) | (1419/6169) | 0.418 | 0.17/0.16 | 0.29 (0.17)* | +0.09+ | 0.27 (0.29) | +0.02 |
| 318 | S | (0/1035) | (0/995) | (6783/11083) | 0.475 | 0.44/0.29 | 0.93 (0.60)* | +0.20+ | 0.94 (0.55)* | +0.34+ |
| 318 | B | (0/2742) | (0/2541) | (2/952) | 0.386 | 0.09/0.08 | 0.41 (0.40) | -0.03 | 0.29 (0.41) | +0.02 |
| 319 | S | (4453/10182) | (5889/12848) | (2221/4551) | 1.302 | 0.82/0.83 | 0.86 (0.40)* | +0.43+ | 0.90 (0.38)* | +0.43+ |
| 319 | B | (0/2537) | (0/2103) | (859/9369) | 0.388 | 0.07/0.07 | 0.45 (0.32)* | +0.04 | 0.57 (0.50)* | -0.07 |
| 320 | S | (0/784) | (0/877) | (6299/11196) | 0.472 | 0.45/0.36 | 0.80 (0.54)* | +0.08+ | 0.89 (0.51)* | +0.49+ |
| 320 | B | (0/2184) | (0/2436) | (448/7480) | 0.381 | 0.13/0.08 | 0.23 (0.25) | -0.02 | 0.39 (0.42) | +0.01 |
| 321 | S | (0/780) | (0/337) | (8926/12112) | 0.492 | 0.45/0.32 | 0.91 (0.67)* | -0.04 | 0.91 (0.67)* | -0.04 |
| 321 | B | (0/1480) | (0/1278) | (193/4720) | 0.385 | 0.11/0.08 | 0.15 (0.22) | +0.04 | 0.26 (0.28) | +0.00 |
| 322 | S | (5083/10462) | (5085/10515) | (0/1174) | 1.141 | 0.78/0.81 | 0.86 (0.58)* | +0.06+ | 0.87 (0.57)* | +0.09+ |
| 322 | B | (0/1209) | (0/1621) | (0/587) | 0.39 | 0.03/0.07 | 0.44 (0.36)* | -0.02 | 0.39 (0.37) | +0.00 |
| 323 | S | (5463/11739) | (5188/11269) | (2378/4550) | 1.305 | 0.8/0.84 | 0.73 (0.24)* | +0.46+ | 0.82 (0.30)* | +0.53+ |
| 323 | B | (0/2117) | (0/2250) | (2397/10831) | 0.402 | 0.07/0.08 | 0.23 (0.35) | +0.04 | 0.39 (0.45) | -0.07 |
| 324 | S | (0/647) | (0/695) | (6443/12583) | 0.451 | 0.53/0.5 | 0.70 (0.52)* | -0.06 | 0.86 (0.58)* | -0.05 |
| 324 | B | (253/6589) | (172/4890) | (177/4163) | 0.406 | 0.19/0.13 | 0.20 (0.24) | -0.09 | 0.22 (0.24) | -0.10 |
| 325 | S | (0/738) | (0/1068) | (6368/12532) | 0.457 | 0.45/0.46 | 0.63 (0.62) | +0.02 | 0.84 (0.64)* | -0.01 |
| 325 | B | (0/2819) | (0/1826) | (398/6981) | 0.382 | 0.16/0.08 | 0.40 (0.40) | +0.00 | 0.42 (0.43) | -0.00 |

Table 80: Machine-learning reconstruction, outward from the team world 517 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (3406/10537) | (3776/12458) | (2240/4263) | 1.091 | 0.73/0.65 | 0.30 (0.11)* | +0.16+ | 0.57 (0.21)* | +0.29+ |
| 307 | S | (5903/14285) | (4574/11673) | (2767/4832) | 1.288 | 0.83/0.76 | 0.51 (0.16)* | +0.13+ | 0.55 (0.21)* | +0.20+ |
| 308 | S | (1098/4104) | (788/2837) | (9150/9799) | 0.687 | 0.54/0.27 | 0.51 (0.18)* | +0.36+ | 0.80 (0.31)* | +0.60+ |
| 309 | S | (0/216) | (0/139) | (12020/12916) | 0.543 | 0.53/0.24 | 0.55 (0.25)* | +0.29+ | 0.97 (0.53)* | +0.57+ |
| 310 | S | (8074/15452) | (6757/12652) | (3473/4602) | 1.659 | 0.88/0.89 | 0.48 (0.15)* | +0.15+ | 0.59 (0.29)* | +0.06+ |
| 311 | S | (1/213) | (0/178) | (12065/13108) | 0.541 | 0.57/0.3 | 0.58 (0.14)* | +0.13+ | 0.88 (0.37)* | +0.58+ |
| 312 | S | (0/215) | (1/181) | (12067/13067) | 0.542 | 0.53/0.23 | 0.56 (0.20)* | +0.10+ | 0.72 (0.37)* | +0.45+ |
| 315 | S | (6787/11912) | (6475/11429) | (3965/5181) | 1.644 | 0.83/0.84 | 0.50 (0.14)* | +0.46+ | 0.69 (0.24)* | +0.46+ |

Table 80: Machine-learning reconstruction, outward from the team world 517 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 316 | S | (10766/15625) | (9365/13476) | (2883/4147) | 2.031 | 0.88/0.91 | 0.40 (0.16)* | +0.22+ | 0.64 (0.23)* | +0.30+ |
| 317 | S | (6401/10573) | (7238/11737) | (4481/5455) | 1.671 | 0.84/0.83 | 0.48 (0.19)* | +0.03 | 0.82 (0.34)* | +0.24+ |
| 318 | S | (95/654) | (186/908) | (10851/12272) | 0.571 | 0.49/0.28 | 0.34 (0.22)* | +0.74+ | 0.85 (0.41)* | +0.49+ |
| 319 | S | (9262/11878) | (8209/10449) | (5718/6568) | 1.886 | 0.86/0.85 | 0.57 (0.21)* | +0.11+ | 0.86 (0.36)* | +0.34+ |
| 320 | S | (4240/8683) | (3970/8205) | (4867/6598) | 1.085 | 0.73/0.68 | 0.63 (0.19)* | +0.65+ | 0.81 (0.24)* | +0.66+ |
| 321 | S | (8119/15100) | (5652/10989) | (2956/4600) | 1.575 | 0.85/0.86 | 0.39 (0.13)* | +0.01 | 0.55 (0.26)* | +0.07+ |
| 322 | S | (7083/13501) | (7592/14061) | (2547/4259) | 1.667 | 0.83/0.64 | 0.19 (0.13)* | +0.08+ | 0.56 (0.22)* | +0.25+ |
| 323 | S | (301/3043) | (280/3178) | (7933/9812) | 0.57 | 0.42/0.17 | 0.62 (0.14)* | +0.52+ | 0.94 (0.34)* | +0.64+ |
| 324 | S | (7856/14070) | (5369/9508) | (3576/4905) | 1.623 | 0.83/0.84 | 0.36 (0.16)* | +0.12+ | 0.64 (0.28)* | +0.28+ |
| 325 | S | (9874/14168) | (9918/13955) | (3269/4035) | 2.159 | 0.88/0.91 | 0.44 (0.16)* | +0.46+ | 0.81 (0.32)* | +0.50+ |

Table 81: Machine-learning reconstruction, half level from the team world 517 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 313 | S | (2909/9260) | (3157/9949) | (3720/5810) | 1.111 | 0.73/0.68 | 0.75 (0.33)* | +0.24+ | 0.86 (0.41)* | +0.20+ |
| 314 | S | (7203/13252) | (7940/14179) | (2732/4116) | 1.683 | 0.86/0.89 | 0.50 (0.20)* | +0.03 | 0.62 (0.28)* | +0.21+ |

Table 82: Machine-learning reconstruction, presence level from world 712 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 310 | S | (0/197) | (0/215) | (12265/13008) | 0.546 | 0.62/0.63 | 0.93 (0.65)* | -0.03 | 0.93 (0.65)* | -0.03 |

Table 83: Machine-learning reconstruction, outward from world 714 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (10014/14901) | (9804/14207) | (3239/3974) | 2.105 | 0.89/0.91 | 0.79 (0.31)* | +0.54+ | 0.79 (0.31)* | +0.55+ |
| 307 | S | (12853/15217) | (12128/14119) | (2715/3433) | 2.517 | 0.89/0.93 | 0.68 (0.28)* | +0.27+ | 0.75 (0.32)* | +0.32+ |
| 308 | S | (13003/15642) | (12332/14367) | (2740/3854) | 2.398 | 0.9/0.93 | 0.81 (0.37)* | +0.24+ | 0.81 (0.37)* | +0.23+ |
| 309 | S | (12813/16548) | (9591/12501) | (3067/4244) | 2.013 | 0.89/0.92 | 0.44 (0.34)* | -0.03 | 0.43 (0.35)* | -0.03 |
| 315 | S | (12692/15324) | (11153/13516) | (2993/3816) | 2.532 | 0.9/0.93 | 0.89 (0.35)* | +0.54+ | 0.89 (0.34)* | +0.53+ |

Table 84: Machine-learning reconstruction, presence level from world 714 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (12306/13876) | (14000/15588) | (3069/3488) | 2.587 | 0.9/0.94 | 0.93 (0.39)* | $+0.54^{+}$ | 0.93 (0.39)* | $+0.53^{+}$ |

Table 85: Machine-learning main set, seeds 512–519 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 512 | S | (0/836) | (0/613) | (9296/12497) | 0.509 | 0.37/0.18 | 0.55 (0.24)* | $+0.35^{+}$ | 1.00 (0.51)* | $+0.73^{+}$ |
| 512 | B | (0/2521) | (0/1361) | (514/7729) | 0.385 | 0.08/0.08 | 0.23 (0.20) | -0.03 | 0.34 (0.36) | +0.01 |
| 513 | S | (0/474) | (0/248) | (9658/12998) | 0.508 | 0.44/0.23 | 0.79 (0.42)* | $+0.52^{+}$ | 0.97 (0.56)* | $+0.57^{+}$ |
| 513 | B | (0/340) | (0/401) | (0/356) | 0.394 | 0.05/0.07 | – | – | – | – |
| 514 | S | (0/378) | (0/491) | (10116/12916) | 0.514 | 0.34/0.17 | 0.88 (0.35)* | $+0.58^{+}$ | 0.98 (0.48)* | $+0.71^{+}$ |
| 514 | B | (0/989) | (0/1123) | (3706/10143) | 0.419 | 0.13/0.08 | 0.12 (0.28) | +0.06 | 0.40 (0.45) | -0.06 |
| 515 | S | (6050/13668) | (5793/12682) | (1804/3879) | 1.464 | 0.73/0.4 | 0.61 (0.20)* | $+0.37^{+}$ | 0.75 (0.25)* | $+0.71^{+}$ |
| 515 | B | (0/818) | (0/525) | (190/3900) | 0.39 | 0.1/0.08 | 0.40 (0.35) | -0.11 | 0.64 (0.52)* | -0.03 |
| 516 | S | (6978/14195) | (7166/13705) | (2509/4006) | 1.551 | 0.87/0.89 | 0.80 (0.40)* | $+0.34^{+}$ | 0.84 (0.35)* | $+0.39^{+}$ |
| 516 | B | (0/2436) | (0/1725) | (1373/9768) | 0.394 | 0.08/0.08 | 0.23 (0.35) | $+0.04^{+}$ | 0.18 (0.46) | $+0.18^{+}$ |
| 517 | S | (4698/12080) | (3155/8634) | (2966/5750) | 1.122 | 0.77/0.78 | 0.61 (0.28)* | $+0.30^{+}$ | 0.87 (0.36)* | $+0.47^{+}$ |
| 517 | B | (0/1642) | (0/1194) | (378/5913) | 0.388 | 0.1/0.08 | 0.51 (0.45)* | -0.01 | 0.52 (0.45)* | -0.01 |
| 518 | S | (7873/14159) | (7168/13246) | (2695/4254) | 1.551 | 0.87/0.9 | 0.67 (0.31)* | $+0.19^{+}$ | 0.67 (0.30)* | $+0.22^{+}$ |
| 518 | B | (0/3064) | (0/2315) | (1096/10043) | 0.39 | 0.11/0.08 | 0.48 (0.52) | -0.08 | 0.46 (0.53) | -0.09 |
| 519 | S | (0/488) | (0/318) | (10449/12861) | 0.513 | 0.52/0.31 | 0.59 (0.31)* | $+0.29^{+}$ | 0.91 (0.55)* | $+0.43^{+}$ |
| 519 | B | (209/6763) | (193/5825) | (206/3938) | 0.405 | 0.16/0.13 | 0.24 (0.29) | -0.03 | 0.22 (0.29) | -0.00 |

Table 86: Machine-learning world, whole at 120 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 512 | S | (0/359) | (0/258) | (10243/12992) | 0.696 | 0.52/0.34 | 0.77 (0.43)* | $+0.39^{+}$ | 0.88 (0.49)* | $+0.43^{+}$ |
| 512 | B | (0/1200) | (0/772) | (1186/11399) | 0.461 | 0.06/0.07 | 0.21 (0.28) | +0.03 | 0.57 (0.43)* | -0.02 |
| 513 | S | (0/349) | (0/373) | (9476/13041) | 0.687 | 0.43/0.21 | 0.81 (0.38)* | $+0.48^{+}$ | 0.96 (0.51)* | $+0.60^{+}$ |
| 513 | B | (0/794) | (0/334) | (0/337) | 0.46 | 0.04/0.07 | 0.38 (0.40) | +0.00 | – | – |
| 514 | S | (0/588) | (0/378) | (9145/12382) | 0.684 | 0.45/0.22 | 0.89 (0.39)* | $+0.49^{+}$ | 0.98 (0.52)* | $+0.61^{+}$ |
| 514 | B | (0/1981) | (0/1967) | (1749/10855) | 0.463 | 0.06/0.08 | 0.54 (0.48) | $+0.07^{+}$ | 0.53 (0.50) | $+0.06^{+}$ |
| 515 | S | (0/490) | (0/406) | (6817/12126) | 0.615 | 0.46/0.48 | 0.52 (0.43)* | -0.03 | 0.69 (0.60)* | -0.04 |
| 515 | B | (0/1352) | (0/1910) | (1415/10534) | 0.463 | 0.06/0.07 | 0.38 (0.29)* | -0.05 | 0.59 (0.44)* | -0.01 |

Table 87: Machine-learning world, births and deaths at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (0/566) | (0/591) | (8162/11969) | 0.494 | 0.55/0.2 | 0.95 (0.49)* | $+0.50^{+}$ | 0.91 (0.41)* | $+0.52^{+}$ |
| 306 | B | (178/4461) | (157/4159) | (48/2775) | 0.405 | 0.24/0.09 | 0.32 (0.33) | -0.03 | 0.47 (0.32)* | -0.09 |

Table 87: Machine-learning world, births and deaths at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 307 | S | (0/859) | (0/686) | (4466/10141) | 0.444 | 0.5/0.32 | 0.75 (0.40)* | +0.22$^{+}$ | 0.77 (0.44)* | +0.39$^{+}$ |
| 307 | B | (0/2290) | (0/1036) | (962/10367) | 0.384 | 0.23/0.08 | 0.38 (0.42) | -0.00 | 0.32 (0.42) | +0.04 |
| 308 | S | (0/902) | (0/727) | (7315/11887) | 0.478 | 0.55/0.33 | 0.88 (0.43)* | +0.39$^{+}$ | 0.93 (0.43)* | +0.50$^{+}$ |
| 308 | B | (3/455) | (15/1329) | (0/803) | 0.391 | 0.29/0.09 | 0.50 (0.36)* | -0.02 | 0.42 (0.36) | -0.02 |
| 309 | S | (925/2310) | (1068/2853) | (445/2474) | 0.597 | 0.43/0.31 | 0.47 (0.40)* | -0.08 | 0.74 (0.42)* | -0.04 |
| 309 | B | (0/1368) | (0/1331) | (34/1213) | 0.383 | 0.3/0.09 | 0.52 (0.38)* | -0.01 | 0.31 (0.47) | +0.09$^{+}$ |
| 412 | S | (0/1293) | (0/1323) | (4946/10539) | 0.45 | 0.5/0.41 | 0.66 (0.42)* | -0.05 | 0.81 (0.49)* | +0.11$^{+}$ |
| 412 | B | (0/2145) | (0/1548) | (2036/11520) | 0.389 | 0.37/0.09 | 0.29 (0.25) | -0.01 | 0.52 (0.39)* | +0.01 |
| 413 | S | (0/1131) | (0/676) | (7917/12280) | 0.485 | 0.45/0.2 | 0.98 (0.57)* | +0.49$^{+}$ | 0.99 (0.56)* | +0.61$^{+}$ |
| 413 | B | (0/6686) | (0/3906) | (0/1435) | 0.379 | 0.2/0.08 | 0.36 (0.41) | +0.01 | 0.37 (0.42) | +0.01 |
| 414 | S | (0/1238) | (0/1771) | (3793/9290) | 0.431 | 0.43/0.31 | 0.73 (0.48)* | +0.06$^{+}$ | 0.75 (0.46)* | +0.11$^{+}$ |
| 414 | B | (0/557) | (0/469) | (3/1006) | 0.388 | 0.18/0.08 | 0.48 (0.45) | -0.02 | 0.35 (0.48) | +0.01 |
| 415 | S | (0/1148) | (0/1544) | (7317/11505) | 0.477 | 0.62/0.22 | 0.83 (0.33)* | +0.39$^{+}$ | 0.89 (0.34)* | +0.46$^{+}$ |
| 415 | B | (0/1762) | (0/1881) | (693/8725) | 0.379 | 0.41/0.09 | 0.46 (0.38)* | +0.03 | 0.53 (0.38)* | +0.07$^{+}$ |

Table 88: Machine-learning capacity arm (8 learners with 2 rows, 8 with 3) at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 320 | S | (0/327) | (0/254) | (11063/12944) | 0.529 | 0.52/0.33 | 0.94 (0.58)* | +0.45$^{+}$ | 1.00 (0.63)* | +0.48$^{+}$ |
| 320 | B | (0/914) | (0/255) | (151/3736) | 0.39 | 0.11/0.08 | 0.34 (0.39) | -0.01 | 0.44 (0.51) | +0.01$^{+}$ |
| 321 | S | (6658/13767) | (6267/12878) | (2182/3628) | 1.509 | 0.85/0.87 | 0.58 (0.28)* | +0.29$^{+}$ | 0.84 (0.42)* | +0.38$^{+}$ |
| 321 | B | (0/1592) | (0/1436) | (471/7351) | 0.384 | 0.13/0.08 | 0.36 (0.35) | -0.01 | 0.37 (0.36) | -0.01 |
| 322 | S | (10977/15398) | (9705/13422) | (2984/4146) | 2.087 | 0.89/0.92 | 0.49 (0.23)* | +0.04 | 0.61 (0.32)* | +0.18$^{+}$ |
| 322 | B | (0/1617) | (0/1195) | (1694/10209) | 0.401 | 0.07/0.07 | 0.48 (0.41)* | -0.08 | 0.47 (0.41) | -0.08 |
| 323 | S | (8278/11207) | (7603/10344) | (0/457) | 1.869 | 0.82/0.85 | 0.77 (0.47)* | +0.15$^{+}$ | 0.80 (0.50)* | +0.16$^{+}$ |
| 323 | B | (0/389) | (0/561) | (0/549) | 0.393 | 0.07/0.07 | 0.35 (0.29)* | +0.01 | 0.17 (0.33) | +0.06 |
| 716 | S | (5702/13556) | (4896/11310) | (2890/4707) | 1.336 | 0.83/0.84 | 0.82 (0.42)* | +0.20$^{+}$ | 0.82 (0.41)* | +0.30$^{+}$ |
| 716 | B | (0/476) | (0/436) | (23/656) | 0.394 | 0.1/0.07 | 0.49 (0.39)* | -0.03 | 0.48 (0.40)* | -0.03 |
| 717 | S | (0/334) | (0/282) | (10892/13217) | 0.52 | 0.54/0.32 | 0.82 (0.44)* | +0.50$^{+}$ | 0.97 (0.60)* | +0.47$^{+}$ |
| 717 | B | (0/1894) | (0/1206) | (2650/11463) | 0.41 | 0.08/0.08 | 0.09 (0.24) | +0.08$^{+}$ | 0.54 (0.47)* | -0.04 |
| 718 | S | (0/400) | (0/320) | (11132/12939) | 0.527 | 0.52/0.32 | 0.99 (0.58)* | +0.49$^{+}$ | 0.99 (0.58)* | +0.49$^{+}$ |
| 718 | B | (0/409) | (0/241) | (0/307) | 0.393 | 0.07/0.08 | 0.61 (0.46)* | -0.03 | 0.39 (0.49) | +0.05 |
| 719 | S | (0/321) | (0/255) | (10186/13038) | 0.512 | 0.56/0.59 | 0.68 (0.50)* | -0.01 | 0.83 (0.66)* | -0.01 |
| 719 | B | (0/1761) | (0/1435) | (983/9664) | 0.392 | 0.14/0.08 | 0.46 (0.27)* | +0.02 | 0.59 (0.39)* | +0.05$^{+}$ |

Table 89: Machine-learning capacity arm, every learner with 2 rows at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (0/607) | (0/944) | (0/569) | 0.394 | 0.26/0.12 | 0.54 (0.39)* | +0.09$^{+}$ | 0.59 (0.54) | -0.00 |
| 306 | B | (0/2123) | (0/1579) | (1973/10371) | 0.404 | 0.08/0.07 | 0.33 (0.28) | +0.19$^{+}$ | 0.57 (0.47)* | -0.01 |
| 307 | S | (0/1023) | (0/577) | (0/296) | 0.392 | 0.14/0.09 | 0.42 (0.41) | +0.02 | 0.67 (0.69) | +0.00 |
| 307 | B | (0/200) | (0/251) | (0/210) | 0.395 | 0.08/0.07 | 0.48 (0.48) | -0.00 | 0.54 (0.45) | -0.02 |
| 308 | S | (0/482) | (0/434) | (6259/11976) | 0.469 | 0.42/0.44 | 0.90 (0.65)* | -0.03 | 0.90 (0.66)* | -0.03 |
| 308 | B | (0/736) | (0/812) | (0/307) | 0.394 | 0.06/0.07 | 0.48 (0.45) | -0.01 | 0.48 (0.45) | -0.01 |
| 309 | S | (4359/11778) | (4636/12500) | (1348/3815) | 1.161 | 0.77/0.48 | 0.51 (0.26)* | +0.25$^{+}$ | 0.61 (0.31)* | +0.35$^{+}$ |
| 309 | B | (0/821) | (0/1402) | (0/806) | 0.39 | 0.06/0.08 | – | – | – | – |

Table 89: Machine-learning capacity arm, every learner with 2 rows at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 310 | S | (6442/13876) | (5422/12010) | (2256/4768) | 1.366 | 0.82/0.68 | 0.61 (0.23)* | +0.33+ | 0.71 (0.26)* | +0.58+ |
| 310 | B | (0/1951) | (0/1585) | (1200/10938) | 0.387 | 0.14/0.08 | 0.26 (0.29) | +0.08+ | 0.46 (0.41) | -0.02 |
| 311 | S | (0/360) | (0/349) | (9752/12679) | 0.516 | 0.52/0.31 | 0.99 (0.62)* | +0.43+ | 1.00 (0.62)* | +0.48+ |
| 311 | B | (0/1739) | (0/1543) | (1286/11107) | 0.396 | 0.07/0.07 | 0.48 (0.45) | -0.04 | 0.48 (0.44) | -0.05 |
| 312 | S | (0/310) | (0/213) | (10305/12906) | 0.518 | 0.43/0.21 | 0.64 (0.34)* | +0.47+ | 0.89 (0.48)* | +0.55+ |
| 312 | B | (0/2392) | (0/1629) | (1402/10454) | 0.395 | 0.07/0.07 | 0.47 (0.37)* | -0.08 | 0.57 (0.49)* | -0.07 |
| 313 | S | (5486/11353) | (6079/12930) | (3181/4779) | 1.277 | 0.85/0.83 | 0.44 (0.27)* | +0.31+ | 0.57 (0.32)* | +0.34+ |
| 313 | B | (0/1820) | (0/1686) | (2473/10623) | 0.408 | 0.09/0.08 | 0.53 (0.45)* | -0.03 | 0.53 (0.46)* | -0.01 |
| 314 | S | (7321/13675) | (7888/13933) | (2664/4533) | 1.598 | 0.87/0.91 | 0.68 (0.20)* | +0.41+ | 0.82 (0.25)* | +0.56+ |
| 314 | B | (0/1031) | (0/1126) | (3173/12083) | 0.417 | 0.08/0.08 | 0.19 (0.33) | -0.06 | 0.25 (0.50) | +0.13+ |
| 315 | S | (0/330) | (0/249) | (10121/12689) | 0.52 | 0.53/0.32 | 0.79 (0.44)* | +0.41+ | 0.80 (0.44)* | +0.40+ |
| 315 | B | (0/1620) | (0/1799) | (2380/10696) | 0.409 | 0.07/0.08 | 0.36 (0.48) | -0.03 | 0.36 (0.49) | -0.03 |
| 316 | S | (0/373) | (0/357) | (9691/12559) | 0.513 | 0.55/0.58 | 0.77 (0.60)* | -0.02 | 0.76 (0.59)* | -0.02 |
| 316 | B | (0/332) | (0/178) | (0/402) | 0.395 | 0.05/0.07 | 0.32 (0.43) | -0.01 | 0.51 (0.44) | -0.02 |
| 317 | S | (0/402) | (0/320) | (10489/12922) | 0.524 | 0.46/0.22 | 0.83 (0.43)* | +0.52+ | 0.91 (0.49)* | +0.56+ |
| 317 | B | (35/3662) | (20/2801) | (1425/9346) | 0.399 | 0.17/0.08 | 0.41 (0.31)* | -0.09 | 0.40 (0.31)* | -0.10 |

Table 90: Machine-learning main set, seeds 712–715 at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 712 | S | (0/403) | (0/273) | (10013/12679) | 0.501 | 0.57/0.6 | 0.76 (0.61)* | -0.01 | 0.76 (0.61)* | -0.01 |
| 712 | B | (0/784) | (0/905) | (208/3919) | 0.388 | 0.11/0.08 | 0.38 (0.34) | -0.01 | 0.38 (0.34) | +0.00 |
| 713 | S | (0/489) | (0/390) | (8631/12394) | 0.501 | 0.41/0.22 | 0.79 (0.42)* | +0.40+ | 0.99 (0.59)* | +0.51+ |
| 713 | B | (0/1884) | (0/1967) | (1208/10593) | 0.392 | 0.07/0.07 | 0.38 (0.34) | -0.06 | 0.45 (0.47) | -0.08 |
| 714 | S | (7178/14083) | (7037/13498) | (2397/4351) | 1.549 | 0.87/0.9 | 0.92 (0.37)* | +0.55+ | 0.91 (0.33)* | +0.56+ |
| 714 | B | (115/3537) | (112/2784) | (0/1330) | 0.404 | 0.07/0.07 | 0.39 (0.39) | -0.01 | 0.44 (0.42) | -0.01 |
| 715 | S | (0/238) | (0/301) | (10945/12701) | 0.517 | 0.59/0.62 | 0.54 (0.47)* | -0.03 | 0.84 (0.63)* | -0.04 |
| 715 | B | (0/997) | (0/139) | (0/451) | 0.394 | 0.04/0.07 | 0.60 (0.54)* | -0.01 | – | – |

Table 91: Blank starts of the machine-learning world beside the reconstructions (seeds 306–325), with their twins at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (0/314) | (0/314) | (10839/12868) | 0.525 | 0.46/0.23 | 1.00 (0.57)* | +0.58+ | 1.00 (0.57)* | +0.58+ |
| 306 | B | (827/8935) | (499/5534) | (0/2862) | 0.477 | 0.26/0.18 | 0.34 (0.42) | +0.01 | 0.31 (0.42) | +0.01 |
| 307 | S | (5962/12527) | (6108/12598) | (2187/4083) | 1.353 | 0.8/0.52 | 0.60 (0.31)* | +0.24+ | 0.67 (0.35)* | +0.35+ |
| 307 | B | (0/2025) | (0/1133) | (1280/11103) | 0.394 | 0.11/0.08 | 0.43 (0.46) | -0.06 | 0.41 (0.47) | -0.05 |
| 308 | S | (0/430) | (0/328) | (9062/12759) | 0.492 | 0.59/0.58 | 0.92 (0.67)* | -0.04 | 0.92 (0.66)* | -0.01 |
| 308 | B | (0/1089) | (0/314) | (0/606) | 0.391 | 0.06/0.07 | 0.60 (0.46)* | +0.02 | – | – |
| 309 | S | (0/286) | (1/279) | (8395/12321) | 0.486 | 0.52/0.53 | 0.66 (0.52)* | -0.04 | 0.90 (0.64)* | -0.04 |
| 309 | B | (0/2192) | (0/1371) | (368/6342) | 0.386 | 0.09/0.07 | 0.33 (0.37) | +0.05+ | 0.32 (0.39) | +0.01 |
| 310 | S | (0/276) | (0/248) | (11367/12847) | 0.528 | 0.53/0.32 | 0.73 (0.42)* | +0.41+ | 0.92 (0.53)* | +0.45+ |
| 310 | B | (0/2387) | (0/1653) | (2128/10413) | 0.404 | 0.05/0.07 | 0.55 (0.48)* | -0.02 | 0.57 (0.49)* | -0.01 |
| 311 | S | (0/312) | (0/237) | (10371/13007) | 0.513 | 0.58/0.6 | 0.92 (0.63)* | -0.04 | 0.92 (0.63)* | -0.04 |
| 311 | B | (0/1356) | (0/1603) | (1605/10974) | 0.395 | 0.05/0.07 | 0.52 (0.44)* | -0.04 | 0.51 (0.48) | -0.08 |
| 312 | S | (0/523) | (0/460) | (8630/12223) | 0.496 | 0.49/0.32 | 0.66 (0.32)* | +0.32+ | 0.98 (0.57)* | +0.49+ |
| 312 | B | (0/1282) | (0/1669) | (1468/9884) | 0.395 | 0.13/0.08 | 0.52 (0.37)* | -0.04 | 0.61 (0.48)* | +0.05 |

Table 91: Blank starts of the machine-learning world beside the reconstructions (seeds 306–325), with their twins at 20,000. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 313 | S | (7730/14033) | (7105/13297) | (738/2534) | 1.547 | 0.81/0.52 | 0.75 (0.41)* | +0.18+ | 0.75 (0.35)* | +0.33+ |
| 313 | B | (0/1329) | (0/571) | (0/399) | 0.392 | 0.06/0.07 | 0.46 (0.36)* | -0.03 | 0.36 (0.40) | +0.00 |
| 314 | S | (0/446) | (0/430) | (9068/12942) | 0.503 | 0.41/0.21 | 0.96 (0.57)* | +0.48+ | 0.98 (0.59)* | +0.50+ |
| 314 | B | (0/2218) | (0/1735) | (1042/9618) | 0.391 | 0.07/0.08 | 0.36 (0.38) | +0.01 | 0.55 (0.53) | +0.01 |
| 315 | S | (0/497) | (0/576) | (6940/12210) | 0.467 | 0.48/0.5 | 0.44 (0.43) | +0.00 | 0.83 (0.61)* | -0.02 |
| 315 | B | (0/565) | (0/952) | (0/374) | 0.393 | 0.04/0.07 | 0.46 (0.41) | -0.07 | 0.64 (0.54) | -0.02 |
| 316 | S | (6909/13790) | (7390/14393) | (1235/2776) | 1.545 | 0.86/0.9 | 0.36 (0.27)* | +0.15+ | 0.62 (0.32)* | +0.38+ |
| 316 | B | (0/228) | (0/155) | (0/414) | 0.394 | 0.05/0.07 | 0.69 (0.46) | -0.05 | 0.69 (0.44) | -0.06 |
| 317 | S | (5587/13735) | (4669/11885) | (2780/4999) | 1.309 | 0.71/0.32 | 0.62 (0.20)* | +0.44+ | 0.73 (0.20)* | +0.60+ |
| 317 | B | (304/8121) | (157/4896) | (1259/7914) | 0.41 | 0.18/0.2 | 0.32 (0.30) | +0.01 | 0.37 (0.30)* | -0.03 |
| 318 | S | (0/422) | (0/372) | (8151/12152) | 0.501 | 0.37/0.19 | 0.98 (0.58)* | +0.49+ | 0.99 (0.58)* | +0.55+ |
| 318 | B | (0/783) | (0/943) | (0/650) | 0.392 | 0.05/0.07 | 0.64 (0.44) | -0.04 | 0.64 (0.42) | -0.05 |
| 319 | S | (7716/14369) | (7813/14311) | (2371/3937) | 1.819 | 0.88/0.91 | 0.84 (0.44)* | +0.35+ | 0.91 (0.50)* | +0.42+ |
| 319 | B | (0/1834) | (0/2429) | (1113/9785) | 0.393 | 0.06/0.07 | 0.24 (0.24) | -0.00 | 0.43 (0.40) | -0.03 |
| 320 | S | (0/366) | (0/418) | (9832/12409) | 0.508 | 0.51/0.33 | 0.86 (0.53)* | +0.43+ | 0.86 (0.53)* | +0.43+ |
| 320 | B | (0/984) | (0/1767) | (0/440) | 0.39 | 0.05/0.07 | 0.27 (0.33) | +0.04+ | 0.37 (0.38) | -0.00 |
| 321 | S | (0/342) | (0/254) | (10350/13378) | 0.513 | 0.51/0.32 | 0.83 (0.53)* | -0.01 | 0.93 (0.60)* | +0.03+ |
| 321 | B | (0/801) | (0/848) | (425/7131) | 0.388 | 0.1/0.08 | 0.45 (0.46) | -0.01 | 0.45 (0.46) | -0.01 |
| 322 | S | (6931/12000) | (7739/13025) | (0/975) | 1.586 | 0.83/0.88 | 0.58 (0.54) | -0.07 | 0.75 (0.67)* | -0.00 |
| 323 | S | (7353/13431) | (7940/14413) | (2929/4334) | 1.532 | 0.86/0.9 | 0.96 (0.42)* | +0.69+ | 0.94 (0.37)* | +0.67+ |
| 323 | B | (0/2145) | (0/1091) | (2158/10607) | 0.403 | 0.06/0.07 | 0.61 (0.43)* | +0.03 | 0.60 (0.42)* | +0.02 |
| 324 | S | (0/491) | (0/279) | (9200/12622) | 0.494 | 0.54/0.56 | 0.77 (0.48)* | +0.01 | 0.92 (0.62)* | -0.05 |
| 324 | B | (288/7269) | (293/6363) | (270/4899) | 0.419 | 0.18/0.19 | 0.29 (0.30) | -0.03 | 0.29 (0.30) | -0.00 |
| 325 | S | (1/574) | (1/752) | (7458/12673) | 0.468 | 0.49/0.52 | 0.77 (0.59)* | -0.03 | 0.88 (0.66)* | -0.04 |
| 325 | B | (0/1961) | (0/1134) | (403/6280) | 0.386 | 0.07/0.07 | 0.33 (0.39) | +0.01 | 0.41 (0.48) | +0.01 |

Table 92: Machine-learning reconstruction, outward from the team world 517 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (6390/13856) | (5747/12718) | (3708/5325) | 1.464 | 0.8/0.44 | 0.34 (0.11)* | +0.34+ | 0.87 (0.24)* | +0.88+ |
| 307 | S | (10553/14636) | (10375/14501) | (2764/3943) | 2.173 | 0.84/0.53 | 0.49 (0.19)* | +0.19+ | 0.58 (0.26)* | +0.30+ |
| 308 | S | (8738/13321) | (9929/14910) | (3690/4226) | 2.128 | 0.89/0.9 | 0.69 (0.23)* | +0.68+ | 0.88 (0.29)* | +0.68+ |
| 309 | S | (0/138) | (0/136) | (12366/13181) | 0.545 | 0.53/0.23 | 0.52 (0.18)* | +0.37+ | 0.83 (0.40)* | +0.45+ |
| 310 | S | (9377/15086) | (9201/14776) | (2492/3185) | 1.982 | 0.89/0.92 | 0.80 (0.30)* | +0.32+ | 0.85 (0.32)* | +0.31+ |
| 311 | S | (0/154) | (0/147) | (12274/13199) | 0.544 | 0.47/0.19 | 0.54 (0.20)* | +0.33+ | 1.00 (0.51)* | +0.77+ |
| 312 | S | (0/120) | (0/115) | (12483/13343) | 0.546 | 0.51/0.22 | 0.60 (0.19)* | +0.19+ | 0.92 (0.42)* | +0.57+ |
| 315 | S | (7567/13737) | (8411/15059) | (3089/3951) | 1.848 | 0.85/0.82 | 0.75 (0.21)* | +0.60+ | 0.88 (0.27)* | +0.86+ |
| 316 | S | (12377/15339) | (12002/14869) | (3002/3551) | 2.461 | 0.84/0.47 | 0.67 (0.20)* | +0.55+ | 0.81 (0.26)* | +0.66+ |
| 317 | S | (12711/15143) | (12370/15011) | (2930/3581) | 2.387 | 0.9/0.94 | 0.51 (0.16)* | +0.11+ | 0.82 (0.26)* | +0.43+ |
| 318 | S | (6616/13074) | (7738/15033) | (3226/3938) | 1.494 | 0.87/0.88 | 0.72 (0.30)* | +0.63+ | 0.92 (0.40)* | +0.54+ |
| 319 | S | (12003/15361) | (10953/13929) | (2654/3098) | 2.336 | 0.9/0.93 | 0.60 (0.13)* | +0.02 | 0.72 (0.19)* | +0.26+ |
| 320 | S | (9416/13565) | (8699/12586) | (3957/4753) | 1.925 | 0.85/0.81 | 0.62 (0.16)* | +0.39+ | 0.75 (0.21)* | +0.52+ |
| 321 | S | (7469/14338) | (7018/13069) | (3034/4382) | 1.551 | 0.83/0.66 | 0.41 (0.13)* | +0.16+ | 0.90 (0.34)* | +0.66+ |
| 322 | S | (10670/15547) | (9686/14333) | (3233/3925) | 2.16 | 0.83/0.46 | 0.60 (0.16)* | +0.55+ | 0.87 (0.24)* | +0.87+ |
| 323 | S | (6212/12985) | (5918/11734) | (3980/4957) | 1.51 | 0.84/0.83 | 0.51 (0.19)* | +0.09+ | 0.88 (0.28)* | +0.46+ |
| 324 | S | (10697/15023) | (10283/14668) | (2989/3525) | 2.223 | 0.89/0.92 | 0.50 (0.16)* | +0.30+ | 0.93 (0.29)* | +0.64+ |
| 325 | S | (9282/15196) | (9587/15213) | (3066/3867) | 2.015 | 0.89/0.92 | 0.66 (0.15)* | +0.60+ | 0.91 (0.29)* | +0.63+ |

Table 93: Machine-learning reconstruction, half level from the team world 517 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 313 | S | (7588/15364) | (7053/14521) | (2832/4020) | 1.716 | 0.89/0.9 | 0.67 (0.29)* | $+0.34^{+}$ | 0.84 (0.38)* | $+0.33^{+}$ |
| 314 | S | (9046/14744) | (9427/15064) | (2713/4101) | 1.873 | 0.89/0.92 | 0.50 (0.15)* | -0.02 | 0.72 (0.25)* | $+0.09^{+}$ |

Table 94: Machine-learning reconstruction, presence level from world 712 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 310 | S | (0/109) | (0/93) | (11699/13020) | 0.541 | 0.61/0.62 | 0.94 (0.65)* | -0.03 | 0.94 (0.65)* | -0.03 |

Table 95: Machine-learning reconstruction, outward from world 714 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (12674/15145) | (12577/14700) | (3007/3483) | 2.601 | 0.9/0.93 | 0.88 (0.33)* | $+0.55^{+}$ | 0.88 (0.33)* | $+0.55^{+}$ |
| 307 | S | (13013/15149) | (12941/14988) | (3024/3559) | 2.638 | 0.89/0.93 | 0.77 (0.26)* | $+0.29^{+}$ | 0.94 (0.34)* | $+0.45^{+}$ |
| 308 | S | (12827/14764) | (12852/14825) | (3104/3740) | 2.546 | 0.9/0.93 | 0.66 (0.26)* | $+0.13^{+}$ | 0.85 (0.37)* | $+0.34^{+}$ |
| 309 | S | (13854/15893) | (12819/14402) | (2934/3595) | 2.62 | 0.9/0.94 | 0.82 (0.38)* | $+0.39^{+}$ | 0.82 (0.39)* | $+0.39^{+}$ |
| 315 | S | (12969/15209) | (12637/14962) | (2775/3423) | 2.521 | 0.9/0.94 | 0.92 (0.35)* | $+0.54^{+}$ | 0.92 (0.35)* | $+0.55^{+}$ |

Table 96: Machine-learning reconstruction, presence level from world 714 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (13789/14975) | (13483/14649) | (2665/3000) | 2.763 | 0.9/0.94 | 0.94 (0.38)* | $+0.57^{+}$ | 0.94 (0.38)* | $+0.57^{+}$ |

Table 97: Machine-learning main set, seeds 512–519 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 512 | S | (0/208) | (0/197) | (11088/12983) | 0.529 | 0.4/0.18 | 0.84 (0.33)* | $+0.56^{+}$ | 0.99 (0.47)* | $+0.67^{+}$ |
| 512 | B | (0/1718) | (0/926) | (452/6711) | 0.388 | 0.09/0.07 | 0.25 (0.32) | +0.03 | 0.49 (0.49) | -0.00 |
| 513 | S | (0/237) | (0/193) | (10346/12675) | 0.523 | 0.44/0.22 | 0.81 (0.45)* | $+0.54^{+}$ | 0.96 (0.60)* | $+0.50^{+}$ |
| 514 | S | (0/227) | (0/188) | (11722/13205) | 0.532 | 0.36/0.15 | 0.83 (0.33)* | $+0.61^{+}$ | 0.99 (0.48)* | $+0.73^{+}$ |
| 514 | B | (0/1174) | (0/1256) | (2815/9861) | 0.42 | 0.07/0.08 | 0.46 (0.45) | -0.07 | 0.48 (0.49) | -0.08 |

Table 97: Machine-learning main set, seeds 512–519 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 515 | S | (6397/13598) | (6289/13131) | (2263/4526) | 1.563 | 0.76/0.37 | 0.86 (0.36)* | $+0.45^{+}$ | 0.92 (0.35)* | $+0.58^{+}$ |
| 515 | B | (0/186) | (0/199) | (0/216) | 0.395 | 0.06/0.07 | 0.42 (0.45) | +0.01 | 0.24 (0.39) | +0.04 |
| 516 | S | (10146/15010) | (9824/14786) | (3000/3954) | 2.06 | 0.89/0.92 | 0.63 (0.23)* | $+0.28^{+}$ | 0.77 (0.30)* | $+0.43^{+}$ |
| 516 | B | (0/2198) | (0/1140) | (1663/10971) | 0.401 | 0.04/0.07 | 0.31 (0.26) | -0.04 | 0.71 (0.49)* | -0.06 |
| 517 | S | (6206/13391) | (6111/13202) | (2141/4039) | 1.567 | 0.79/0.54 | 0.66 (0.25)* | $+0.46^{+}$ | 0.85 (0.31)* | $+0.67^{+}$ |
| 517 | B | (0/916) | (0/1229) | (437/6480) | 0.389 | 0.08/0.07 | 0.65 (0.55)* | -0.02 | 0.65 (0.54)* | -0.02 |
| 518 | S | (8784/15173) | (8777/15465) | (2757/4017) | 1.755 | 0.9/0.93 | 0.59 (0.32)* | $+0.13^{+}$ | 0.58 (0.31)* | $+0.11^{+}$ |
| 518 | B | (0/1781) | (0/1343) | (416/6207) | 0.389 | 0.08/0.07 | 0.19 (0.28) | +0.03 | 0.57 (0.46)* | -0.03 |
| 519 | S | (0/170) | (0/207) | (11300/12991) | 0.526 | 0.54/0.34 | 0.57 (0.41)* | $+0.37^{+}$ | 1.00 (0.62)* | $+0.49^{+}$ |
| 519 | B | (269/6316) | (217/5612) | (271/4485) | 0.417 | 0.17/0.15 | 0.32 (0.46) | $+0.03^{+}$ | 0.32 (0.46) | $+0.03^{+}$ |

Table 98: Machine-learning world, whole at 120 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 512 | S | (0/49) | (0/42) | (2243/2638) | 0.713 | 0.52/0.34 | 0.77 (0.43)* | $+0.32^{+}$ | 0.84 (0.47)* | $+0.37^{+}$ |
| 512 | B | (0/585) | (0/298) | (333/3869) | 0.458 | 0.06/0.07 | 0.34 (0.29) | -0.09 | 0.64 (0.44)* | -0.03 |
| 513 | S | (0/149) | (0/128) | (3841/5253) | 0.686 | 0.42/0.2 | 0.84 (0.41)* | $+0.55^{+}$ | 1.00 (0.55)* | $+0.66^{+}$ |
| 513 | B | (0/399) | (0/179) | (0/227) | 0.46 | 0.04/0.07 | 0.32 (0.42) | +0.02 | – | – |
| 514 | S | (0/310) | (0/179) | (5208/6478) | 0.69 | 0.47/0.23 | 0.82 (0.36)* | $+0.42^{+}$ | 0.97 (0.53)* | $+0.59^{+}$ |
| 514 | B | (0/1688) | (0/1559) | (1408/7782) | 0.468 | 0.06/0.08 | 0.58 (0.47)* | -0.03 | 0.57 (0.48)* | -0.04 |
| 515 | S | (0/165) | (0/153) | (2422/3971) | 0.613 | 0.48/0.49 | 0.52 (0.45)* | -0.00 | 0.70 (0.59)* | -0.04 |
| 515 | B | (0/392) | (0/770) | (685/4832) | 0.465 | 0.07/0.07 | 0.50 (0.34)* | +0.03 | 0.68 (0.47)* | +0.04 |

Table 99: Machine-learning world, births and deaths at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (3400/10837) | (3000/9834) | (2680/5188) | 1.02 | 0.75/0.38 | 0.74 (0.21)* | $+0.48^{+}$ | 0.78 (0.24)* | $+0.66^{+}$ |
| 306 | B | (0/1141) | (0/2125) | (43/2637) | 0.386 | 0.33/0.09 | 0.25 (0.33) | +0.03 | 0.23 (0.31) | $+0.13^{+}$ |
| 307 | S | (0/1155) | (0/657) | (5601/11651) | 0.45 | 0.6/0.39 | 0.79 (0.46)* | $+0.14^{+}$ | 0.85 (0.46)* | $+0.38^{+}$ |
| 307 | B | (0/1540) | (0/1293) | (261/5603) | 0.376 | 0.42/0.09 | 0.15 (0.23) | +0.02 | 0.15 (0.29) | -0.00 |
| 308 | S | (0/999) | (0/1006) | (3850/9949) | 0.44 | 0.5/0.3 | 0.74 (0.55)* | -0.06 | 0.84 (0.59)* | $+0.07^{+}$ |
| 308 | B | (0/719) | (0/797) | (337/5077) | 0.389 | 0.31/0.1 | 0.20 (0.34) | $+0.07^{+}$ | 0.19 (0.34) | $+0.06^{+}$ |
| 309 | S | (0/7744) | (0/6274) | (4118/8815) | 0.423 | 0.42/0.29 | 0.70 (0.35)* | $+0.17^{+}$ | 0.79 (0.37)* | $+0.35^{+}$ |
| 309 | B | (0/2083) | (0/1092) | (1059/9802) | 0.386 | 0.23/0.08 | 0.31 (0.34) | +0.00 | 0.51 (0.49) | -0.02 |
| 412 | S | (0/402) | (1/472) | (7318/12182) | 0.487 | 0.67/0.51 | 0.80 (0.44)* | $+0.28^{+}$ | 0.80 (0.45)* | $+0.26^{+}$ |
| 412 | B | (0/2046) | (0/1792) | (1853/10132) | 0.397 | 0.31/0.08 | 0.40 (0.42) | -0.02 | 0.53 (0.49) | -0.08 |
| 413 | S | (0/599) | (0/433) | (5540/11174) | 0.458 | 0.48/0.45 | 0.56 (0.66) | +0.01 | 0.67 (0.68) | -0.02 |
| 413 | B | (135/4652) | (126/4105) | (740/7770) | 0.401 | 0.33/0.11 | 0.17 (0.26) | -0.10 | 0.19 (0.24) | -0.05 |
| 414 | S | (2505/10520) | (2256/9502) | (2777/5570) | 0.768 | 0.77/0.7 | 0.58 (0.23)* | $+0.33^{+}$ | 0.75 (0.28)* | $+0.47^{+}$ |
| 415 | S | (0/660) | (0/1084) | (6581/11782) | 0.467 | 0.51/0.19 | 0.78 (0.45)* | $+0.33^{+}$ | 0.83 (0.43)* | $+0.48^{+}$ |
| 415 | B | (0/345) | (0/522) | (162/3799) | 0.384 | 0.31/0.09 | 0.24 (0.15)* | -0.02 | 0.37 (0.29)* | -0.00 |

Table 100: Machine-learning capacity arm (8 learners with 2 rows, 8 with 3) at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 320 | S | (0/125) | (0/123) | (12280/12849) | 0.544 | 0.54/0.34 | 0.98 (0.57)* | $+0.45^{+}$ | 0.98 (0.57)* | $+0.45^{+}$ |
| 320 | B | (0/193) | (0/121) | (40/840) | 0.394 | 0.1/0.07 | 0.36 (0.26)* | +0.00 | 0.42 (0.42) | -0.00 |
| 321 | S | (10489/15155) | (10278/14806) | (2731/3613) | 2.016 | 0.9/0.92 | 0.57 (0.22)* | $+0.30^{+}$ | 0.88 (0.37)* | $+0.41^{+}$ |
| 321 | B | (0/998) | (0/1539) | (397/6158) | 0.389 | 0.09/0.07 | 0.42 (0.46) | +0.01 | 0.44 (0.47) | +0.00 |
| 322 | S | (13249/15722) | (11542/13913) | (2337/3502) | 2.427 | 0.91/0.94 | 0.53 (0.30)* | -0.00 | 0.79 (0.45)* | $+0.38^{+}$ |
| 322 | B | (0/1928) | (0/2177) | (1881/9999) | 0.405 | 0.09/0.07 | 0.20 (0.45) | $+0.06^{+}$ | 0.21 (0.46) | $+0.07^{+}$ |
| 323 | S | (9810/11418) | (8862/10308) | (0/227) | 2.036 | 0.82/0.87 | 0.88 (0.60)* | -0.06 | 0.88 (0.60)* | -0.06 |
| 323 | B | (0/199) | (0/199) | (0/220) | 0.396 | 0.07/0.07 | 0.35 (0.40) | +0.01 | 0.41 (0.43) | +0.00 |
| 716 | S | (7722/14829) | (7582/15159) | (2471/3608) | 1.821 | 0.88/0.91 | 0.49 (0.29)* | $+0.27^{+}$ | 0.50 (0.28)* | $+0.27^{+}$ |
| 717 | S | (0/168) | (0/169) | (11887/13078) | 0.535 | 0.55/0.33 | 0.74 (0.43)* | $+0.34^{+}$ | 1.00 (0.61)* | $+0.47^{+}$ |
| 717 | B | (0/2056) | (0/1860) | (2450/10865) | 0.414 | 0.03/0.07 | 0.13 (0.24) | $+0.07^{+}$ | 0.30 (0.50) | $+0.08^{+}$ |
| 718 | S | (0/143) | (0/184) | (11655/12993) | 0.537 | 0.52/0.31 | 0.87 (0.42)* | $+0.35^{+}$ | 1.00 (0.60)* | $+0.50^{+}$ |
| 719 | S | (0/180) | (0/150) | (10125/12902) | 0.524 | 0.56/0.58 | 0.90 (0.63)* | -0.04 | 0.90 (0.63)* | -0.03 |
| 719 | B | (0/1758) | (0/1408) | (1068/11045) | 0.393 | 0.11/0.07 | 0.13 (0.29) | -0.01 | 0.31 (0.39) | $+0.02^{+}$ |

Table 101: Machine-learning capacity arm, every learner with 2 rows at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (0/377) | (0/134) | (6001/9339) | 0.454 | 0.33/0.33 | 0.58 (0.41)* | -0.02 | 0.88 (0.65)* | -0.03 |
| 306 | B | (0/1567) | (0/1340) | (2401/10552) | 0.413 | 0.05/0.07 | 0.30 (0.32) | -0.05 | 0.53 (0.48) | -0.04 |
| 307 | S | (0/206) | (0/168) | (0/143) | 0.396 | 0.13/0.09 | 0.75 (0.41)* | $+0.22^{+}$ | 0.69 (0.60)* | -0.02 |
| 308 | S | (0/176) | (0/196) | (11025/12823) | 0.53 | 0.58/0.61 | 0.94 (0.71)* | -0.03 | 0.94 (0.71)* | -0.03 |
| 308 | B | (0/232) | (0/316) | (0/184) | 0.397 | 0.02/0.07 | – | – | – | – |
| 309 | S | (7114/14157) | (7182/13818) | (2312/4587) | 1.694 | 0.83/0.55 | 0.47 (0.18)* | $+0.15^{+}$ | 0.62 (0.24)* | $+0.35^{+}$ |
| 309 | B | (0/489) | (0/306) | (0/229) | 0.394 | 0.07/0.08 | 0.41 (0.49) | +0.01 | – | – |
| 310 | S | (8934/14095) | (9051/14061) | (3374/4278) | 1.895 | 0.82/0.48 | 0.73 (0.22)* | $+0.73^{+}$ | 0.84 (0.25)* | $+0.83^{+}$ |
| 310 | B | (0/2136) | (0/1841) | (1589/10426) | 0.398 | 0.1/0.08 | 0.44 (0.50) | +0.02 | 0.46 (0.51) | +0.04 |
| 311 | S | (0/158) | (0/164) | (11634/13031) | 0.539 | 0.54/0.32 | 0.76 (0.45)* | $+0.36^{+}$ | 0.91 (0.55)* | $+0.42^{+}$ |
| 311 | B | (0/1424) | (0/1197) | (1185/9789) | 0.399 | 0.05/0.07 | 0.25 (0.30) | +0.01 | 0.41 (0.45) | -0.09 |
| 312 | S | (0/74) | (0/113) | (12001/12949) | 0.541 | 0.48/0.22 | 0.71 (0.33)* | $+0.45^{+}$ | 0.96 (0.51)* | $+0.62^{+}$ |
| 312 | B | (0/1776) | (0/1133) | (1897/11015) | 0.404 | 0.06/0.07 | 0.18 (0.25) | -0.08 | 0.25 (0.36) | -0.05 |
| 313 | S | (11610/16615) | (9287/13472) | (2872/3803) | 1.997 | 0.9/0.93 | 0.65 (0.36)* | $+0.14^{+}$ | 0.69 (0.38)* | $+0.22^{+}$ |
| 313 | B | (0/1273) | (0/1073) | (2818/10750) | 0.42 | 0.06/0.08 | 0.71 (0.49)* | $+0.12^{+}$ | 0.69 (0.48)* | $+0.10^{+}$ |
| 314 | S | (10783/15087) | (10694/14540) | (3219/3997) | 2.347 | 0.89/0.93 | 0.91 (0.30)* | $+0.63^{+}$ | 0.92 (0.30)* | $+0.64^{+}$ |
| 314 | B | (0/1008) | (0/727) | (3259/10676) | 0.428 | 0.06/0.07 | 0.57 (0.38)* | -0.01 | 0.69 (0.49)* | -0.01 |
| 315 | S | (0/190) | (0/164) | (11828/12828) | 0.54 | 0.54/0.33 | 0.68 (0.42)* | $+0.34^{+}$ | 0.99 (0.60)* | $+0.46^{+}$ |
| 315 | B | (0/1446) | (0/1087) | (2516/10290) | 0.416 | 0.06/0.07 | 0.38 (0.31)* | -0.01 | 0.54 (0.50) | -0.05 |
| 316 | S | (0/187) | (0/187) | (10815/13225) | 0.52 | 0.58/0.6 | 0.43 (0.42) | -0.02 | 0.86 (0.64)* | -0.02 |
| 317 | S | (0/181) | (0/158) | (11977/12975) | 0.54 | 0.47/0.23 | 0.76 (0.38)* | $+0.47^{+}$ | 0.93 (0.53)* | $+0.60^{+}$ |
| 317 | B | (45/3765) | (30/2009) | (1625/9518) | 0.405 | 0.07/0.07 | 0.35 (0.30)* | +0.02 | 0.36 (0.31) | +0.03 |

Table 102: Machine-learning main set, seeds 712–715 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 712 | S | (0/95) | (0/165) | (11882/13188) | 0.533 | 0.61/0.65 | 0.87 (0.62)* | -0.04 | 0.87 (0.62)* | -0.04 |

Table 102: Machine-learning main set, seeds 712–715 at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 712 | B | (0/322) | (0/4) | (0/452) | 0.395 | 0.05/0.07 | 0.62 (0.49)* | -0.03 | 0.79 (0.48) | -0.05 |
| 713 | S | (0/234) | (0/326) | (9970/12748) | 0.515 | 0.44/0.24 | 0.96 (0.57)* | $+0.51^{+}$ | 1.00 (0.60)* | $+0.53^{+}$ |
| 713 | B | (0/1776) | (0/1413) | (1207/10356) | 0.394 | 0.05/0.07 | 0.64 (0.47)* | $+0.07^{+}$ | 0.65 (0.51)* | $+0.07^{+}$ |
| 714 | S | (10174/14749) | (10402/15089) | (2924/3897) | 2.027 | 0.89/0.93 | 0.93 (0.35)* | $+0.52^{+}$ | 0.93 (0.35)* | $+0.52^{+}$ |
| 714 | B | (0/593) | (0/1000) | (0/556) | 0.392 | 0.07/0.07 | 0.31 (0.45) | $+0.03^{+}$ | 0.34 (0.43) | +0.02 |
| 715 | S | (0/201) | (0/155) | (11156/12585) | 0.529 | 0.59/0.62 | 0.92 (0.69)* | -0.03 | 0.92 (0.69)* | -0.03 |
| 715 | B | (0/199) | (0/371) | (0/180) | 0.396 | 0.04/0.07 | 0.53 (0.50) | +0.00 | – | – |

Table 103: Blank starts of the machine-learning world beside the reconstructions (seeds 306–325), with their twins at the end of the run. Wholes: consumers' whole draws that scored (scored/drawn) on the first and second corpus in the last span; labelled emb.: the same for labelled embeddings on the first corpus; utility: last window; Gini and top-1: learners' income from consumers; categories by product and by head as in the cipher tables.

| seed | side | wholes A | wholes B | lab. emb. | utility | Gini/top-1 | product acc. | product MI | head acc. | head MI |
|---|---|---|---|---|---|---|---|---|---|---|
| 306 | S | (0/124) | (0/183) | (10992/12825) | 0.533 | 0.46/0.23 | 0.96 (0.54)* | $+0.60^{+}$ | 0.96 (0.54)* | $+0.60^{+}$ |
| 306 | B | (1424/10845) | (1317/9952) | (0/2604) | 0.587 | 0.47/0.24 | 0.28 (0.39) | -0.04 | 0.29 (0.44) | +0.02 |
| 307 | S | (7748/14325) | (6656/13251) | (2180/4144) | 1.613 | 0.82/0.46 | 0.56 (0.26)* | $+0.28^{+}$ | 0.54 (0.25)* | $+0.24^{+}$ |
| 307 | B | (0/2044) | (0/1167) | (1226/10961) | 0.394 | 0.06/0.07 | 0.22 (0.30) | -0.05 | 0.29 (0.49) | -0.00 |
| 308 | S | (0/214) | (0/232) | (10957/12837) | 0.518 | 0.59/0.63 | 0.59 (0.40)* | $+0.07^{+}$ | 0.83 (0.56)* | -0.00 |
| 309 | S | (0/213) | (0/150) | (9213/12958) | 0.496 | 0.53/0.57 | 0.30 (0.41) | +0.00 | 0.91 (0.66)* | -0.03 |
| 309 | B | (0/1635) | (0/869) | (413/6676) | 0.388 | 0.11/0.08 | 0.41 (0.32)* | +0.01 | 0.44 (0.33)* | +0.02 |
| 310 | S | (0/105) | (0/166) | (11924/12813) | 0.541 | 0.55/0.33 | 1.00 (0.59)* | $+0.53^{+}$ | 1.00 (0.60)* | $+0.53^{+}$ |
| 310 | B | (0/1226) | (0/908) | (2389/11039) | 0.411 | 0.08/0.08 | 0.46 (0.38)* | -0.10 | 0.46 (0.41) | -0.10 |
| 311 | S | (0/197) | (0/158) | (11029/13345) | 0.517 | 0.59/0.61 | 0.74 (0.54)* | -0.05 | 0.86 (0.65)* | -0.05 |
| 311 | B | (0/1937) | (0/2316) | (1799/9949) | 0.402 | 0.05/0.07 | 0.36 (0.51) | +0.03 | 0.38 (0.51) | +0.03 |
| 312 | S | (0/212) | (0/180) | (10919/12743) | 0.527 | 0.53/0.34 | 0.84 (0.46)* | $+0.47^{+}$ | 0.97 (0.59)* | $+0.45^{+}$ |
| 312 | B | (0/1896) | (0/1510) | (1751/11091) | 0.4 | 0.08/0.07 | 0.53 (0.42)* | $+0.08^{+}$ | 0.55 (0.47)* | $+0.06^{+}$ |
| 313 | S | (10948/15102) | (11162/15228) | (1523/3219) | 2.146 | 0.84/0.52 | 0.86 (0.41)* | $+0.45^{+}$ | 0.86 (0.41)* | $+0.44^{+}$ |
| 314 | S | (0/211) | (0/227) | (9702/12769) | 0.511 | 0.44/0.23 | 0.99 (0.59)* | $+0.49^{+}$ | 0.99 (0.59)* | $+0.51^{+}$ |
| 314 | B | (0/1188) | (0/1311) | (447/6874) | 0.39 | 0.08/0.08 | 0.26 (0.26) | -0.07 | 0.57 (0.44)* | -0.04 |
| 315 | S | (0/191) | (0/132) | (8978/12957) | 0.495 | 0.53/0.56 | 0.89 (0.69)* | -0.02 | 0.90 (0.68)* | -0.02 |
| 315 | B | (0/17) | (0/1) | (0/375) | 0.393 | 0.07/0.07 | 0.60 (0.50)* | -0.08 | – | – |
| 316 | S | (10150/15449) | (9958/14523) | (1986/3755) | 1.929 | 0.9/0.94 | 0.63 (0.33)* | $+0.27^{+}$ | 0.84 (0.38)* | $+0.36^{+}$ |
| 317 | S | (8469/13798) | (8682/14004) | (3057/4042) | 1.786 | 0.75/0.32 | 0.61 (0.15)* | $+0.61^{+}$ | 0.80 (0.21)* | $+0.87^{+}$ |
| 317 | B | (427/8422) | (312/6792) | (928/5399) | 0.439 | 0.27/0.28 | 0.17 (0.34) | $+0.07^{+}$ | 0.18 (0.35) | +0.04 |
| 318 | S | (0/272) | (0/224) | (9981/12452) | 0.519 | 0.44/0.23 | 0.83 (0.37)* | $+0.48^{+}$ | 1.00 (0.55)* | $+0.62^{+}$ |
| 318 | B | (0/199) | (0/47) | (0/228) | 0.395 | 0.06/0.07 | 0.35 (0.49) | $+0.03^{+}$ | 0.57 (0.54) | -0.01 |
| 319 | S | (10014/15146) | (9346/14492) | (3451/4277) | 2.096 | 0.9/0.92 | 0.65 (0.35)* | $+0.33^{+}$ | 0.90 (0.50)* | $+0.47^{+}$ |
| 319 | B | (0/1894) | (0/1413) | (1334/11071) | 0.394 | 0.06/0.07 | 0.53 (0.39)* | +0.01 | 0.53 (0.41)* | +0.01 |
| 320 | S | (0/244) | (0/187) | (11708/12789) | 0.534 | 0.55/0.33 | 0.81 (0.46)* | $+0.37^{+}$ | 0.99 (0.61)* | $+0.46^{+}$ |
| 321 | S | (0/182) | (0/164) | (10870/12908) | 0.529 | 0.52/0.34 | 0.72 (0.39)* | $+0.41^{+}$ | 0.99 (0.58)* | $+0.49^{+}$ |
| 321 | B | (0/1901) | (0/1610) | (447/7428) | 0.387 | 0.07/0.08 | 0.45 (0.39)* | -0.07 | 0.58 (0.48)* | -0.03 |
| 322 | S | (8083/13880) | (7839/13543) | (0/577) | 1.706 | 0.85/0.9 | 0.54 (0.53) | $+0.07^{+}$ | 0.60 (0.65) | -0.01 |
| 323 | S | (10002/14736) | (10015/14830) | (3256/3908) | 2.072 | 0.89/0.93 | 0.89 (0.25)* | $+0.56^{+}$ | 0.89 (0.25)* | $+0.57^{+}$ |
| 323 | B | (0/1024) | (0/1147) | (2318/10707) | 0.407 | 0.07/0.07 | 0.36 (0.30) | -0.02 | 0.59 (0.46)* | +0.00 |
| 324 | S | (0/250) | (0/240) | (10299/12652) | 0.518 | 0.58/0.6 | 0.62 (0.54)* | -0.02 | 0.77 (0.60)* | -0.03 |
| 324 | B | (369/8140) | (261/6514) | (223/4015) | 0.425 | 0.21/0.2 | 0.37 (0.24)* | -0.02 | 0.33 (0.19)* | -0.01 |
| 325 | S | (0/232) | (0/240) | (7563/12701) | 0.484 | 0.5/0.52 | 0.92 (0.67)* | -0.03 | 0.92 (0.67)* | -0.03 |
| 325 | B | (0/2027) | (0/1312) | (398/6516) | 0.388 | 0.08/0.08 | 0.56 (0.51) | -0.01 | 0.56 (0.52) | -0.01 |

## 11 Other configurations whose records are published with these

The repository holds the readings of several other configurations of the same market. Two of them bear on the designs of record, and one is the world in which the holding ceiling was checked.

**An item read as a single number over the whole tree.** Instead of one number per level, a name is a single number read coarse to fine over the whole tree, which makes a leaf's slice very narrow — $0.656/(4K)$ for a cipher key, which is 0.041 at $K = 4$ — and allows two names that differ only at the leaf to lie far apart as numbers. Plans have 6 lines, and prices are learned as bands rather than listed, with a second check after the draw that the buyer's and seller's bands overlap. This is the configuration in which the holding ceiling of Eq. 1 was checked (Section 6.6); the check was made at a slice width of 0.082.

**A machine-learning world with one corpus and six classifiers.** Consumer weights of 30, 60 and 120 over the goods. The first two goods of the chain formed on every seed, but the prices the agents agreed did not follow the consumers' weights: a whole's mean agreed price stayed between 0.85 and 1.53 times a labelled embedding's, where the weights implied a factor of 2 to 4. A hired whole was therefore selected against, because a part cost its buyer more than the whole it went into paid — the failure of Eq. 2 again. Pricing the goods by a list price times the cube of the score, as the designs of record do, is the answer to that reading.

**Five four-seed variants.** Each changes one thing and was run with its own registered prediction: two cipher layers per class; a judge under which a decoder applied to a class that the good leaves in plain text makes the whole delivery wrong, rather than costing nothing; a floor of 0.03 on the step size of half the learners; half the learners with 3 lines and half with 16; and classifier votes in the machine-learning world. Their readings are in the repository.

## 12 What a run costs

A world costs no money to run and takes about 14 hours for 40,000 rounds on one shared processor core. The sets were run on rented 16-core machines, sixteen worlds to a machine, so a set of eight pairs takes about a day of one machine. Nothing in the model calls a paid service, and a run can be repeated exactly from its seed.

## 13 What is in the repository

In the companion material published alongside:

**the engine** the market, the agents, the draw by sign, settlement, payment and learning.

**the world generators** the programs that write each world's configuration, one per set.

**the readers** the programs that compute every measurement defined in Section 7 from a record.

**the reading files** one text file per world and reading. A name ending `_b`$N$ is the reading from the record cut at round $N$; `_final` is the reading of the whole record at the end of the run; `mlcomp_` files are the composition readings that follow each delivered program back to the agents that made its parts. Readings taken from a record as it stood rather than cut at a round are kept apart under `reads/_lagged` (Section 7).

**the sources of this supplement** and the scripts that build the tables from the reading files.